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PHILIPS FM92E MANUAL

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PHILIPS

FM92E

Conversion

To

6 Meters

Ver 3.9 - **21 Apr 2007**

Rod McCosker VK2DOT

Philips FM92E Conversion to 6 Meters:

Version 3.8 (14 Apr 2007) Rod McCosker VK2DOT

1: Introduction to Project:

This paper is a rough guide for the conversion of a Philips FM92E band transceiver to 6 meters. This project is for the Central Coast Amateur Radio Club's Tuesday Nite Group. The information contained within this document is derived from other Australian 6 meter conversion projects plus our own experimentation. The outcomes of this document, is to enable the easiest method to perform the conversion project. The project will take approximately 12 hours for a component technician or longer for other amateurs. The document should be able to be used by external amateurs to the club who intend to embark on this project.

On the first nite it will be determined if conversion members need a soldering refresher course and, if there are enough numbers to warrant the soldering refresher course, then the Tuesday Nite Group will conduct such a course to enable members to be able to complete this conversion. It is estimated that the Tuesday nite group will take approximately 3 months to complete this project. After each Tuesday nite, this documentation will be updated and uploaded to the VK2AFY web site, so look at the version number above. This should result in the experiences of members being included into this document.

The "Tuesday Nite Group" convenes each Tuesday nite of the month from 7:30pm till 9:00pm. The philosophy and schedule for the group is promulgated on the **vk2afy.dyndns.org** web site under Tuesday Nite Projects & development - **Philosophy**. The meeting Schedule is under the heading – **Meetings**.

The EPROM for the Philips FM92E has been setup for either local or remote heads. The number of channels and the frequency range of the EPROM is 77 channels and from 52.525Mhz to 53.975Mhz. The frequency range has been setup to transmit & receive on the FM allocation within the 6 meter band. There are repeater channels, reverse repeater channels and repeater simplex channels. EPROM's will be supplied to club members who are on the course. At the end of this document is information on programming the EPROM, for external users who intend to program their own EPROM. The software for setting up the firmware is also located on the web site.

This document [**6m_con.doc or 6m_con.pdf**], the 6 meter frequency verses channel list [**6m_fre.rtf**] and the EPROMs [**6m?loc.bin, 6m?rem.bin, 6mtest?l.bin, 6mtest?r.bin** etc] are located on the VK2AFY web site at **http://vk2afy.dyndns.org/6mconv/6mfiles.htm**. Go to the projects section, and click on the **FM92E Conversion to 6 meters**. These files will also be available on the main clubs computer, under the directory "**D:FM92E Conversion to 6 meters**". You will be able to transfer them to your USB memory stick, or print out the documentation on the club's duplex laser printer; You can also burn a CDROM or DVD with the files.

At the end of the 6 meter conversion, Bruce VK2ZAD will be conducting 6 meter antenna building nights. The 6 meter antenna will be a companion for the Philips FM92E we has just converted to 6 meters. He will be supplying a goods required paper – for building the antenna. The building of the antennas will be conducted on the following Tuesday nights, after the conversion is finished.

The FM92E conversion to 6 Meters should start on Tuesday the 6th of March 2007 at the club rooms at 7:30pm till 9:00pm. The club has a number of Philips FM92E's available for conversion. These transceivers will be free to members of the club.

Information for this conversion has been gleaned from and thus acknowledgements to Amateur Radio May 1997, Amateur Radio June 1997, VK2KFJ, VK2TMF, VK2TPH, VK6ZTJ, VK4KTP, VK4KCS, VK2XMD, VK1 Technical Group, VK3GK, VK3BYY, Blue Mountains Amateur Radio Club, SKINNY'S HOME OF RADIO, VK7ZRO plus other Australian amateurs. The information has been collected over 15 years in anticipation to conducting this conversion.

If you need to add extra information or change this document, please send the information to Rod at **vk2dot@iinet.net.au** and we will insert or change your information.

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3: Files Associated with this Project:

6m_con.doc = Documentation to convert Philips FM92E to 6 Meters. **Note:** Approximately 10Mb.

6m_con.pdf = Documentation to convert Philips FM92E to 6 Meters. **Note:** Approximately 3.5Mb.

6m_fre.rtf = Frequency list of channels of EPROM's below.

EPROM's for 6 Meters transmitter & receiver after modification.

6m1loc.bin = EPROM for Philips FM92E with **Local Head**.
Hardware Code: = **9502** 9221 0000 25.0KHz channel, 25W & 10.7 MHz IF.
Software Code = 110400200000000 Check Sum = **C2FF**

6m2rem.bin = EPROM for Philips FM92E with **Remote Head**.
Hardware Code: = **9502** 9222 0000 25.0KHz channel, 25W & 10.7 MHz IF.
Software Code = 110400200000000 Check Sum = **09FF**

6m3loc.bin = EPROM for Philips FM92E with **Local Head**
Hardware Code: = **9502** 9223 0000 25.0 KHz channel, 25W & 21.4 MHz IF.
Software Code = 110400200000000 Check Sum = **B8FF**

6m4rem.bin = EPROM for Philips FM91E changed to FM92E with **Remote Head**
Hardware Code: = **9502** 9224 0000 25.0 KHz channel, 25W & 21.4 MHz IF.
Software Code = 110400200000000 Check Sum = **FFFF**

6m5loc.bin = EPROM for Philips FM92E with **Local Head**.
Hardware Code: = **9502** 9225 0000 12.5 KHz channel, 25W & 10.7 MHz IF.
Software Code = 110400200000000 Check Sum = **AAFF**

6m6rem.bin = EPROM for Philips FM92E with **Remote Head**.
Hardware Code: = **9502** 9226 0000 12.5 KHz channel, 25W & 10.7 MHz IF.
Software Code = 110400200000000 Check Sum =

6m7loc.bin = EPROM for Philips FM92E with **Local Head**.
Hardware Code: = **9502** 9227 0000 12.5 KHz channel, 25W & 21.4 MHz IF.
Software Code = 110400200000000 Check Sum = **2FFF**

6m8rem.bin = EPROM for Philips FM92E with **Remote Head**.
Hardware Code: = **9502** 9228 0000 12.5 KHz channel, 25W & 21.4 MHz IF.
Software Code = 110400200000000 Check Sum =

EPROMs for testing FM92E 66-88 MHz transmitter & receiver before modification.

6mtest1l.bin = EPROM for Philips FM92E with **Local Head**.
Hardware Code: = **9502** 9221 0000 25.0KHz channel, 25W & 10.7 MHz IF.
Software Code = 110400000000000 Check Sum = **AFFF**

6mtest2r.bin = EPROM for Philips FM92E with **Remote Head**.
Hardware Code: = **9502** 9222 0000 25.0KHz channel, 25W & 10.7 MHz IF.
Software Code = 110400000000000 Check Sum = **F6FF**

6mtest3l.bin = EPROM for Philips FM92E with **Local Head**.
Hardware Code: = **9502** 9223 0000 25.0KHz channel, 25W & 21.4 MHz IF.
Software Code = 110400000000000 Check Sum = **ADFF**

6mtest4r.bin = EPROM for Philips FM91E converted to FM92E with **Remote Head**.
Hardware Code: = **9502** 9224 0000 25.0KHz channel, 25W & 21.4 MHz IF.
Software Code = 110400000000000 Check Sum = **F4FF**

NOTE: You must determine what IF your FM92E is using. This will determine what EPROM to use. The IF frequency is normally located on the side panel of the VCO under RF/BW. It will be either 10.7 or 21.4 MHz.

OTHER FILES:

Fpp.zip = Software for generating channel verses frequencies for Philips FM92E for 6 Meters.

4: Philips FM92E Background:



Remote Head for FM92E

This is a general information section on the Philips transceiver model FM-92 before we commence to convert its operation from commercial VHF low band (70-85 MHz) to 6 metres HAM band
There are several versions of the FM-92, for high Band VHF and UHF, this page primarily covers the 99 channel low Band VHF, 25 watt output (E band, 66-88 Mhz model)
The FM-92 is part of the Philips FM-900 series, there are also FM-91 and FM-93 models. It comes in E, A, T, U, W1 & W2 band versions.
This transceiver is EPROM programmable, normally a 2764 (8kbyte) EPROM.

The FM92 is characterised as a mobile base unit, it comes into two variations, a "Local" unit, with front panel fixed to the base unit and the "Remote" unit, which has the front, control panel connected to the main unit by a remote cable with DB-15 connectors at each end approx 25cm L, 20cm W, 6cm D, with a diecast body with heat-sink ribbing and sides and rear panel. Connectors are a 15 pin DB-15 for remote head connection, a BNC antenna socket and 13.8V DC power lead.

The remote Head and Local front panel, includes twin 7 segment displays for channel indication and several buttons for UP/DOWN channel selection, SEND button, AUX button, POWER on/off button, plus Volume and Mute controls, Power, TX & RX indication LEDS.

For those considering using a Phillips FM900 series for amateur radio, and are confused by the band codes used, the Philips FM900 series band plan is as follows:

E Band radios are Lo Band VHF covering 68-88 MHZ: This is the one to convert to 6M
B Band radios are Hi Band VHF covering 132-153 MHZ: These are fine for 2M amateur
A Band radios are Hi Band VHF covering 148-174 MHZ
T Band radios are UHF covering 403-420 Mhz
U Band radios are UHF covering 440-470 Mhz
W1 Band radios are UHF covering 470-500 Mhz: These are the ones for UHF CB use
W2 Band radios are UHF covering 500-520 Mhz

5: Basic Specifications:

- Mode - FM only
- Channels - 99, simplex and/or duplex
- Power output - 1 Watt to 25 Watts
- Power supply - 13.8VDC, (-ve) ground
- Power consumption - 6.5A with Transmitter at 25W
- Scanning - 2 groups of 10 channels
- Channel frequencies, simplex/duplex, scan groups and power levels programmable by EEPROM
- Economiser mode for low power consumption
- IF – 20.4 MHz. Earlier versions 10.7 MHz.



6: A little History:

In the early eighties, Philips Australia started producing a commercial radio for the Australian and international markets. The radios came in various configurations and frequency bands. They where constructed from a one piece die-cast enclosure, with ether a local control face plate or remote control head. The circuitry is contained on 3 main boards (RF, CPU and PA), PCB's are double sided FRG4 fibreglass. All of the components are through hole except the SMT parts mounted on ceramic hybrid SIP's. The whole construction is extremely ruggedised and very solid. The FM900 series came in 3 basic models (FM91, FM92 and FM93). It also came in some special/custom variants and also a waterproof model FM97.

The FM91 is intended as the high end 120 channel model and only came in a remote head version. It has a numeric keypad, function, scan, site, send keys, digital mute and volume, it also has an eight digit seven segment red LED display, most options are programmable from the head.

The FM92 was the standard 40 channel radio (up to 99 CH's), it came in 2 variants, local head and remote head. It has channel up/down, aux and send buttons, analogue mute and volume. The display consists of two seven segment red LED display. The radio can be programmed with almost the same options as the FM91 but it's not changeable from the front panel. The remote head also has a built in speaker. This is the most common FM900 series radio.

The FM93 is the low end 10 channel baby. It has channel up, aux and send buttons, analogue mute and volume. The display is a single seven segment green LED display. It only came in a local head version and differs considerably from the other FM900 series radios, in that it's a servery striped down version, utilising cheaper receiver section and different PLL. It also lacks an A-D chip, therefore it can't facilitate some functions, mainly voting. Sometimes these units actually have FM92 PCB's installed, but they can only display channels 0-9.

The FM97 is basically a waterproof FM92 with an FM91 CPU board. It only came in a remote version, the head was made from a white die-cast jiffy box, it uses Mil-spec bayonet connectors and high quality push button switches with waterproof boots. It has volume up/down, channel up/down, mute, a, b, c function buttons. The display consists of 4, 7 segment red LED displays. According to information from an ex-Philips employee they were designed for police motor bikes!!

The only real deference between the FM91 / FM97 and FM92 is that the FM92 has a standard volume control pot, but the FM91 / FM97 has a 4 bit digital audio attenuator hybrid instead.

All models are based on the Motorola MC146805 CMOS micro controller, and normally have an 8k (2764) EPROM. The EPROM contains the program code and channel data.

All models have provisions for add on modules or "options" as Philips calls them.

```
Selcall                 Type 1
Selcall                 Type 2
CTCSS                   Encode
CTCSS                   Decode
CTCSS                   Encode / Decode
CTCSS                   Encode / Decode and Reverse tone burst
RTB                     ( Reverse tone burst )
Battery Backup          ( CPU current data backup, long duration )
Cap Backup              ( CPU current data backup, short duration )
Extended B/W            for MK2 only
TXCO                    Temperature controlled crystal oscillator Crystal Heater
```

7: Just a few notes, hints, tips and tricks which don't fit anywhere else:

Before swinging back the PC boards, and especially before removing the EPROM, ensure you disconnect the DC supply from the unit. Remember, although the "PWR" switch may be off, there is still DC supplied to the EPROM circuitry and many parts of the boards. On swinging back the boards with DC applied, it is easy to short the 13.8V and 5V supplies. And obviously, removing and replacing EPROMS with power applied, is NOT a good idea!

When reassembling the transceiver unit after disassembly, first spray the panel screws with WD-40, then replace all screws lightly in their respective holes. Then, while maintaining a little pressure on the screw, unscrew slightly until a click is felt, then tighten most of the way. Repeat for all screws, then go around and finally tighten them all. This avoids "second-starting" (cross-threading), which is otherwise easily achieved.

Miscellaneous intermittent problems caused by dry Joints. As mentioned previously, these units have seen service with many commercial operators. Presumably, servicing has been carried out by technicians with various degrees of skill, and possibly in a hurry. As a result of these factors, some of the units have been found to have intermittent problems of all kinds. It has been suggested to me on a number of occasions that all solder joints on each board should be re-soldered!

Be careful with this, however, as some of the pads are very close together, and it is possible to melt the low-temp solder inside the front-end and VCO blocks, causing more problems.

Given the sheer number of joints, and in some cases the extreme proximity of some of these joints (eg, IC connectors) this is not a particularly easy task, but close examination of the circuit boards is certainly warranted.

8: Cleaning up an FM-900 to make it look good:

This assumes, of course, that you don't actually want your new transceiver to have that "lived-in" look. Remember, a clean mind makes you radiate more and be the talking point of your shack.

Since most of these units have seen "active service" in commercial operations including fire brigade tankers and the like, it is unusual to find one which is not absolutely filthy dirty. No problem.

The transceiver unit

- Cover all electrical connectors with tape. Do not remove the top and bottom covers at this time.
- Get a paint brush, maybe 1 or 2cm in width, and brush out the worst of the dust from the grooves and channels.
- It may be necessary to cut the length of the bristles by half, to make a stiffer brush.
- Use a dampened sponge or cloth to finish cleaning as far as possible. Dampen your brush to clean in crevices.
- If the unit is not scratched or badly marked, spray with furniture cleaner (Mr Sheen or similar) and wipe. Another option is Armour-all or similar.
- If the unit is badly marked, obtain a can of auto touch-up spray (Holts or similar) and lightly re-spray the unit. You may like to remove the covers for this, remembering to cover the electrical parts, of course. :)
- If you do remove the covers, place the screws on a cloth and spray them with WD-40 or similar.
- When the unit is dry, remove the tape from the connectors and lightly spray them with WD-40 or similar and wipe off the excess.

The Control (Head) unit

Disassembly

- For this you are going to dismantle the unit.
- fill a sink (with a strainer in the plughole) or any container with warm water and detergent.
- If fitted, remove the mounting bracket from the control unit, and throw the bits in the sink.
- If the microphone connects to the front of the unit (directly under the "PWR" switch), prise it straight off towards you.
- Remove the vol and mute knobs from the front panel. Just pull them straight off. Throw them in the sink.
- Flip the unit over, and undo the four screws from the back corners. Put them on a cloth and spray them with WD-40 or similar.
- Lift off the back cover and throw it in the sink.
- There are two circuit boards in the control unit. They are connected by ribbon cables and need to be removed together, along with the speaker.
- Undo five screws from the upper (back) board, and two screws from the lower (front) board, on either side of the "Ext Spk" switch. (If fitted.)
- Remove the nut holding the "Ext Spk" switch (if fitted) and pull the switch back into the unit. Gently lift the two boards out of the front panel, but only as far as is necessary to reach the screws holding the speaker brackets.
- Remove the screws holding the speaker brackets.
- Remove the circuit boards and the speaker as one unit. On some units, note the position of the hard-wired microphone connector cable.
- From the front circuit board, gently remove the "channel up" and channel down" buttons, as well as the three blue buttons. Just pull them off. Throw the buttons and the front panel in the sink.
- using your cut-down paint brush, scrub all the plastic parts with warm water and detergent, in particular removing all the gunk from the crevices in the front panel, then rinse and dry with a clean cloth.
- It may be necessary to brush the circuit boards with a clean paint brush.

- You may like to spray and wipe the front panel with "Mr Sheen" or similar, or "Armour-all" or similar, or even apply a light coat of clear (matt) varnish (ensure it is safe for plastics!).
- Ensure you now have, nice and clean:
    - two "channel" buttons
    - three blue buttons
    - two knobs
    - the back panel
    - the front panel
    - and all the screws and hardware you started with. :)

Reassembly

- Replace the channel change buttons (arrows pointing out) on the front circuit board switches. They should snap lightly into place.
- Replace the three blue buttons on their switches.
- Ensure the three LEDs are still vertical from the circuit board.
- Re-mount the speaker.
- Replace the circuit boards into the front panel, ensuring that:
    - the three LEDs and five buttons fit properly into their respective holes
    - the white bracket between the circuit boards is correctly positioned (for the mounting screws to locate correctly)
    - the hard-wired microphone cable (if fitted) is correctly positioned.
    - the speaker wires are not caught up
- Replace the screws holding the circuit boards.
- Replace the "Ext Spk" switch (if fitted).
- Replace the knobs (they are different!)
- Replace the rear panel and screws.

Ensure you have no bits left over.

The Microphone

- Undo the four screws at the back.
- Separate the back from the microphone
- blow or brush out any dust
- spray the PTT contacts with WD-40 or similar, wipe off any excess
- Reassemble and wipe with "Mr Sheen" or similar.

Finished!

9: 6 METER FM92E FM FREQUENCIES

Ch	Scan	Rx	Tx					
1	1	52.525Mhz	52.525Mhz	**Voice Main Calling Frequency**				
2		52.550Mhz	52.550Mhz	Voice Simplex for Repeater # 42				
3		52.575Mhz	52.575Mhz	Voice Simplex for Repeater # 43				
4		52.600Mhz	52.600Mhz	Voice Simplex for Repeater # 44				
5		52.625Mhz	52.625Mhz	Voice Simplex for Repeater # 45				
6		52.650Mhz	52.650Mhz	Voice Simplex for Repeater # 46				
7		52.675Mhz	52.675Mhz	Voice Simplex for Repeater # 47				
8		52.700Mhz	52.700Mhz	Voice Simplex for Repeater # 48				
9		52.725Mhz	52.725Mhz	Voice Simplex for Repeater # 49				
10		52.750Mhz	52.750Mhz	Voice Simplex for Repeater # 50				
11		52.775Mhz	52.775Mhz	Voice Simplex for Repeater # 51				
12		52.800Mhz	52.800Mhz	Voice Simplex for Repeater # 52				
13		52.825Mhz	52.825Mhz	Voice Simplex for Repeater # 53				
14		52.850Mhz	52.850Mhz	Voice Simplex for Repeater # 54				
15		52.875Mhz	52.875Mhz	Voice Simplex for Repeater # 55				
16		52.900Mhz	52.900Mhz	Voice Simplex for Repeater # 56				
17		52.925Mhz	52.925Mhz	Voice Simplex for Repeater # 57				
18		52.950Mhz	52.950Mhz	Voice Simplex for Repeater # 58				
19		52.975Mhz	52.975Mhz	Voice Simplex for Repeater # 59				
20		53.000Mhz	53.000Mhz	**Data**				
21		53.025Mhz	53.025Mhz	**Data**				
22		53.050Mhz	53.050Mhz	**Data**				
23		53.075Mhz	53.075Mhz	**Data**				
24		53.100Mhz	53.100Mhz	**Data**				
25		53.125Mhz	53.125Mhz	Voice				
26		53.150Mhz	53.150Mhz	Voice				
27		53.175Mhz	53.175Mhz	Voice				
28		53.200Mhz	53.200Mhz	Voice				
29		53.225Mhz	53.225Mhz	Voice				
30		53.250Mhz	53.250Mhz	Voice				
31		53.275Mhz	53.275Mhz	Voice				
32		53.300Mhz	53.300Mhz	Voice				
33		53.325Mhz	53.325Mhz	Voice				
34		53.350Mhz	53.350Mhz	Voice				
35		53.375Mhz	53.375Mhz	Voice				
36		53.400Mhz	53.400Mhz	Voice				
37		53.425Mhz	53.425Mhz	Voice				
38		53.450Mhz	53.450Mhz	Voice				
39		53.475Mhz	53.475Mhz	Voice				
40		53.500Mhz	53.500Mhz	Voice				
41		53.525Mhz	53.525Mhz	Voice				
42		53.550Mhz	52.550Mhz	Repeater	**NSW**	Albury & Lismore	**Vic**	Melbourne
43	1	53.575Mhz	52.575Mhz	Repeater	**NSW**	Walcha & Snowy	**Vic**	Dandenong
44	1	53.600Mhz	52.600Mhz	Repeater	**NSW**	Narrabri	**Vic**	Gippsland
45	1	53.625Mhz	52.625Mhz	Repeater	**NSW**	Newcastle	**Vic**	Melbourne
46	1	53.650Mhz	52.650Mhz	Repeater	**NSW**	Wollongong		
47	2	53.675Mhz	52.675Mhz	Repeater	**NSW**	Terry Hills Sydney	**Vic**	MelbourneEast & NE Vic
48	1	53.700Mhz	52.700Mhz	Repeater	**NSW**	Goulburn	**Qld**	Sunshine Coast
49	2	53.725Mhz	52.725Mhz	Repeater	**Qld**	KroombitTops/C Qld, Woodridge/BrisbaneSouth		
50	2	53.750Mhz	52.750Mhz	Repeater	**SA**	Summertown/Adelaide		
51	2	53.775Mhz	52.775Mhz	Repeater	**SA**	Crafers/Adelaide	**Qld**	Bundaberg, Atherton Tland
52	1	53.800Mhz	52.800Mhz	Repeater	**WA**	Roleystone/Perth		
53	1	53.825Mhz	52.825Mhz	Repeater	**Tas**	Hobart, Lonah Central N/W Coast		
54	1	53.850Mhz	52.850Mhz	Repeater	**NSW**	Dural/Sydney		
55	1	53.875Mhz	52.875Mhz	Repeater	**NSW**	Lawson/Blue Mts	**Tas**	Mt Barrow/N Tas
56	2	53.900Mhz	52.900Mhz	Repeater	**Vic**	Melbourne		
57	2	53.925Mhz	52.925Mhz	Repeater	**NSW**	ACT & SE NSW	**Qld**	Ipswich
58	2	53.950Mhz	52.950Mhz	Repeater	**Qld**	Brisbane South		
59	2	53.975Mhz	52.975Mhz	Repeater	**Vic**	NE Vic	**Qld**	Mt Gravatt Brisbane

Ch	Scan Rx	Tx	
60	52.550Mhz	53.550Mhz	Voice Reverse for Repeater # 42
61	52.575Mhz	53.575Mhz	Voice Reverse for Repeater # 43
62	52.600Mhz	53.600Mhz	Voice Reverse for Repeater # 44
63	52.625Mhz	53.625Mhz	Voice Reverse for Repeater # 45
64	52.650Mhz	53.650Mhz	Voice Reverse for Repeater # 46
65	52.675Mhz	53.675Mhz	Voice Reverse for Repeater # 47
66	52.700Mhz	53.700Mhz	Voice Reverse for Repeater # 48
67	52.725Mhz	53.725Mhz	Voice Reverse for Repeater # 49
68	52.750Mhz	53.750Mhz	Voice Reverse for Repeater # 50
69	52.775Mhz	53.775Mhz	Voice Reverse for Repeater # 51
70	52.800Mhz	53.800Mhz	Voice Reverse for Repeater # 52
71	52.825Mhz	53.825Mhz	Voice Reverse for Repeater # 53
72	52.850Mhz	53.850Mhz	Voice Reverse for Repeater # 54
73	52.875Mhz	53.875Mhz	Voice Reverse for Repeater # 55
74	52.900Mhz	53.900Mhz	Voice Reverse for Repeater # 56
75	52.925Mhz	53.925Mhz	Voice Reverse for Repeater # 57
76	52.950Mhz	53.950Mhz	Voice Reverse for Repeater # 58
77	52.975Mhz	53.975Mhz	Voice Reverse for Repeater # 59

10: Equipment Supplied: [at no cost to members]

- Philips FM92E band transceiver
- Remote Control Head or Speaker [for local control head].
- 2764 EPROM [already programmed for 6 meters – 77 FM channels].
- VHF Signal Generator
- RF Power Meter
- De-Soldering station
- Soldering Iron *
- 12 volt meter.
- Tuning stick.
- Grid Dip Oscillator.
- 50ohm RF cables.
- This document plus frequency v channel page.
- 50 watt plus soldering iron with big flat tip.
- Large File [rasp] *
- Methylated Spirits
- Tin Snips *
- Philips Screwdriver *

Note: * = Bring your own if you have one.

11: Equipment Required: [to be supplied by each member]

- Soldering Iron [if needed]
- Solder wick [if needed]
- Stanley Knife or Scapple or Box Cutter
- Tin snips [if needed]
- Pair of small tweezers
- Engraver with grinder [if needed]
- Hot glue gun [if needed]
- 12 volt meter [if needed]
- Small plastic GLAD bag for holding screws.
- A4 plastic GLAD bag for holding covers.
- Can of WD40.
- Mr Sheen, Armour-all or similar.
- 1cm or 2cm paint brush for cleaning.
- Sponge or cloth for cleaning.
- Masking Tape for cleaning.
- Tooth brush for cleaning.
- Can of black paint [if needed].
- In-Line Fuse holder & 8 or 10 amp fuse.
- 5 pin DIN plug [for speaker] or 3.5mm plug.
- **0.80mm [B&S 20] Enamelled Copper Winding Wire. [Jaycar WW-4020] will cover aprox 10 sets.**
- **0.63mm [B&S 22] Enamelled Copper Winding Wire. [Jaycar WW-4018] will cover aprox 20 sets.**
- 12 Volt 10 Amp Power cable.
- 1 of 100K ohm potentiometer. Jaycar # **RT-4366**

Receiver Capacitors:

- 4 of 33pF ceramic NPO Capacitors [Jaycar RC-5318] for Receiver. Dick Smith Part No R2247

Transmitter Capacitors:

The following capacitors have to be purchased.

If original capacitors reused, then the following are necessary.

		Jacar Part No	**Dick Smith Part No**
1 of 18 pF	***1 of 18 pF***	***RC5315***	R2241
3 of 33 pF	***3 of 33 pF***	***RC5318***	R2247
2 of 47 pF		RC5320	R2251
4 of 56 pF	***4 of 56 pF***	***RC5321***	R2253
2 of 68 pF		RC5322	R2257
5 of 82 pF	***3 of 82 pF***	***RC5323***	R2259
1 of 100 pF		RC5324	R2259
3 of 120 pF	***3 of 120 pF***	***RC5325***	R2287
1 of 220 pF		RC5328	R2293
1 of 270 pF	***1 of 270 pF***	***RC5329***	R2295

NOTE: We will cover the "Equipment Required" at the night of the first lecture, so do not purchase anything before attending this lecture.

12: Initial Check:

- Unfortunately this can not be completed easily with the CCARC FM92E's supplied. So for the Tuesday Nite Group, the Initial Check will be conducted by installing the 6 Meter Test EPROM. All other users of this document should have a EPROM installed in their FM92E transceiver. Thus they should commence the "initial check" as the first thing done. However be careful when checking the transmitter, do test it into a shielded "Dummy Load", for you do not want to interfere with a commercial operator on the transmitter frequency.
- When you remover the covers from transceiver, do not loose or get mixed up with other transceiver top's & bottom's. These covers normally have information on the EPROM's.
- Plug in Test EPROM into FM92E [only for club members]. See chapter 38 "FM92E Test EPROM 68-88 Mhz Channel verses Frequency List:" and chapter 16 "EPROM Change:".

  6mtest1l.bin = Local Head Test EPROM 10.7 MHz IF.
  6mtest2r.bin = Remote Head Test EPROM 21.4 MHz IF.
  6mtest3l.bin = Local Head Test EPROM 10.7 MHz IF.
  6mtest4r.bin = Remote Head Test EPROM 21.4 MHz IF.
- Check under one of the top or bottom covers for the list of frequencies, that was used originally by the transceiver.
- Switch on Transceiver and check that it works ok. [ie lights work & mute works, etc].
- Go to the nearest Mhz of one of the frequencies [68 – 88 Mhz].
- Go to the channel Number for that Mhz on the chapter 38 "FM92E Test EPROM 68-88 Mhz Channel verses Frequency List:" by the up/down buttons.
- With signal generator check that the receiver works.
- With the RF Power Meter check that RF power is being generated into dummy load & power meter.
- Switch off set if working ok.
- Remove EPROM.
- Plug in EPROM **6m1loc.bin**, **6m2rem.bin**, **6m3loc.bin**, **6m4rem.bin**, **6m5.bin**, **6m6.bin**, **6m7loc.bin** **or** **6m8rem.bin** See page 3 on "Files associated with this project". to EPROM socket. EPROM notch points away from hinges of Synthesiser board.
- Replace EPROM tin box over EPROM, 4 screws.
- Screw down the Synthesiser board.
- Replace cover for Synthesiser board side.
- Clean up FM92E transceiver. * This can be done at any time.

13: Faults:

- If all led's are lit when you switch on, then there may be no EPROM in the transceiver.
- If the transceiver switched off after a few seconds of turn on, then it is expected that a new EPROM is needed.
- If the left dot of the 7 segment display is on, this indicates a programming or memory error.
- If the right dot of the 7 segment display is on, this indicates a power out of range problem.

  The red light comes on when the output doesn't reach a set level, either due to antenna mismatch or PA stage failure.
  It also comes on if the PA over-heats, and automatically backs off the output power.
  Setting the output power trimmer pots on the control PCB too low, or the 25W set trimmer in the PA too low will also bring up the red dot.

  I can't remember precisely, but low battery voltage and out of lock VCO may also turn on the red dot???
  In any case, I'd take a close look at the 13.8 volt supply voltage and current, because it could be a cooked power supply (that may explain why the squelch is open....).
  There is also the infamous internal regulator chip problem (a fix is described elsewhere.).

  Otherwise it's most likely that you've cooked an output transistor if the right dot comes on, since these radios are only made to run with a duty cycle of about 5%.

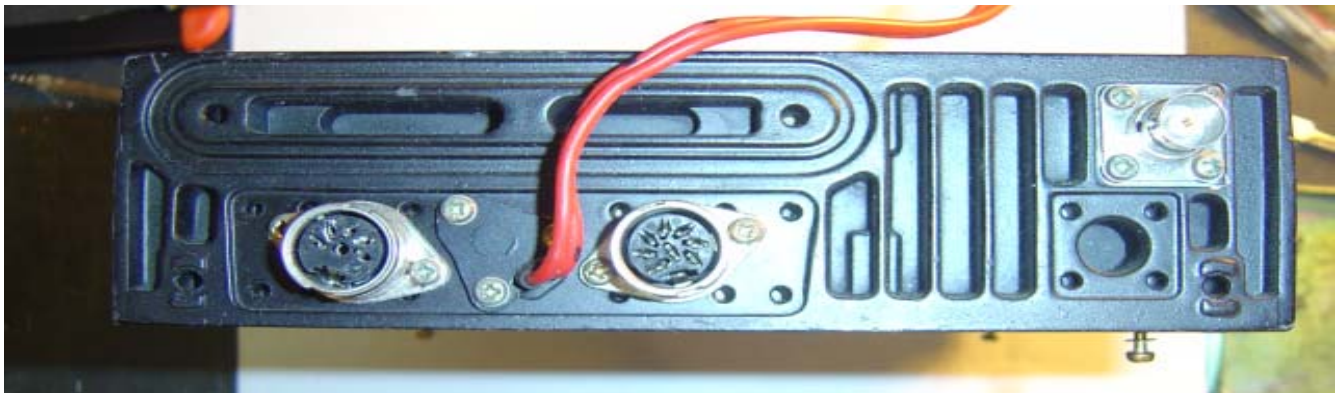
14: Speaker Socket Change:

This change is only conducted if you are converting a Philips FM92E local control head and do not intend to use the rear 5 pin for the speaker socket.
NOTE: The FM900 remote head has a built in speaker, thus this change is not necessary.

5 pin DIN -> Earth - Black Pin # 2
Speaker - Grey Pin # 4
Cradle - Mauve Pin # 1

You only need to change the speaker 5 pin DIN output socket, if you intend to keep all of your speaker leads standard as a 3.5mm socket.

Remove 5pin din socket from rear of chassis.
Replace with flat piece of metal or similar material approximately 20mm by 30mm.
Mark mounting holes from removed 5 pin din socket mounting points.
Mark center from above two marked holes.
Drill 2 holes, the outer two for the two mounting screws.
Drill 1 hole in center to hold 3.5mm socket.
Mount 3.5mm socket.
Black wire to outer shell & **grey** wire to center pin.
Isolate the **move** wire by shrink tubing.

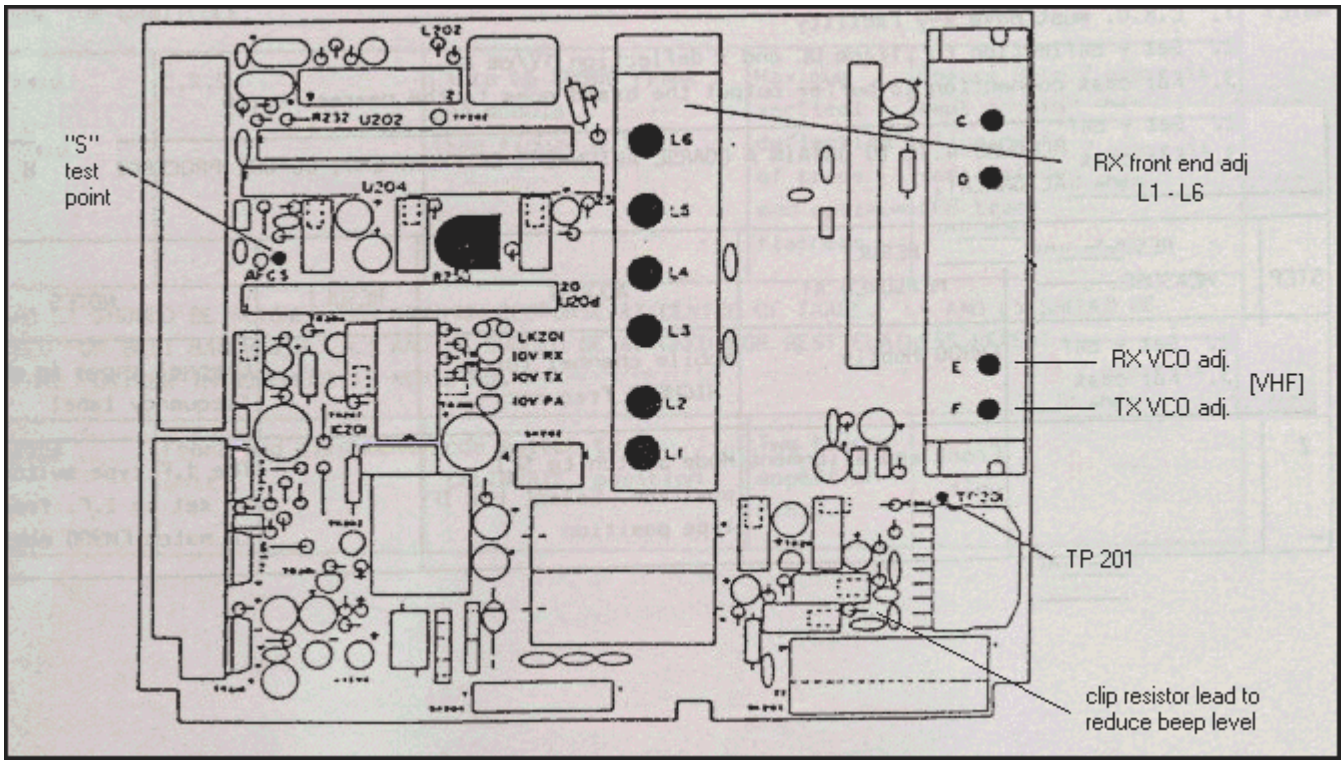


FM92E Rear View.

15: Initial Pull-Apart:

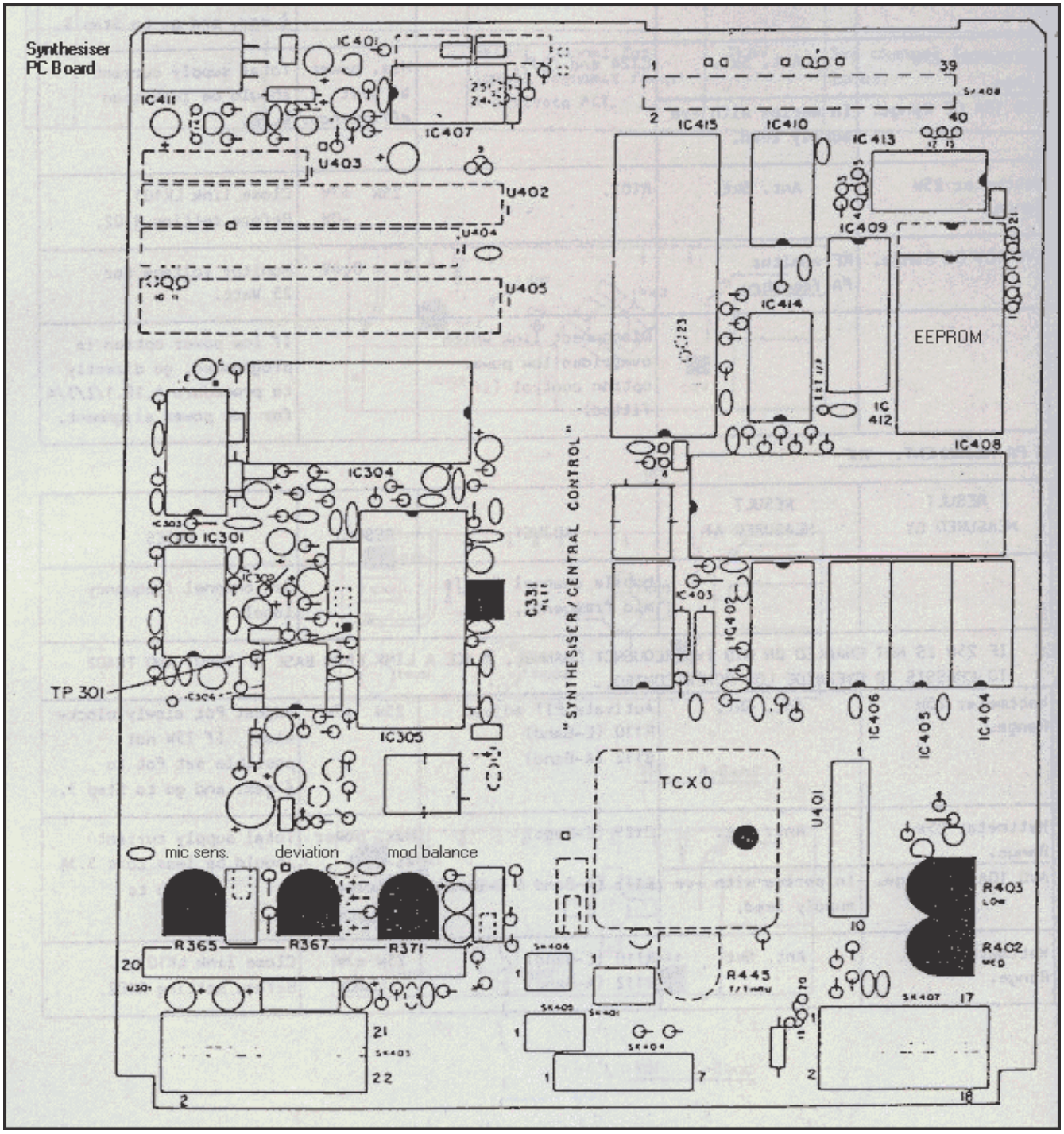
- Remove top & bottom covers from Philips FM92E transceiver.
- Place top & bottom covers into A4 plastic GLAD bag.
- Place all screws into small plastic GLAD bag.

16: Board Layouts:



RF Board:

- All boards are hinged from the front.
- To swing boards upright to allow access to the component side, unscrew the screws on the blank side of the boards, that have an arrow pointing to those screws.
- The screws do not screw all the way out.
- The RF Board has normally 6 screws to undo for the board to swing out.
- The Synthesiser Board has normally 4 screws to undo for the board to swing out.



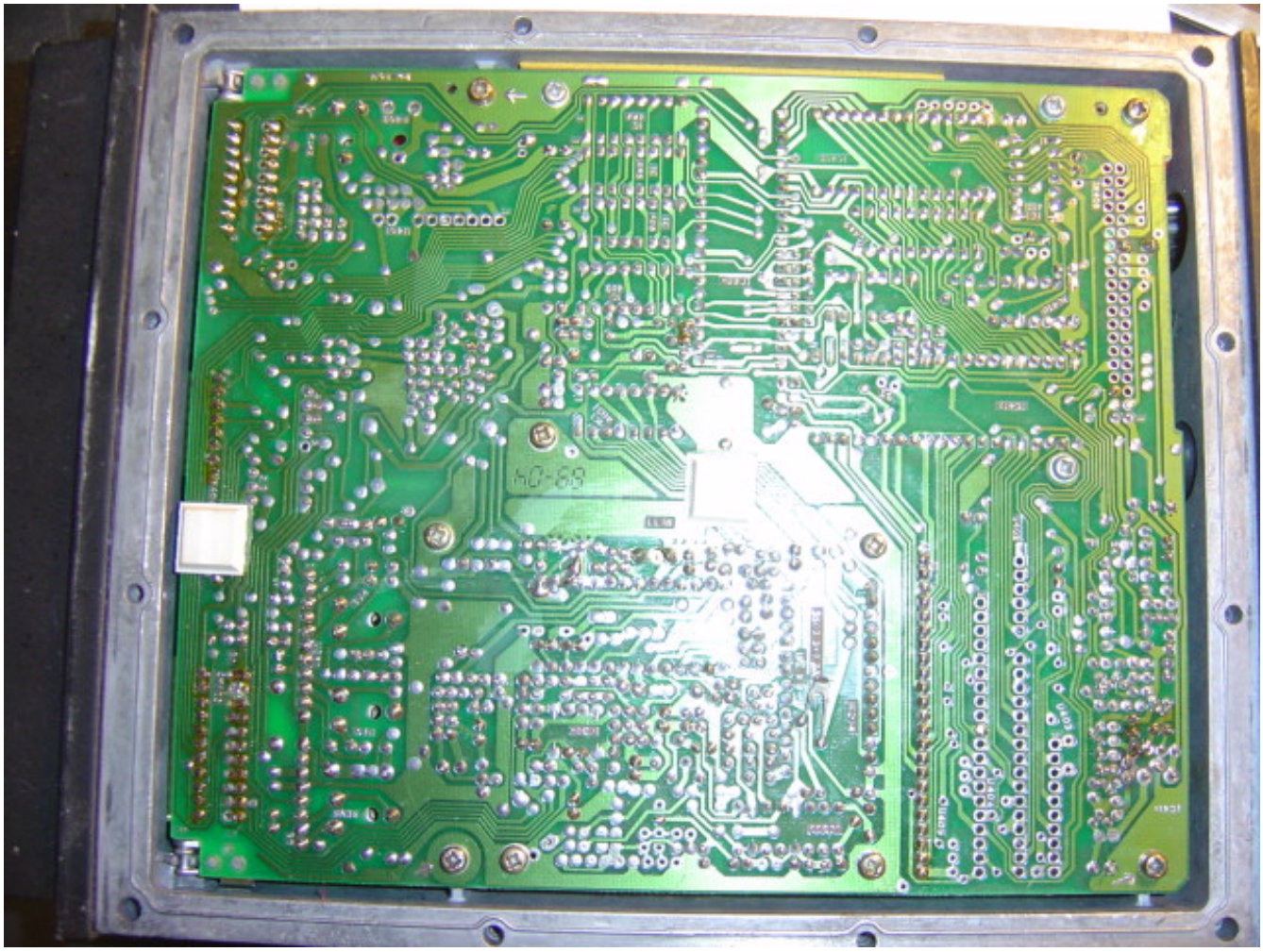
Synthesiser Board:

17: EPROM Change:

- Remove cover on Synthesiser board side from the FM92E Transceiver, this is the side closest to the two sockets at the rear of the FM92E transceiver. Then mark the cover with your call-sign, then place it & the cover's screws into their GLAD bags & mark your Glad bags with your call sign. Should be a minimum of 5 screws.
- Remove cover from receiver side of FM92E transceiver. This is the side furthest away from the rear two connectors; Side closest to the antenna connection. Unscrew the six screws from the Receiver PCB and swing out.
- Check out and note the IF frequency on the side of the RF/Mixer module. [It is normally hand written.] It should be 10.7 MHz or 21.4 MHz.
- Note if the FM92E transceiver is remotely or locally controlled.
- Note the center frequency or single frequency your transceiver was working on.
- Undo screws, holding the Synthesiser board down. Swing Synthesiser board out.
- Remove can covering the EPROM socket. [4 screws] Place Can & Screws into Glad bag.
- Solder single fuse holder onto positive power lead [red cable from transceiver; **NOT** red lead with black dashes]. Or; remove fuse section on earth lead from transceiver, then connect this section via fuse to the positive section of the red wire from the transceiver. * This can be done before EPROM can removal.
- Solder 12 Volt black cable power cable onto the earth lead from the transceiver [red with black dashes] and red onto end of fuse. * This can be done before EPROM can removal.
- Plug in Test EPROM **6mr912t.bin** [EPROM for Philips FM91E converted to FM92E with remote head] to EPROM socket; For testing the FM92E Transceiver from 68-88 MHz. EPROM notch points away from hinges of Synthesiser board.
- Apply 12 Volts to the FM92E Transceiver. Check that audio emanates from the speaker
- With a Signal Generator check that FM92E receiver mute works under 4 micro Volt on test frequency and note this reading.
- Connect VHF Power Meter and press PTT button so a power out is registered on the Power Meter. Note the power reading.
- Disconnect Power.
- Remove EPROM.
- Plug in EPROM **6mr3.bin** [EPROM for Philips FM91E converted to FM92E with remote head] to EPROM socket. EPROM notch points away from hinges of Synthesiser board.
- Replace EPROM tin box over EPROM, 4 screws.
- Screw down the Synthesiser board.
- Replace cover for Synthesiser board side.
- Go to Section 3, and check out which **6m?t?.bin** file meets the above criteria for testing. eg The FM92E is a **remotely controlled** transceiver with a **21.4 MHz IF**. Then download the test file, to initially check the transceiver. [ie download file **6m?t?.bin**.]
- Download the test file that represents your Philips FM92E on the web site, or on the clubs main PC; [http://vk2afy.dyndns.org/6mconv/6mfiles.htm] for burning the 2764 EPROM for your Philips FM92E.
- Check that your FM92E transceiver works by using the information on **Section 38: FM92E Test EPROM 68-88 MHz**.
- Then again go to Section 3, and check out which **6m??.bin** file meets the above criteria for running on 6 meters. eg The FM92E is a **remotely controlled** transceiver with a **21.4 MHz IF**. Then download the actural file, to run the transceiver on 6 meters. [ie download file **6m??.bin**.]
-
- CCARC members will be given a burnt 6 Meter 2764 EPROM's to be placed into their transceiver for testing and running on 6 meters..

- See below for pictures of Synch board [top & bottom] where to place EPROM.

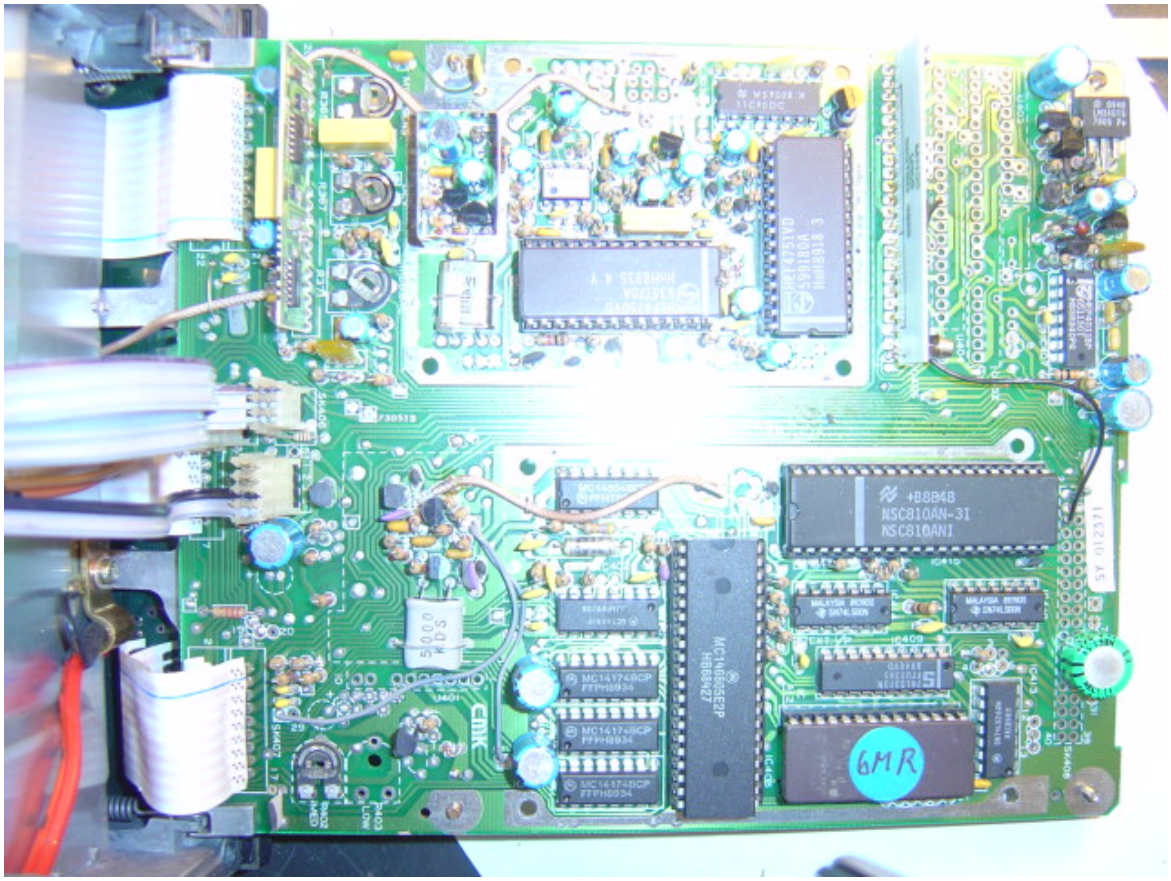




Synthesiser Board [Bottom View]

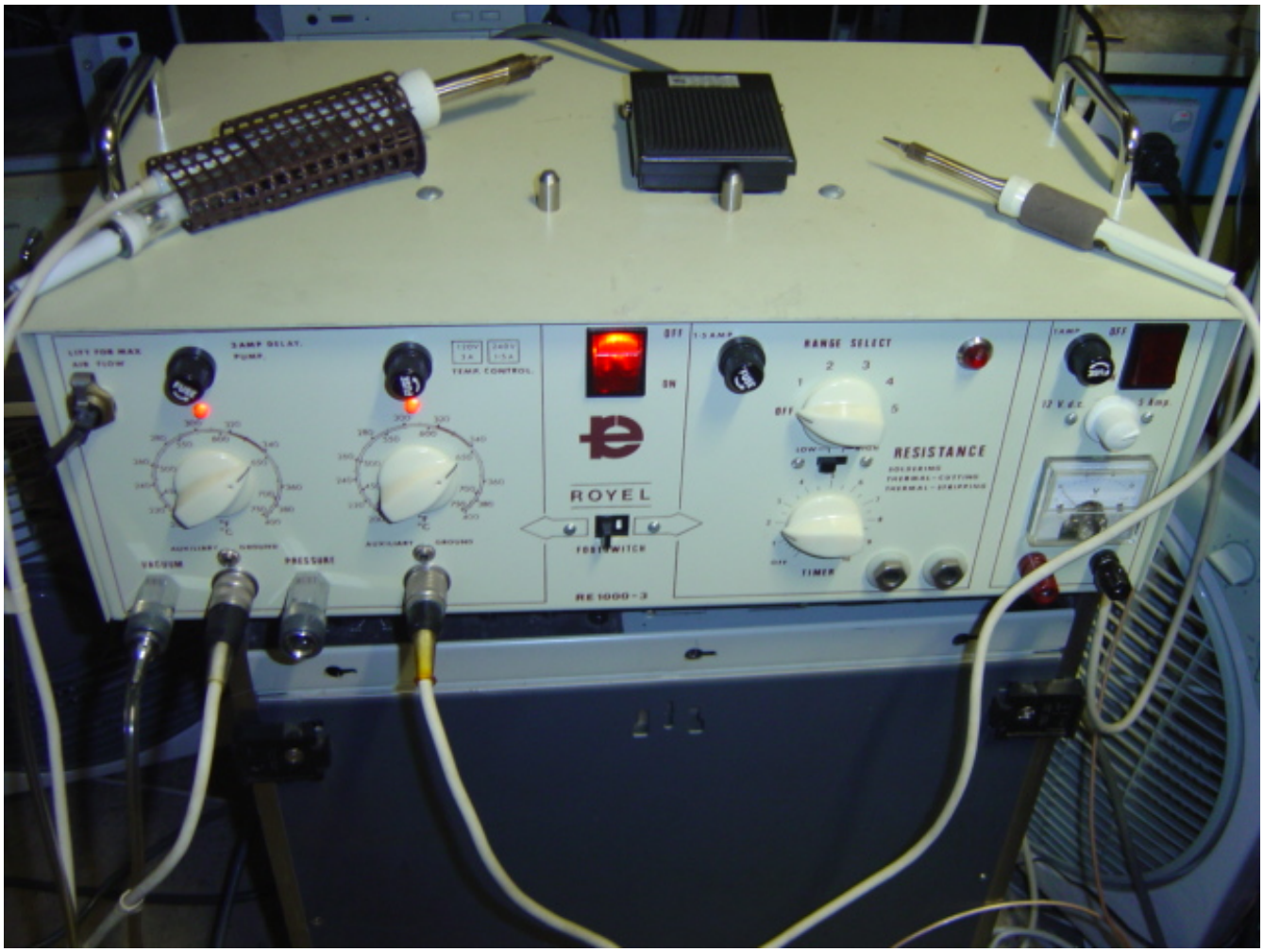


Synthesiser Board [Top View with cans on]



Synthesiser Board [Top View with cans off]

18: Using the Club's De-soldering Station:



Plug in de-soldering tool [Vacuum & power] (left top in above picture) and soldering iron (right top in above picture) into the **Royel de-soldering** Station, then plug into 240V and turn on. Set both to approximately 340 degrees Centigrade. Place vacuum fool control (center top in above picture) on floor and make sure that the foot-switch [on front panel of de-soldering station] is in the left position. Make sure that the de-soldering iron sucks ok when heated up [ie solder is sucked into barrel when foot switch is pressed, (which leaves a hole at the end of the iron tip, barrel)].

The instructions below are based on a right handed person, reverse the instructions if you are left handed. If you have problems using two hands, then get another person to help you with one of their hands.

With left hand pick up soldering iron and place onto the side of the pin to be extracted, then with the right hand feed solder onto the pin [ensure solder flows around pin] then;

Keep the left hand & soldering iron on the side of the pin and, pick up the de-soldering iron with the right hand and place over the pin to be extracted.

With a foot [left or right], press the suction [vacuum] pedal to enable solder to be extracted from around the pin by the de-soldering iron [tool].

Check that the solder has been extracted from around the pin; If not redo the above operation.

Make sure pin is loose and not stuck to the side of the hole. If it is stuck, gently with the de-soldering iron move it to the center of the hole and free.

Do all pins for each module. Then unscrew the two holding screws for each module and gently lever each module off the receiver board. Note: Too much pressure can break a pin or several pins.

“Bugs Juice” The original brew from Bruce VK2ZAD but text modified by Rod VK2DOT.

The following concoction is recommended for users of this document to brew up, this brew will enable your de-soldering to become easier.

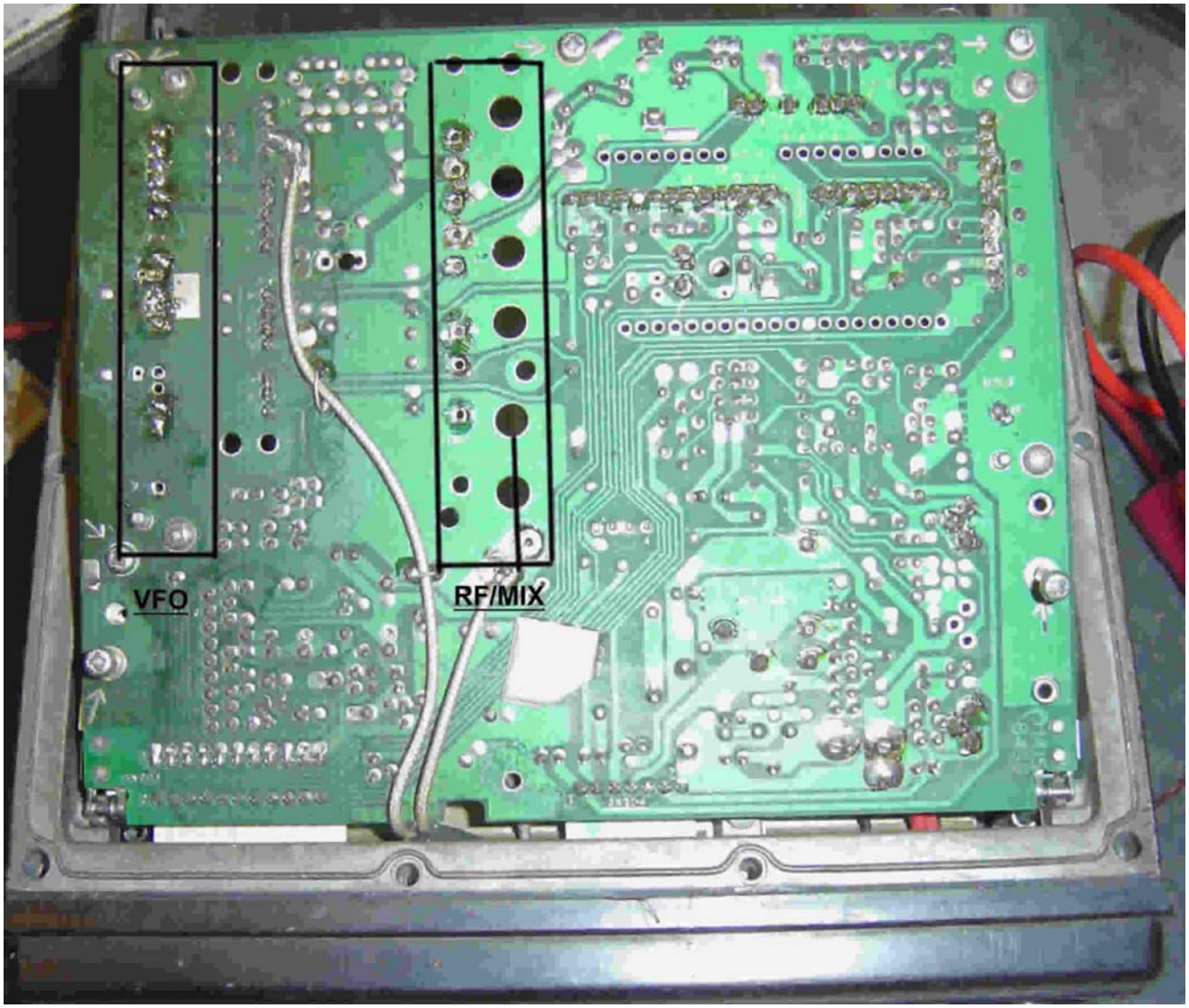
Rosin and Methylated Spirits are purchased from your local hardware store.

To brew up this “**Bugs Juice**”:

- Crush lumps of rosin and place in a jar or a small tin can with an air tight lid.
- Cover the rosin in the jar or can with methylated spirits.
- Apply the air tight lid and allow to dissolve.
- If the final solution is too thick, then thin with more methylated spirits.
- If the final solution is too thin, then thicken by adding more rosin.
- Apply to the surface to be de-soldered with a small brush, old toothbrush or icy pole stick. We have found that a wooden **tooth pick** has been the most successful method of applying the Bugs Juice to the pin requiring de-soldering.

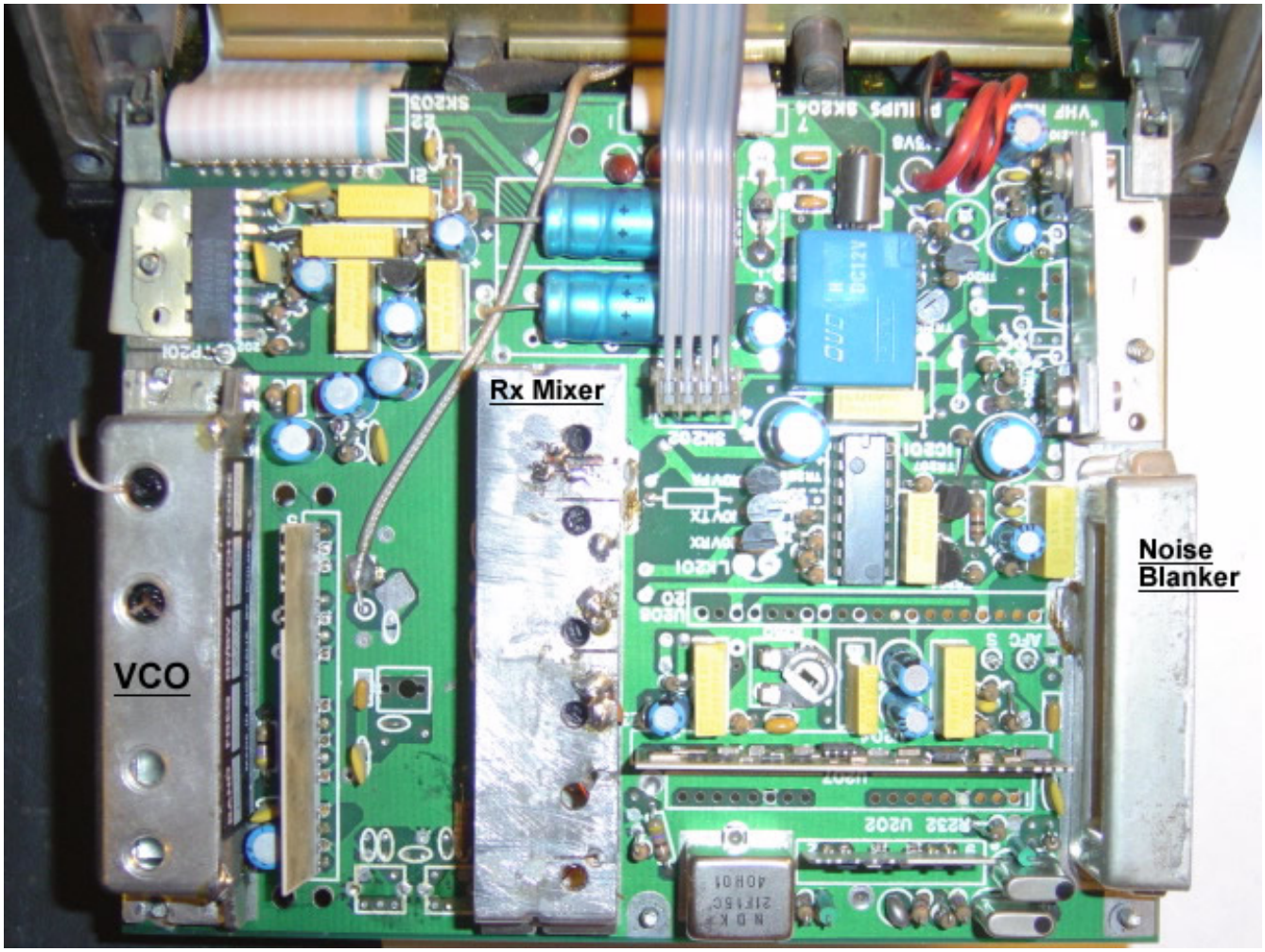
19: VCO Change:

Turn upside down, your FM92E. Remove the base plate. The PCB should be the small one, 12x15cm. Looking down at the board from the front [non-component side], the left hand side back is where the VCO is located on the Receiver board. Note: **VFO** below should be **VCO**.



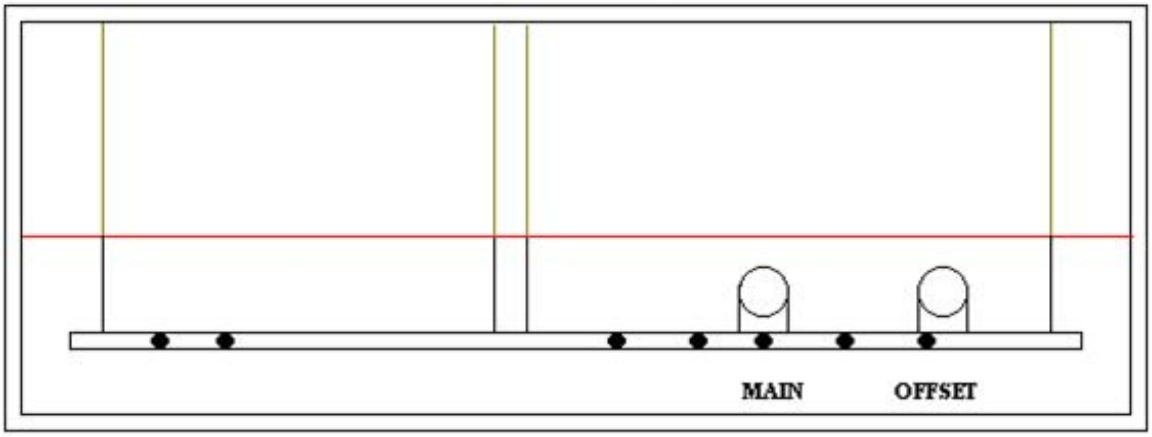
Receiver Board [bottom view]

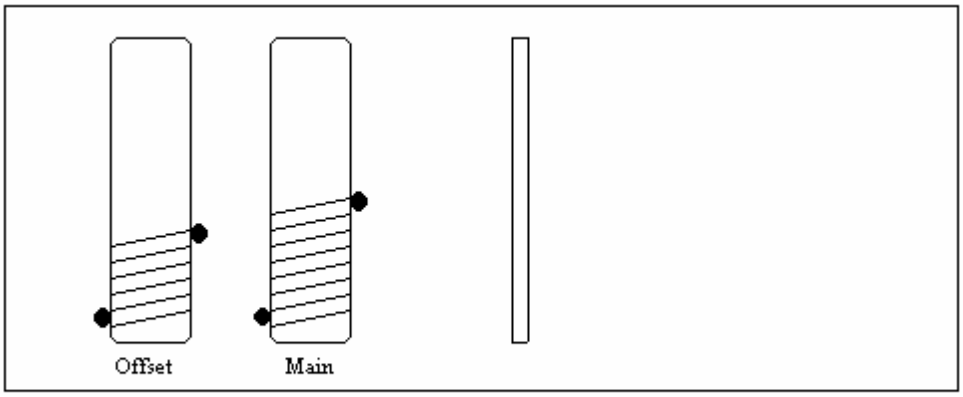
NOTE: You must determine what IF your FM92E is using. This will determine how to wind the VCO Main coil. The IF frequency is normally located on the side panel of the VCO under RF/BW. It will be either 10.7 or 21.4 MHz. In the following examples, the 10.7 MHZ & 21.4 MHz coils are the same.



Receiver Board.

VCO Board





VCO Board

1: Remove the VCO block from the Receiver board by de-soldering the pins and removing the 2 screws. Use a solder sucker or solder wick if you do not have access to a de-soldering station, then ensure that all the solder is removed and the holes in the board are clean (this makes it easy to refit later). The club has a de-soldering station, you will be trained in how to use it during one of the Tuesday night sessions. Be very, very careful of this operation, you can easily break the pins from the module to the "Receiver PCB" – Be **care-full**. You may have to heat the pins up that are holding and gently pry module, this is a three hand job.

   There will be two de-soldering irons at the club rooms, the Rovel De-Soldering station and a Single Shot De-Soldering iron. If you can use the Single Shot De-Soldering iron to remove solder from the non-earth pins and, the Rovel De-Soldering station de-soldering iron for the earth connections of the VCO module. This will enable two users at a time to be de-soldering.

   Firstly, use a wooden **tooth pick** [supplied by the club] to drop VK2ZAD Bruce's **Bug Juice** onto each pin to be de-soldered.

   When you have removed the VCO module, with an old tooth brush, clean the Receiver PCB with methylated spirits. Then, make sure that the VCO ceramic PCB pin holes are clear; if they are not then de-solder them clear. Then re-brush with methylated spirits.

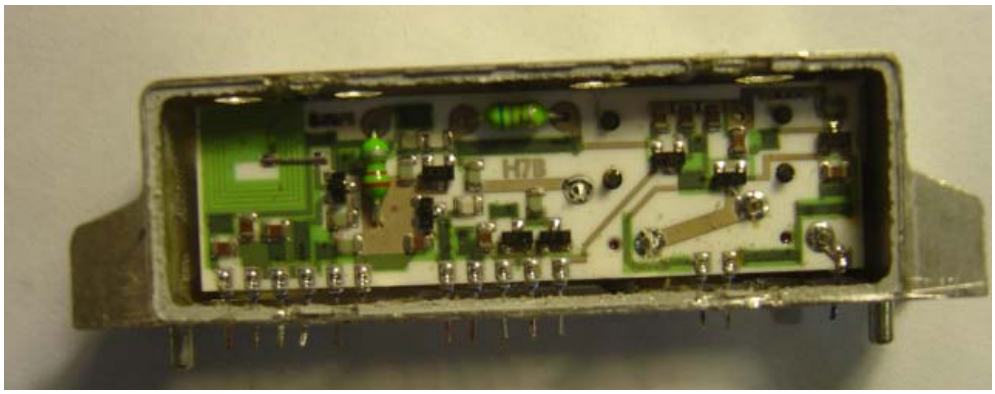


Note the IF frequency [21.4MHz] in the photo above.

2: Once the module is free of the board, put the radio to the side for the time being and concentrate on the VCO. Using a large soldering iron, you need to remove the cover of the VCO. The cover is removed by heating the case along the join and gently prying off the cover whilst ensuring it is not distorted too much, as it needs to be re-sealed later. With the soldering iron on the side of the panel, use a scalpel, small screwdriver, etc, to lift the side panel.

   NOTE: Too much heat and you can unsolder components on the ceramic board, inside of the VCO module.

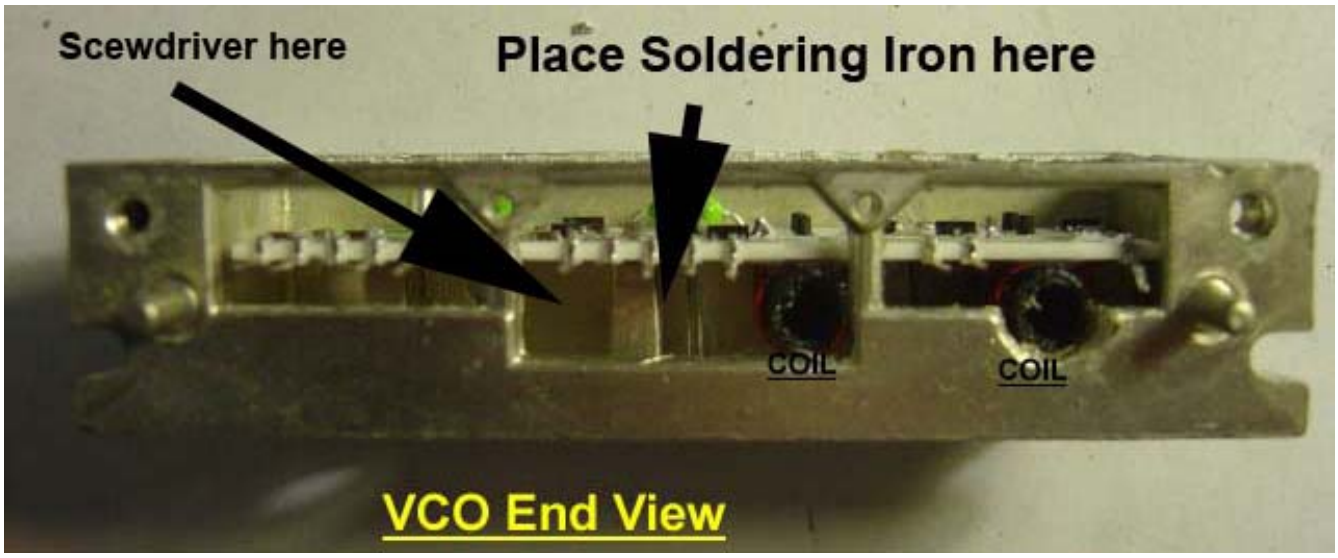
   Or, a better method is to score the solder on four sides of the VFO block with a scalpel; then remove the four sides by prying with the side of the scalpel or a very small screwdriver. As you score the sides, watch your fingers. The scoring causes small slivers of solder to be formed, dispose of these slivers as soon as possible.

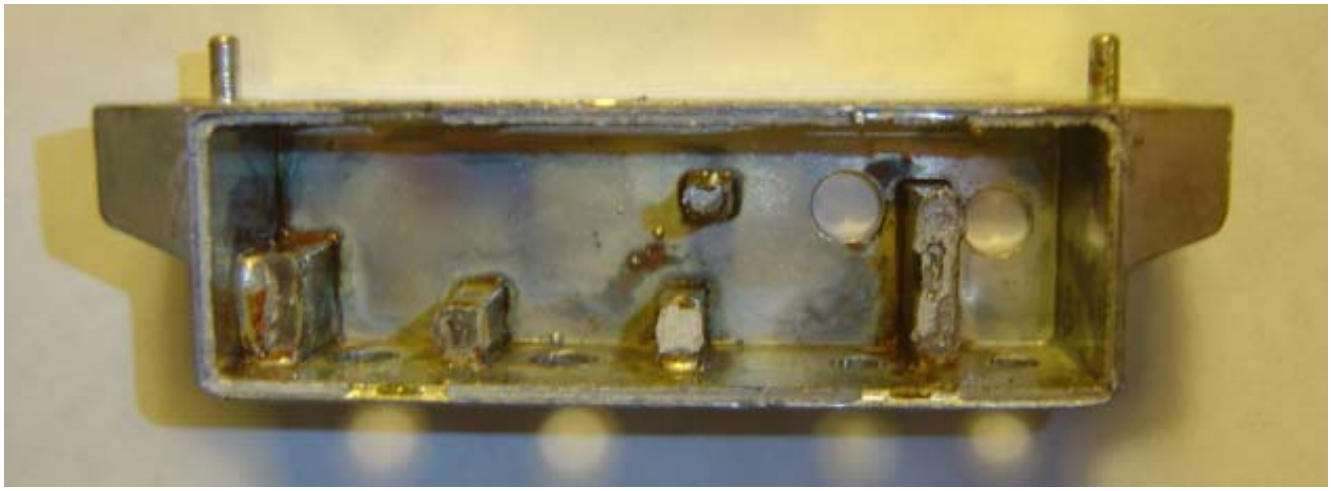


3: Now comes the fun bit, to remove the ceramic board from the VCO module, you need to heat the module case with a soldering iron placed on the side of the center metal post. [see photo below] Where the soldering iron connects to the side of the metal post, which is soldered to the ceramic board, the heat from the soldering iron will travel down the post and the solder will melt, allowing the ceramic board to come free.

You will need to use a pair of pliers or some other device to hold the module whilst this is happening, it will be far too hot to touch! I used a large flat bladed screw driver inserted on the other side of the center post, between the ceramic board and the VCO module case; Then gently twisting the screwdriver as the heat flows into the solder connection between the center post and the ceramic board.

Remember – **gently twist the screwdriver**, you do not want to break the ceramic PCB. You can use the right hand soldering iron of the de-soldering station with success. Too much heat and components on the ceramic board will start to fall off.

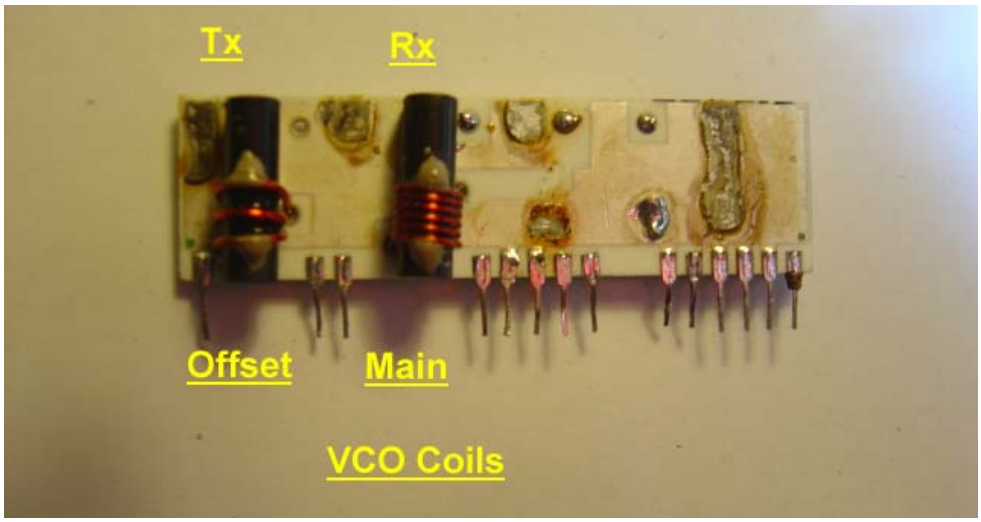




VCO Ceramic PCB Removed above:

4: Once the board is removed from the module, with a smaller soldering iron, de-solder the coils from the board, then with a pair of long nose pliers, gently wobble the coil formers free of the board.

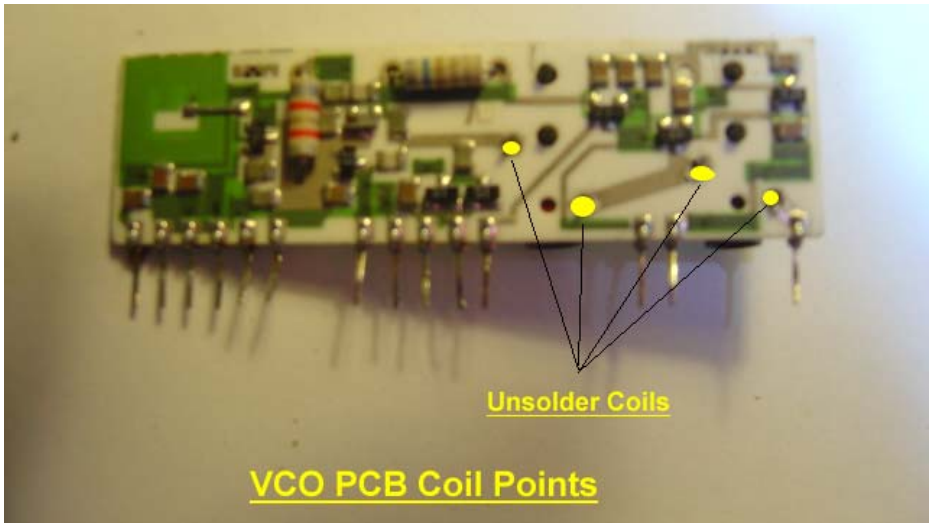
Note: That the coils are normally glued down.



When the 2 formers are free, take the old "main" coil and fit it into the "offset" position, then strip the old Offset coil [remember which way the old offset coil was wound].

If the **IF frequency is 21.4 MHz** then wind a new one of **8 turns** using wire [**150 mm** of enamel coated wire, 0.63mm] [or 146mm of Leon's enamel coated wire, 0.??mm] of the same size (or as near to it as possible); plus wound in the same direction as the old offset coil. This coil needs to be wound tightly and be aligned as per the old one, you may have to distort the ends of the coils to fit them back through the holes. You should remove the enamel coating on the wire and "pre-tin" the wire with solder before fitting, this will ensure a good electrical connection for the coils.

If the **IF frequency is 10.7 MHz** then wind a new one of **8 turns** using wire [**150 mm** of enamel coated wire, 0.63mm] [or 146mm of Leon's enamel coated wire, 0.??mm] of the same size (or as near to it as possible); plus wound in the same direction as the old offset coil. This coil needs to be wound tightly and be aligned as per the old one, you may have to distort the ends of the coils to fit them back through the holes. You should remove the enamel coating on the wire and "pre-tin" the wire with solder before fitting, this will ensure a good electrical connection for the coils.







ie:

- Unsolder the **main** & **offset** coils from the VCO board. Mark each coil as **main** and **offset**.
- When Un-soldering, look at de-soldering points above; do not un-solder **former** mounting points.
- Remember to cut glue off former mounting points. These are on the opposite side to the coil side. You may have to heat up the glue to remove it.
- Be careful of PCB pins, they bend and break easily.
- Refit the old **main** coil into the **offset**'s coil position.
- Strip all wire from the old offset coil. Remember which way the coil was wound.
- Scrape off old glue from former with scalpel and fine file.
- If the **IF frequency is 21.4 MHz** then rewind with **8 turns** of same size of wire [150 mm length, 0.63mm]. [or 146mm of Leon's enamel coated wire, 0.??mm] The same way as the old offset coil.
- If the **IF frequency is 10.7 MHz** then rewind with **8 turns** of same size of wire [150 mm length, 0.63mm]. [or 146mm of Leon's enamel coated wire, 0.??mm] The same way as the old offset coil.
- Refit the old **offset** coil [now 8 turns] into the **main**'s coil position.
- Don't forget to use a hot glue gun or other method of securing the two coil formers to the ceramic board.

Move Main Coil From----------v here
To----------v here





5: As you fit these coils back to the board, we recommend the use of a hot glue gun to both fix the wire to the former, and the former to the board to make sure it can't float about [on the non solder side]. **The next thing to do is to swap the cores between the coils**, and once that is done, it is time to commence fitting it all back together.

At this point in time, check that the VCO PCB pins fit into the "Receiver Board"; if they do not, then **gently** bend them to fit.

You have two options from this point on.

The ***first method*** is the traditional one, where the VCO PCB is replaced into the VCO module, the side of the module is affixed, then the VCO module including the VCO PCB with coils are fitted back to the "Receiver Board". This method is probably the best for mobile operations, however if you intend to use your FM92E converted to 6 meters, then the second method is easier.

The ***second method*** is to cut down the VCO module [without PCB], so the VCO PCB with coils will slip vertically into the VCO module case. This method enables a smooth replacement of the VCO PCB & the VCO module case. This method is the simplest.

Method 1:

6: It's back to the big soldering iron, this time to refit the board back into the module. To do this we recommend that you carefully add some solder to the pads where the module connects to ensure a good joint. Find something raised to rest the board on (a rectangular 9 volt battery is ideal) and place the module over it in the correct place, then apply heat to the module. Again the heat will travel down the posts of the module melting the solder on the board, make sure the board doesn't move at this time, and that the coils align with the holes in the module so you can gain access later for tuning.

Or, glue the PCB with new coils back into the module.

7: When the board has been re-fitted and you are comfortable with the solder joints to the module, refit the cover by sweating the solder around the joints and slowly work your way around the edge. Be careful to overheat the case as the internal board may come loose.

8: Now you need to remount the module back to the board. Carefully fit the pins back through the holes and refit the screws, then solder all the pins making sure that there are no short circuits as a result. When it is all refitted, we can do a quick VCO tune up to make sure what we have done so far is all okay.

ie:

- Replace VCO block back onto main PCB. Screw in 2 screws, then solder pins to board.

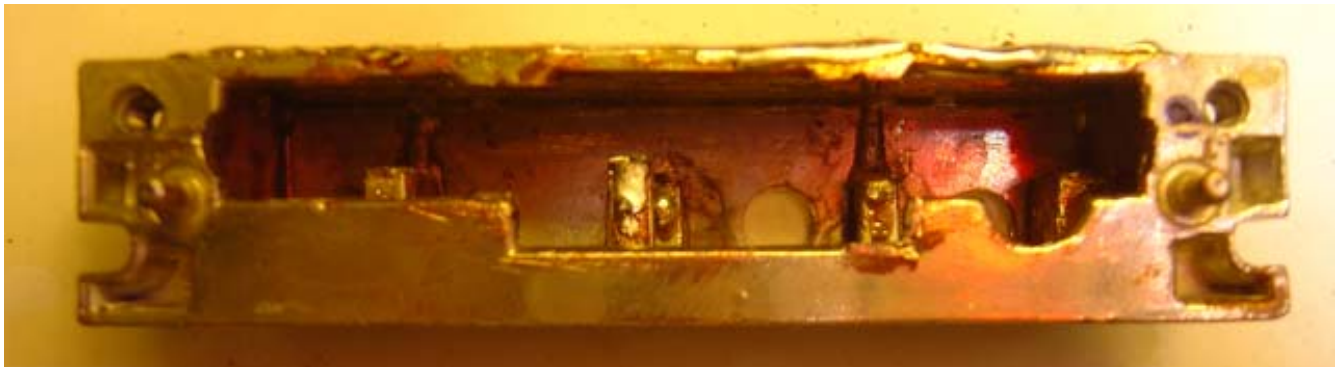
NOTE: In the above operation be very careful of the PCB pins, they bend & break very easily.

Method 2:



VCO Module Case Uncut

6: Cut VCO Module Case so the case will fit over the PCB, after the VCO PCB has been soldered into the Main Receiver Board. Use tin snips, files or/and Dremell tool with grinder tip, use a round file to enable coils to be accommodated through case when placing case over PCB. See picture below.

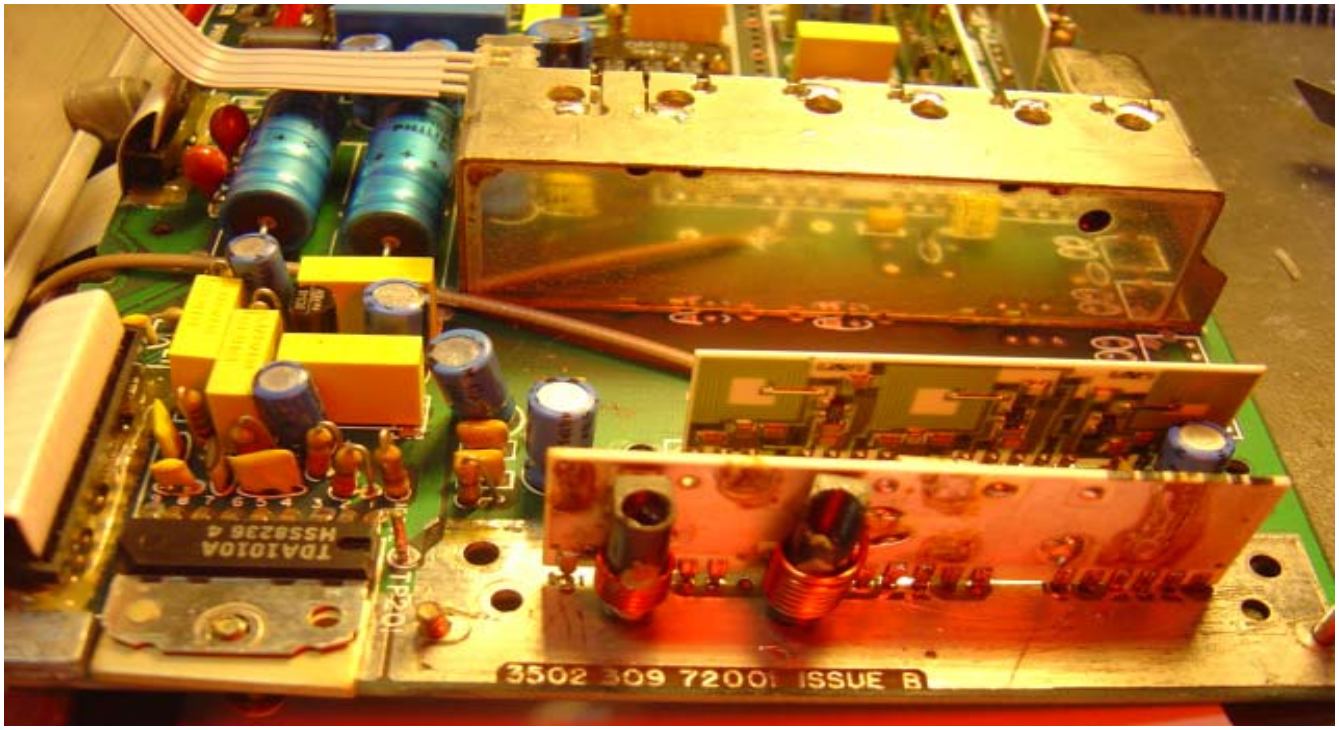


VCO Module Case Modified

7. Clean the receiver board [where the VCO PCB plugs in] with mentholated spirits and a tooth brush. If the holes are not clear, clear with a solder sucker, solder wick, etc.

8: Plug in VCO PCB into the Receiver Board and check that it is upright. Solder the VCO PCB to the board. Check that the soldering is ok. Clean the back of the receiver board with mentholated spirits where you have just soldered.

See picture below.



VCO PCB soldered into Receiver Board

9: Solder side to VCO module can.

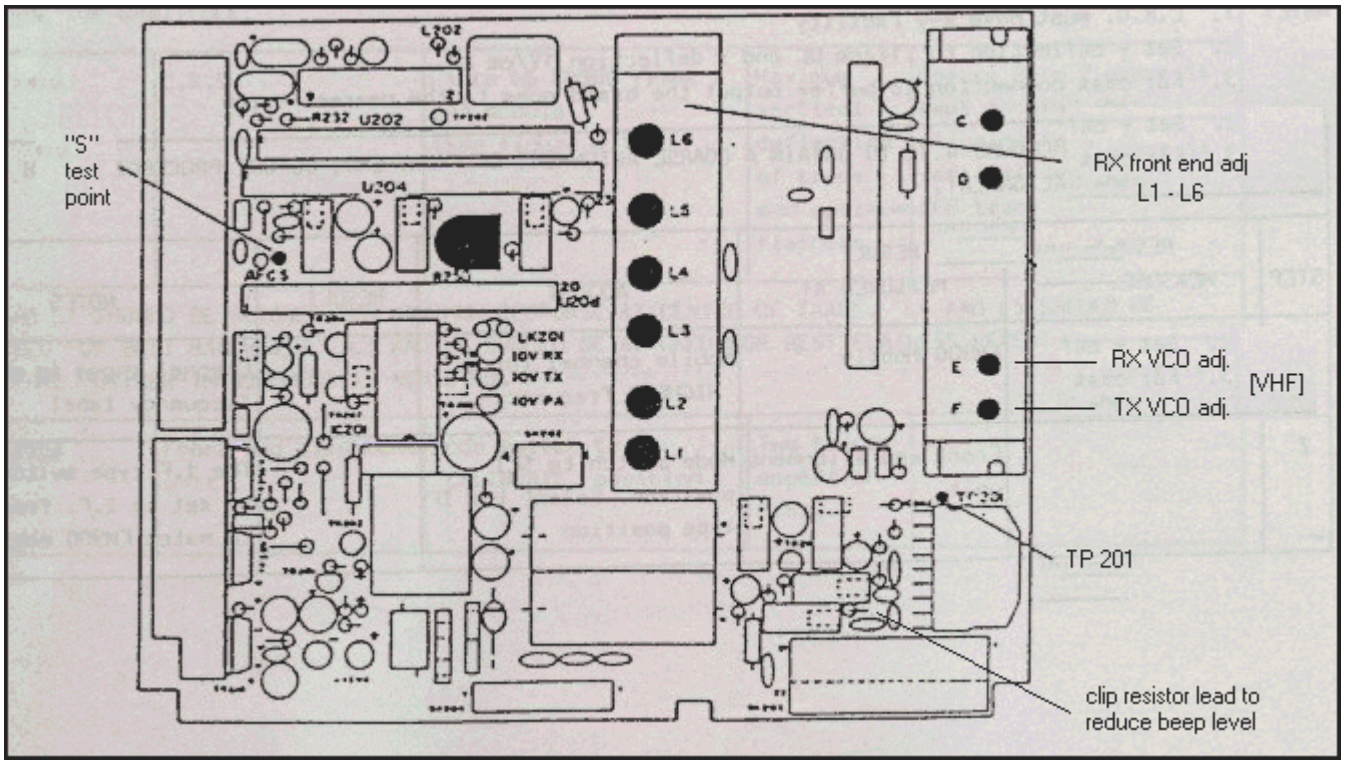
10: You can now check the VCO Tuning [next section] then return back here.

11: Place VCO module over VCO PCB, which has been soldered into receiver board. Do this gently.

12: Screw VCO module to Receiver board with the two screws.

13: The tuning will have changed when you replace the VCO module can. So continue on with “VCO Tune” on next page.

20: VCO Tune:



Rx Board:

Make sure that you have the 6 metre EPROM installed, connect the speaker, microphone, dummy load (through a power meter, however you should not see any TX power yet) and 12 volt DC power, then switch on. Connect a multimeter to the test point below the VCO module, this is TP 201 (positive on pin, negative on case, care should be taken with your test lead, that it does not short to ground on the module case). If you have a push-on socket, then that will even be better.



Set the radio to the highest frequency [channel 59], set the multimeter to a 12 volt DC range and with a plastic, or ceramic tuning tool, adjust the core of the **MAIN coil** until you get a reading of around **11.7 Volts** (not above 12V !). When this has been achieved, key the transmitter and adjust the core of the **OFFSET coil** until you get a matching voltage reading. When satisfied that both RX & TX are about the same, select the lowest frequency [channel 1] and check to make sure the VCO test point voltage is above **4 Volts** DC on both RX & TX (we found ours to be around **5.6 - 7 volts**).

NOTE: Be care-full, when you press the microphone button, you can transmit 5 watts.

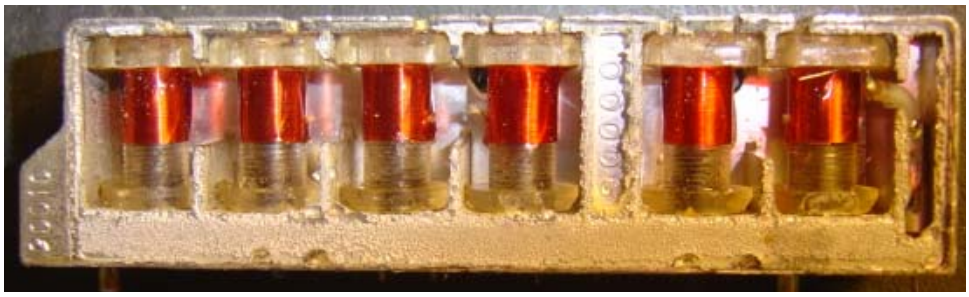
As a final test, with channel 20 setup on the FM92E connect your signal generator to 53.000 MHz. You should hear a signal through the FM92E with approximately 100 mVolt of signal.

NOTE: If the VCO will not tune properly or the transmitter side will not come within voltage range, then check that pin 19 of the VCO module is 0 Volts on receive and 10 Volts on transmit.

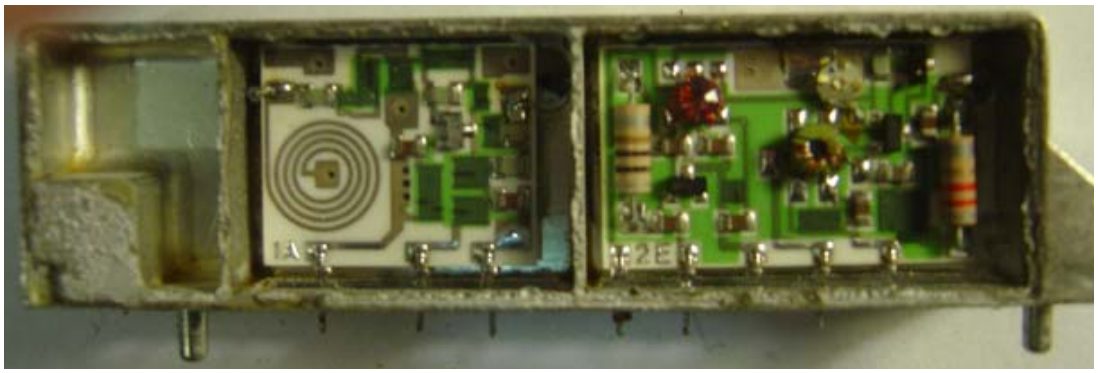
VCO Tune Check References:

Ch #	Rx Freq	Injection Frequency		VCO Voltage		
		21.4MHz	10.7MHz	Rx	Tx	
3	70.000MHz		80.7MHZ	5.3v	4.7V	Not Converted
10	77.000MHz	98.4MHz		9.6V	12.0V	Not Converted
20	53.000MHz	74.4MHz				
20	53.000MHz		63.7MHz			

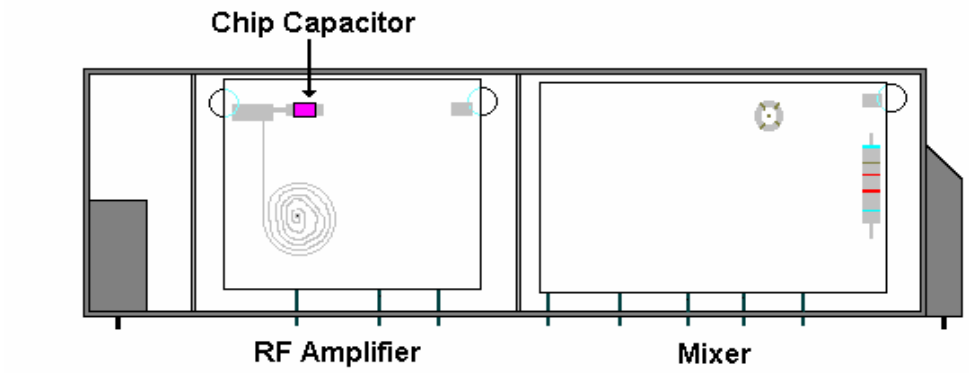
21: Rx RF Amp/Mixer Change:



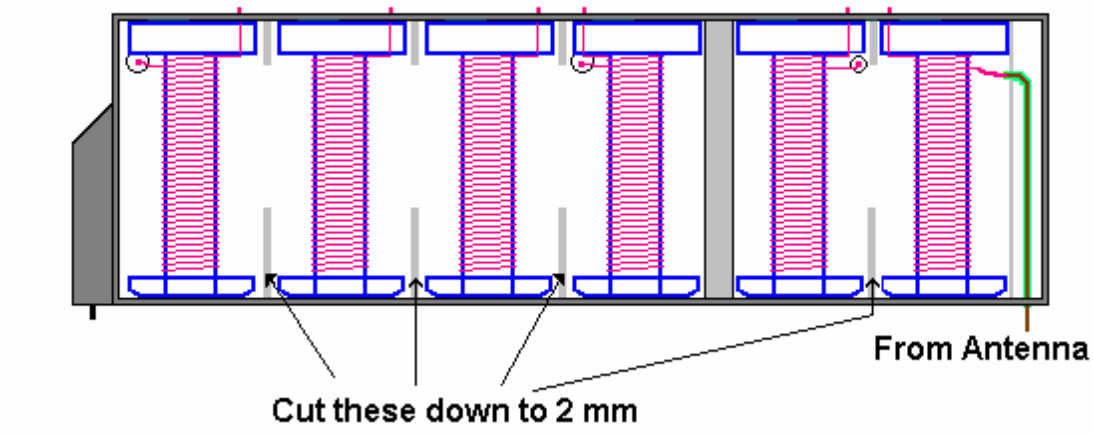
Rx/Mixer Coil Side.



Rx/Mixer PCB Side.



Rx/Mixer PCB Side.



Coil # 6 5 4 3 2 1

Rx/Mixer Coil Side.

There are two methods available to recoil the “RF amplifier / Mixer”. The first example is recommended by other amateurs and clubs. I have found this method very cumbersome and time consuming, so I will recommend my second method.

Rewinding Coils Method #1:

It is not recommended to wind the coils in this first example, but it may suit some people. All previous examples by other clubs and amateurs have used this method.



Coil # 1:

This is RF antenna input coil.
Move coil slug to center of coil.
Remove by unsoldering antenna input wire and wire going to ground at top of coil former.
Gently remove former from housing.
Pry off bottom 1 cm of wire & remove enamel from that 1cm of wire.
Cut enamelled wire [0.125mm] to 250mm length & remove enamel from that 1cm from one end of wire.
Solder two pieces of wire.
Wind additional 12 turns of wire onto coil former, in same direction as other wire.
When 12 turns completed, hold down wire with glue from hot glue gun.
Cut wire at the 12 turn mark. [just added]
Or; strip all wire & wind on 54.5 turns. Earth bottom of coil. With tap 2.5 turns from earth.

Coil # 2:

This is RF amplifier input coil.
Move coil slug to center of coil.
Remove by unsoldering amplifier input wire [wire on rear PCB near chip capacitor].
Remove wire going to ground at top of coil former.
Gently remove former from housing.
Pry off bottom 1 cm of wire & remove enamel from that 1cm of wire.
Cut enamelled wire [0.125mm] to 250mm length & remove enamel from that 1cm from one end of wire.
Solder two pieces of wire.
Wind additional 12 turns of wire onto coil former, in same direction as other wire.
When 12 turns completed, hold down wire with glue from hot glue gun.
Cut wire at the 12 turn mark. [just added]
Or; strip all wire & wind on 54 turns turns. Earth bottom of coil. With tap 2 turns from earth.

Cut down spacer between coil 1 & 2 to height of approximately 2mm. Then grind down sharp edges.

Coil # 3:

This is RF amplifier output coil.
Move coil slug to center of coil.
Remove by unsoldering amplifier input wire [wire on rear PCB near chip capacitor].
Remove wire going to ground at top of coil former.
Gently remove former from housing.
Pry off bottom 1 cm of wire & remove enamel from that 1cm of wire.
Cut enamelled wire [0.125mm] to 250mm length & remove enamel from that 1cm from one end of wire.
Solder two pieces of wire.
Wind additional 12 turns of wire onto coil former, in same direction as other wire. Total of 54 turns.
When 12 turns completed, hold down wire with glue from hot glue gun.
Cut wire at the 12 turn mark. [just added]
Or; strip all wire & wind on 54 turns. Earth bottom of coil. With tap 2 turns from earth.

Coil # 4:

This is RF amplifier 1st output bandpass coil.
Move coil slug to center of coil.
Remove wire going to ground at top of coil former.
Gently remove former from housing.
Pry off bottom 1 cm of wire & remove enamel from that 1cm of wire.
Cut enamelled wire [0.125mm] to 250mm length & remove enamel from that 1cm from one end of wire.
Solder two pieces of wire.
Wind additional 11 turns of wire onto coil former, in same direction as other wire. Total of 53 turns.
When 11 turns completed, hold down wire with glue from hot glue gun.
Cut wire at the 11 turn mark. [just added]
Or; strip all wire & wind on 53 turns. Earth bottom of coil.

Coil # 5

This is RF amplifier output 2nd bandpass coil.
Move coil slug to center of coil.
Remove wire going to ground at top of coil former.
Gently remove former from housing.
Pry off bottom 1 cm of wire & remove enamel from that 1cm of wire.
Cut enamelled wire [0.125mm] to 250mm length & remove enamel from that 1cm from one end of wire.
Solder two pieces of wire.
Wind additional 11 turns of wire onto coil former, in same direction as other wire. Total of 53 turns.
When 11 turns completed, hold down wire with glue from hot glue gun.
Cut wire at the 11 turn mark. [just added]
Or; strip all wire & wind on 53 turns. Earth bottom of coil.

OR;

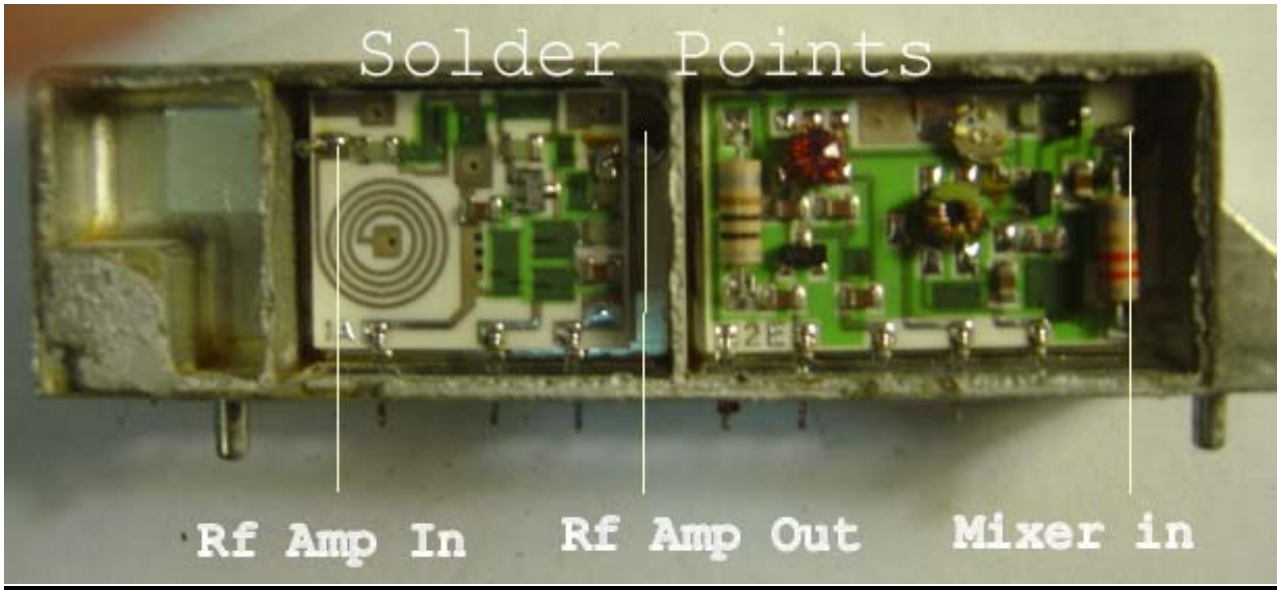
Rewinding Coils Method #2:

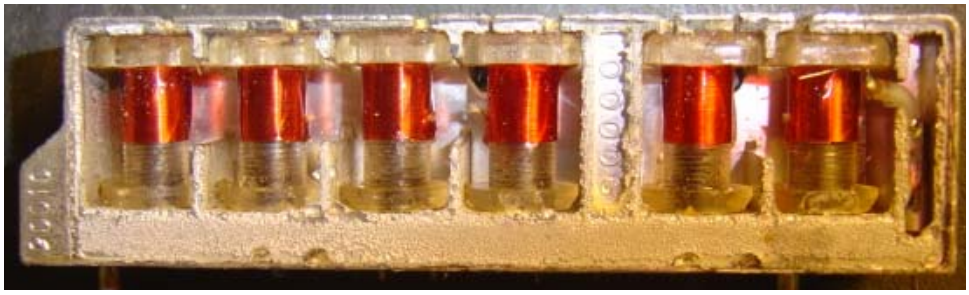
- Rasp off all solder from top of the module with a file. This will break the coils earth wire points, you will not be using the wire for earthing by this method again.



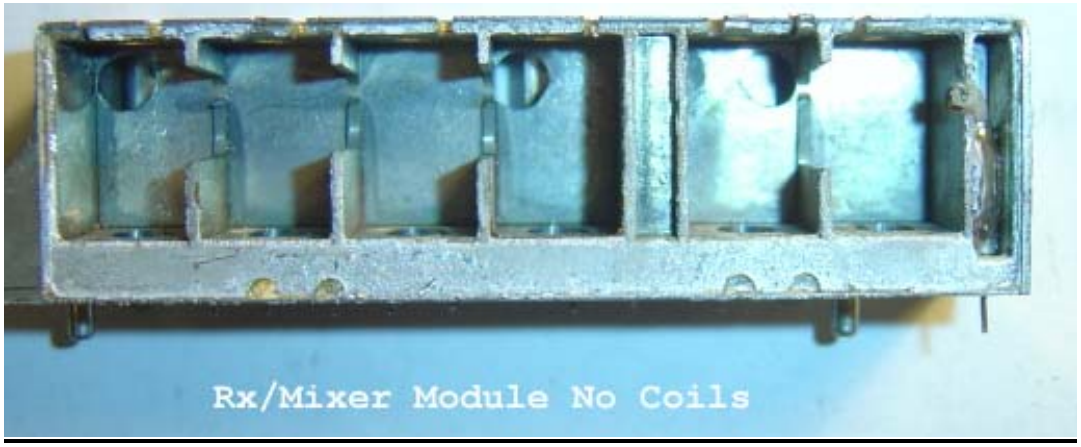


- Unsolder coil wire from RF amplifier input & RF amplifier output, plus mixer input and antenna input.
- Move coil ferrites to middle of coil formers.
- Gently remove all coils from block.
- Strip all wire from all coils and, keep the wire previously stripped.
- Clean coil tops & bottoms, plus clean module coil inserts from old glue.
- File down and clean all earth connections on block.

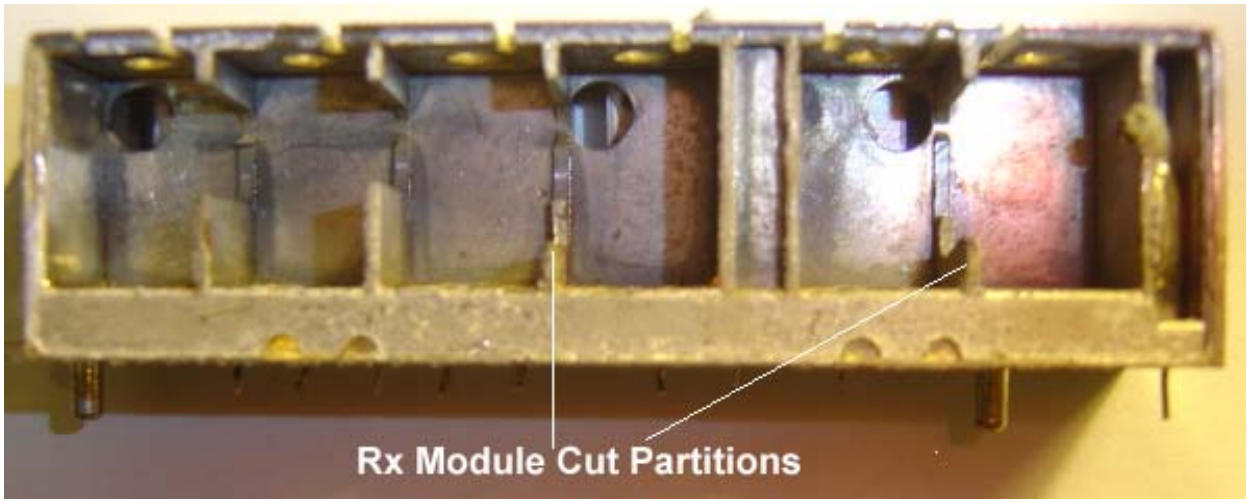




Original Rx Coils



Cut out 50% of the two partitions between coil #1 & coil #2; Plus between coil #3 & coil #4. With a pair of tin snips.



NOTE: All coils have the same fundamental start for winding:-

- Wind all the following coils in the same direction.
- Cut 4 lengths of 160mm of 0.8mm enamel coated wire.
- After cutting wire lengths, clean off all enamel from two ends of wire, one end of 5mm in length cleaned, the other earth end 10mm to be cleaned.
- Solder both ends of 0.8mm wire.
- Bend 8mm of the 0.8mm wire at [10mm tinned end] at right angles.



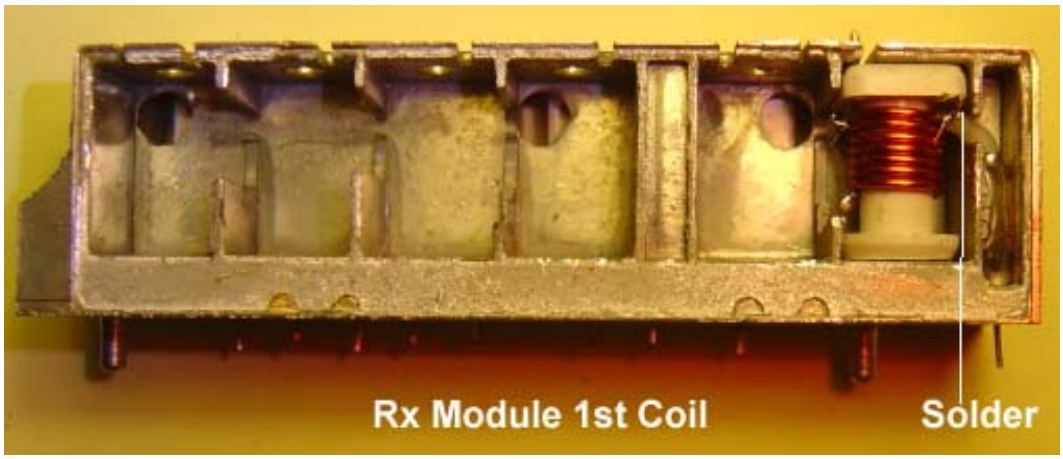






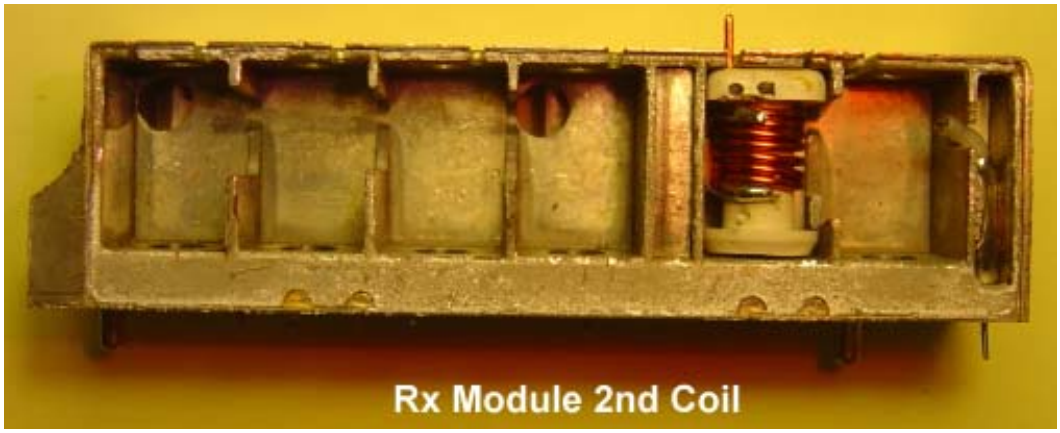
Coil # 1

- Cut a **160mm length** of 0.8mm enamel coated wire.
- Clean off all enamel from two ends of wire, one end of 5mm in length to be cleaned, the other earth end 10mm to be cleaned.
- Solder both ends of 0.8mm wire.
- Bend 8mm of the 0.8mm wire at [10mm tinned end] at right angles.
- Cut **33 mm length** from the original wire; Remove 5mm of enamel from both ends of the original wire; Tin both ends of the original wire; Solder one end of the original wire to the 0.8mm wire at earth end; See picture above labelled “Winding Wire”.
- Push one end of the bent 0.8mm wire through a left hand hole of the square end of the coil former [for earth connection]; Leave a short section [approximately 2-3mm] of the 10mm tinned section on the winding side of the former. Wind tightly, 7 turns of 0.8mm of wire [160mm] onto former.
- With the 33mm of the original wire, which has been soldered to the earth end of the 0.8mm wire; Wind 1.5 turns of from earth connection in same direction as 0.8mm wire; solder other end of the original wire to the antenna input.
- Solder a 33pf capacitor across the 7 turns of 0.8mm wire. Capacitor goes under coil [via left].
- If you are conducting this conversion remote to the club, then check with GDO that the coil resonates at 50.00 Mhz. Adjust the ferrite in the coil to go to that frequency. [look at note below, if this point is applicable]
- Place coil back into coil #1 position in module.
- Solder 0.8mm earth wire to module & solder original wire [33mm] to **antenna connection**. [see solder points on page 28]
- If you are conducting this conversion remote to the club, then check with GDO that the coil resonates at 53.25 Mhz. Adjust the ferrite in the coil to go to that frequency. [look at note below, if this point is applicable]



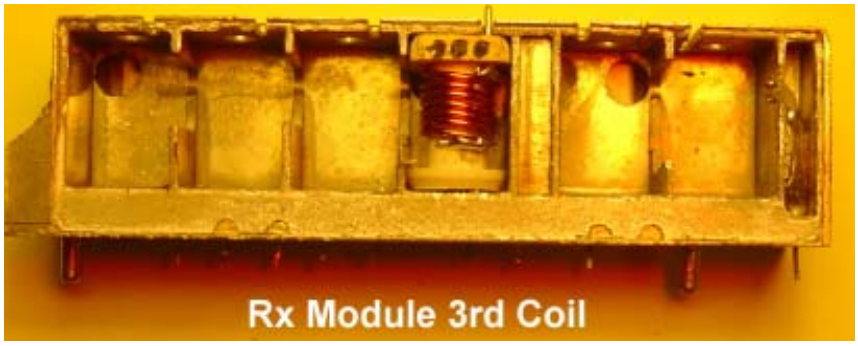
Coil # 2

- Wind coil #2 the same as coil #1, with the exceptions of:-
- The original wire length is **45mm** for coil #2.
- With free end of the original wire, solder to the **amplifier input** on the PCB side of the Rx Module.
- 33 pF Capacitor is placed on left hand side of coil. [alongside of shield]; place capacitor in first, then coil, then solder.
- Place into coil **position #2** of Rx module.
- Solder original wire to **RF amplifier input connection** on rear PCB side of Rx Module. Use a pair of tweezers to help you. [see solder points on page 28]



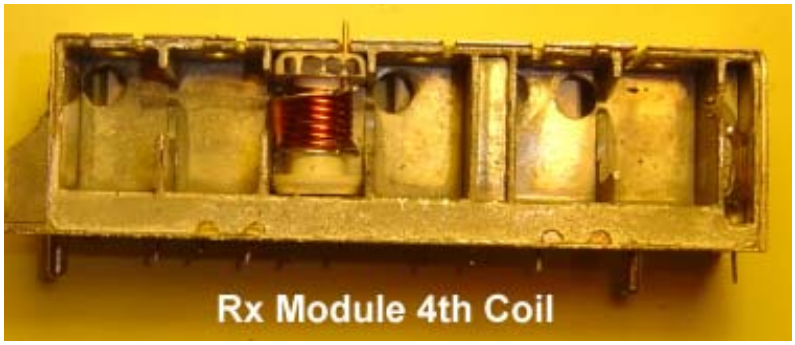
Coil # 3

- Wind coil #3 the same as coil #1, with the exceptions of:-
- The original wire length is **50mm** for coil #3.
- With free end of the original wire, solder to the amplifier output on the PCB side of the Rx Module.
- Push one end of the bent 0.8mm wire through a **right hand hole** of the square end of the coil former [for earth connection]
- 33 pF Capacitor is placed on right hand side of coil. [alongside of shield]
- Place into coil **position #3** of Rx module.
- Solder original wire to **RF amplifier output connection** on rear PCB side of Rx Module. Use a pair of tweezers to help you. [see solder points on page 28]

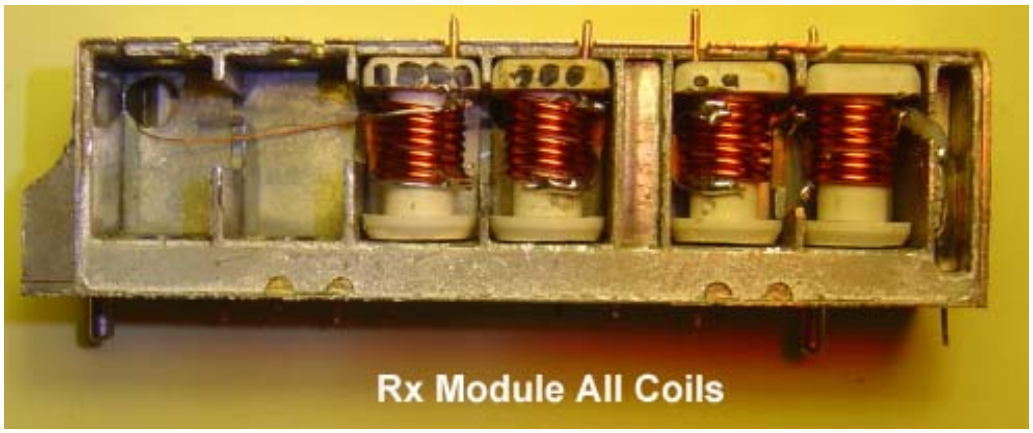


Coil # 4

- Wind coil #4 the same as coil #1, with the exceptions of:-
- The original wire length is **75mm** for coil #4.
- With free end of the original wire, solder to the **mixer input** on the PCB side of the Rx Module..
- Push one end of the bent 0.8mm wire through a **right hand hole** of the square end of the coil former [for earth connection]
- 33 pF Capacitor is placed on left hand side of coil.
- Place into coil **position #4** of Rx module.
- Solder original wire to **RF mixer input connection** on rear PCB side of Rx Module. Use a pair of tweezers to help you. [see solder points on page 28]


Rx Module 4th Coil

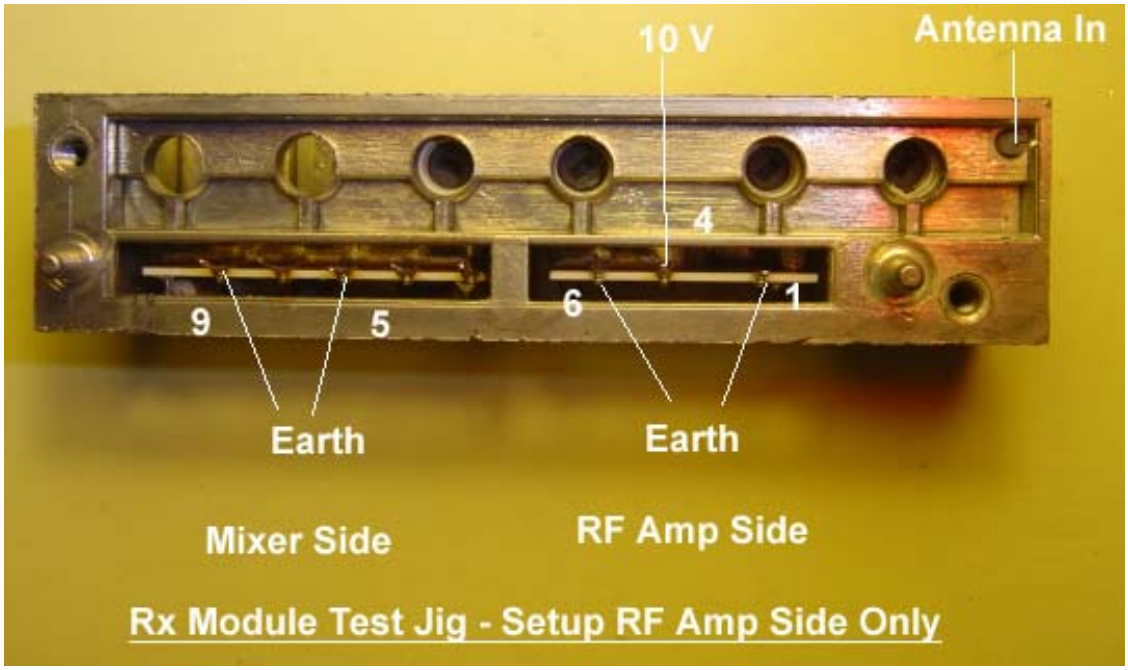
NOTE: A GDO should not be necessary if you are using 0.8mm enamelled coated wire and a NPO 33 pF capacitor. However if you are using different dimensioned wire or a different sized capacitor, then you should check everything with a GDO. Remember, when you place the coil into the Rx Module, the resonant frequency of the coil changes.


Rx Module All Coils

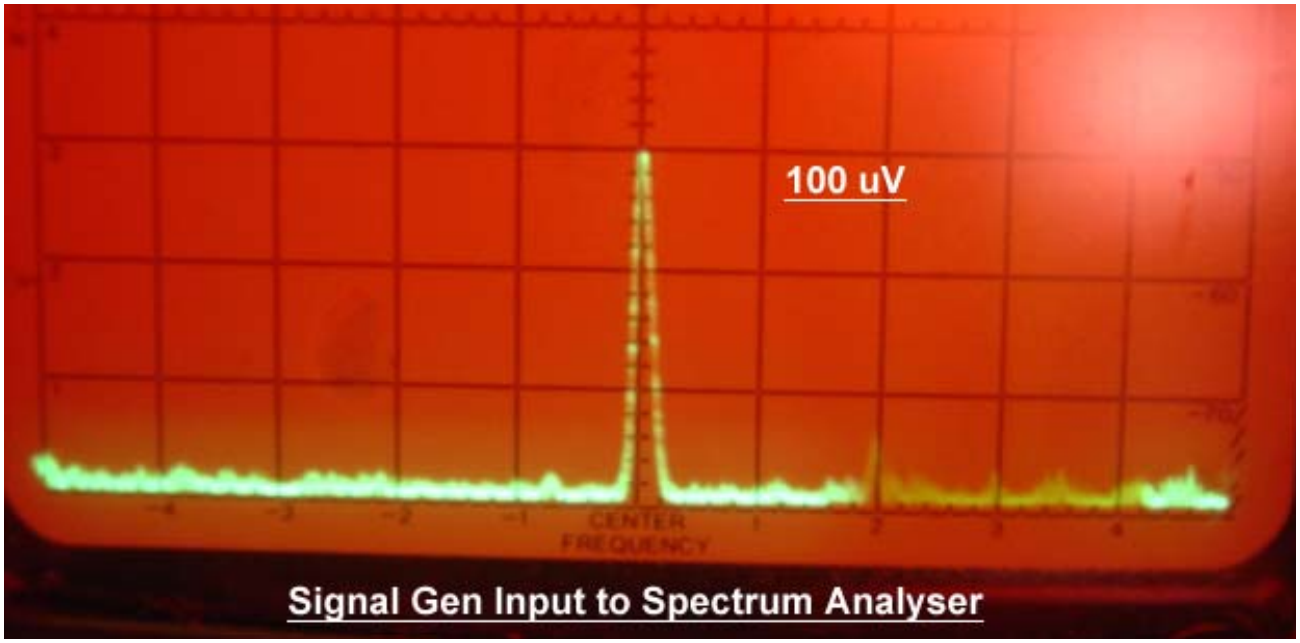
Replacing Module into Board:

- This is the real tricky bit. Check all of your soldering, have you any shorts, is there any shorts to ground.
- Remember, at this point on, when the sides are sealed and you resolder the Rx module back into the Receiver PCB and it does not work, then you will have to de-solder Rx Module and the module sides and recheck your work again.
- At this point, if you are not really sure, then before you even solder the sides on of the Receiver Module, with a signal generator and receiver or spectrum analyser, you can check that the Rx amplifier is amplifying, by setting up a test jig.

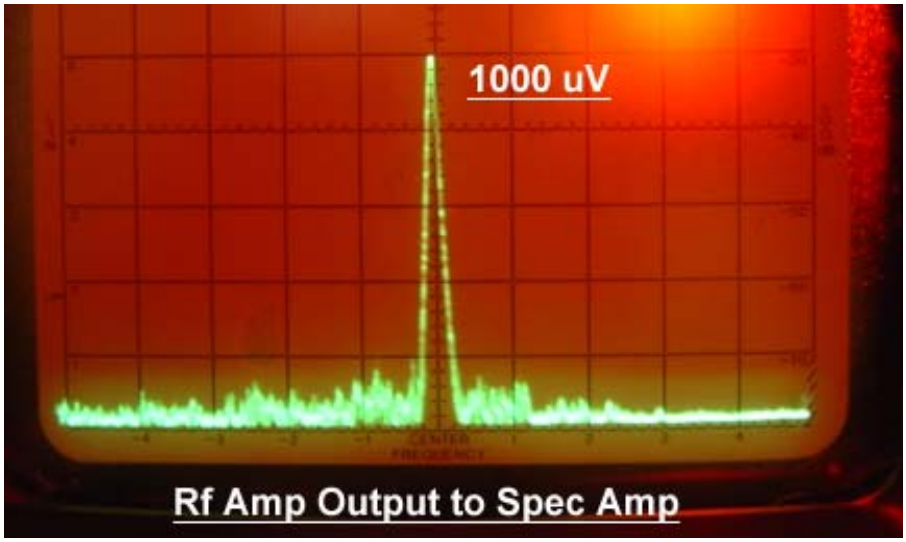
Rx Module Test Jig:



- Earth pins 1 & 6 of RF Amplifier side of RX Module. Be very careful of module pins, they can easily break when soldering wire to them.
- Apply a 10 Volt supply to pin 4 of the RF Amplifier.
- Connect a 53.25 MHz signal generator to a 53.25 MHz receiver or Spectrum Analyser and note the measurement out of the signal generator. [ie setup the signal generator to say Micro Volts.]



- Connect a 53.25 MHz signal generator to the Antenna In pin with the coax earthed to the module.
- Disconnect original wire from Coil 4 to the Mixer Input pin on the Mixer PCB.
- Connect the removed wire to a 53.25 MHz receiver or Spectrum Analyser.
- Check that the RF amplifier is working. See Below:



-

22: Rx RF Amp/Mixer Tune:

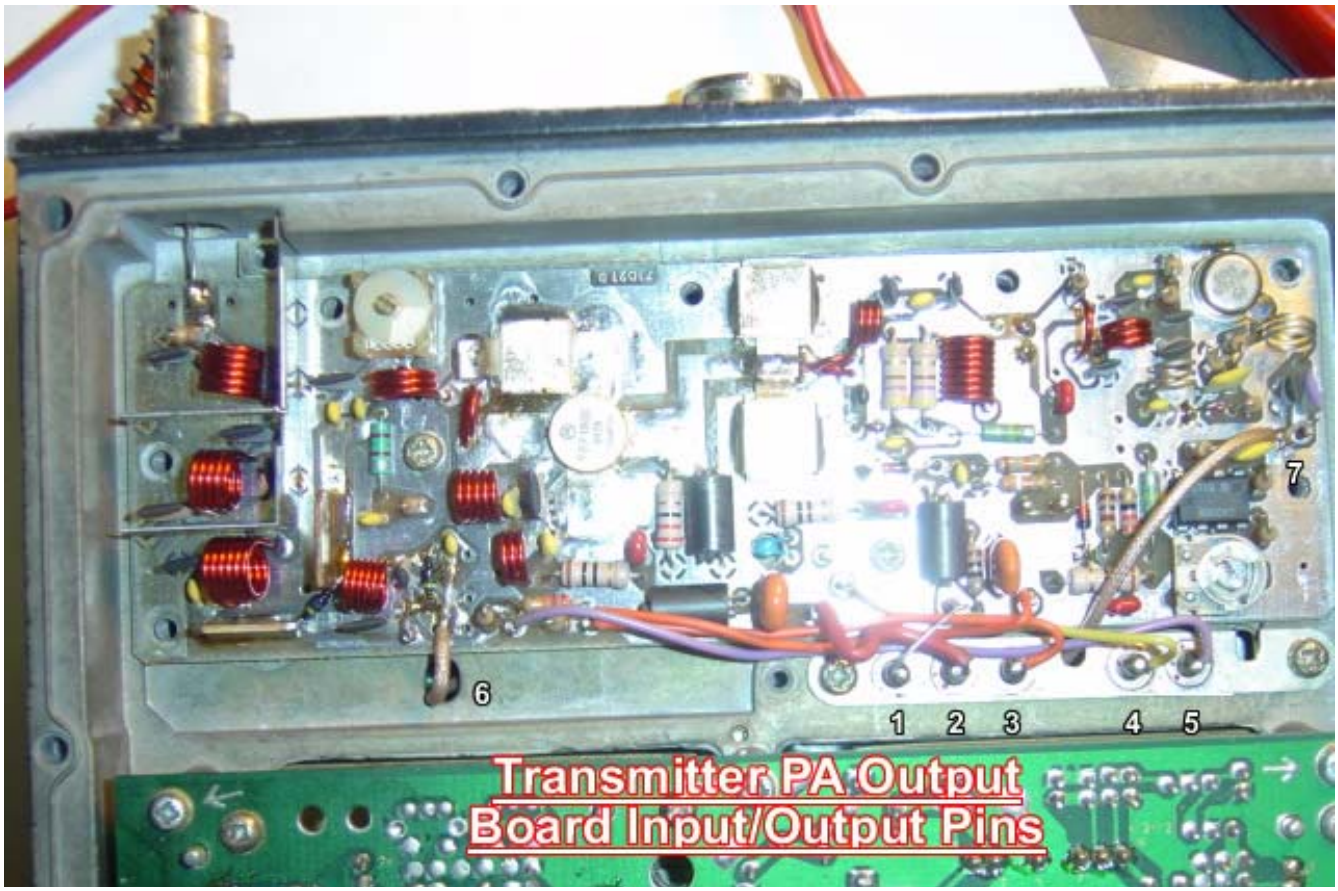
Apply signal at a frequency of 53.250 MHz from you signal generator.

Adjust the RF coil for the best signal to noise ratio. Keep the signal generator at the threshold of noise during the adjustment.

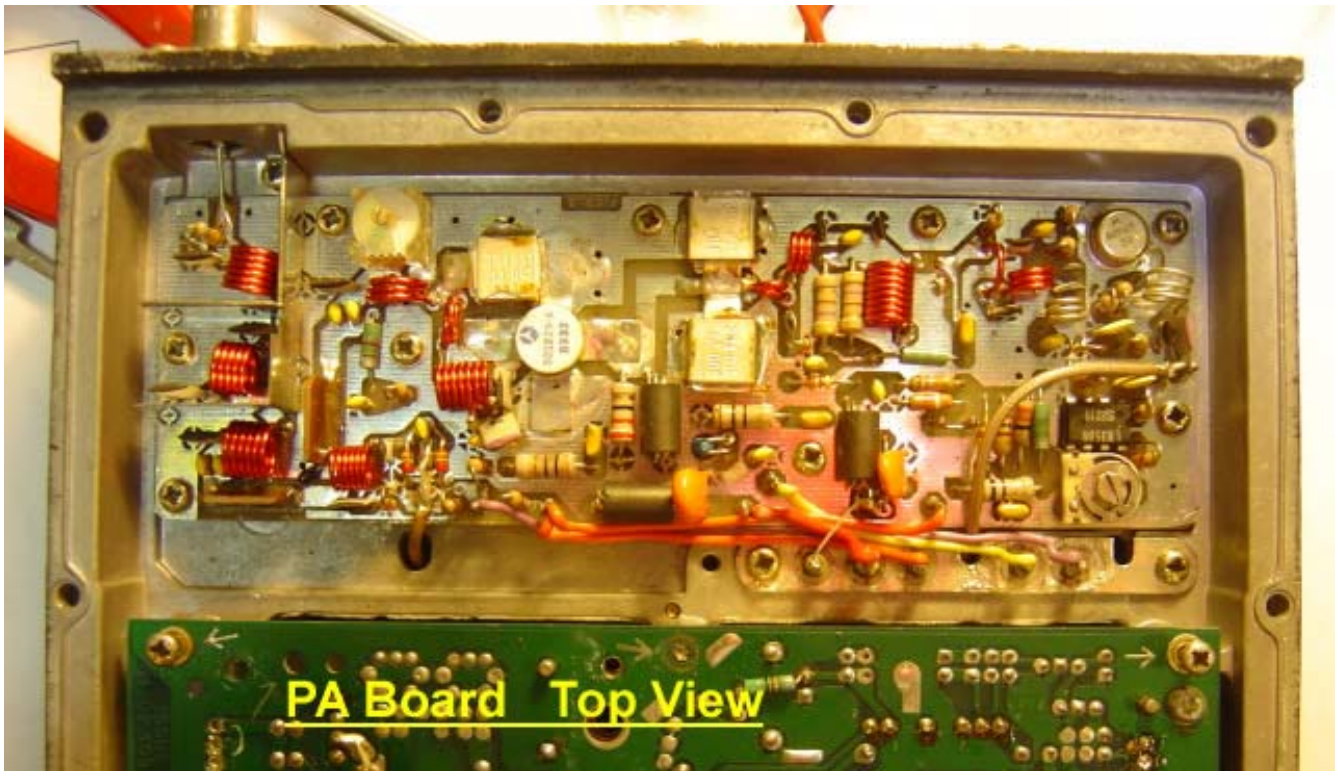
This adjustment should result in the mute lifting for a signal above 0.5 uVolts.

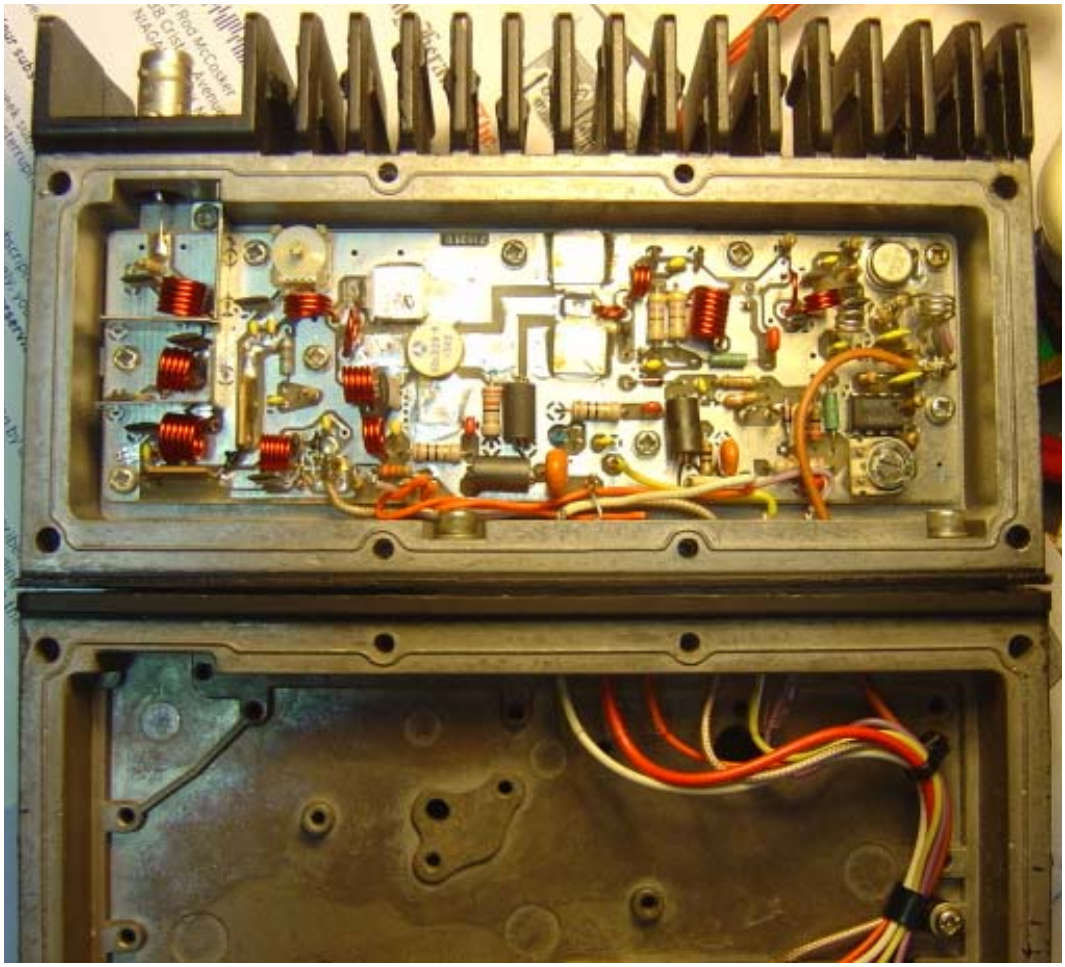
23: PA Changes:

Without any changes to the PA stage you should get between 5 & 7 watts.

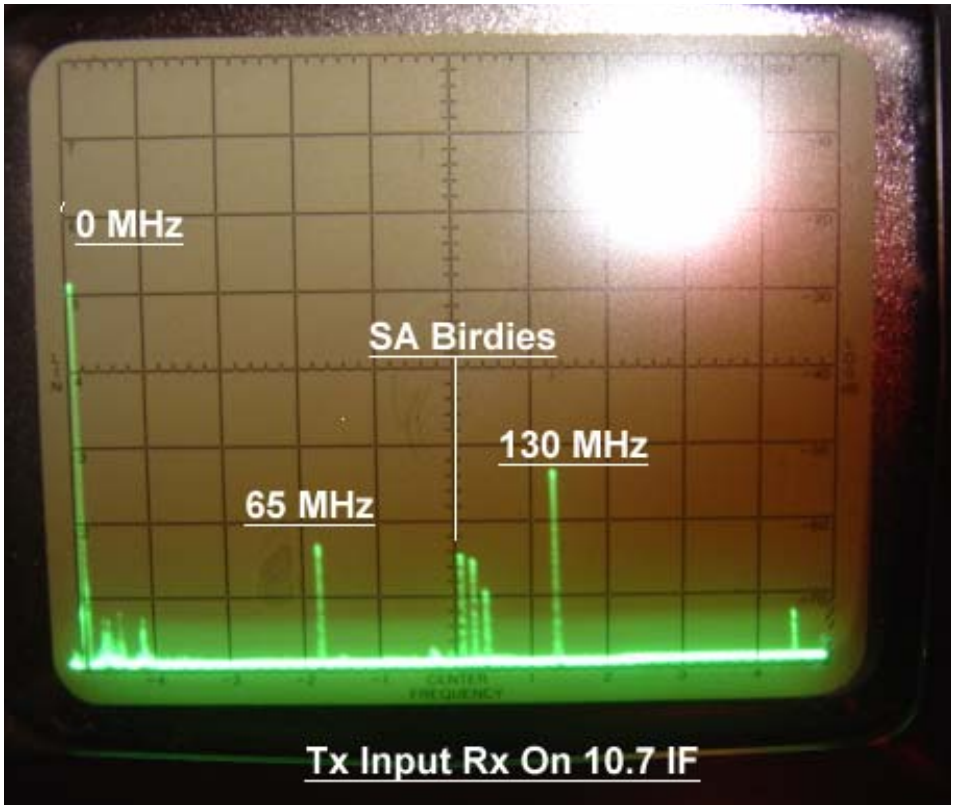


Pin	1 = 12 Volts	Move	
	2 = 13.8 Volts	Yellow	
	3 = 10 Volts Transmitter	Orange	
	4 = 10 Volts PA	Red	
	5 = RF Monitor	Bare Wire	
	6 = Rx to Receiver amplifier input.	Coax	
	7 = Tx PA input from VCO Buffer.	Coax	[25 mW In @ 50 ohms]
	8 = Tx PA output (Top Left Corner)	Bare Wire	[25 W output @ 50 ohms]

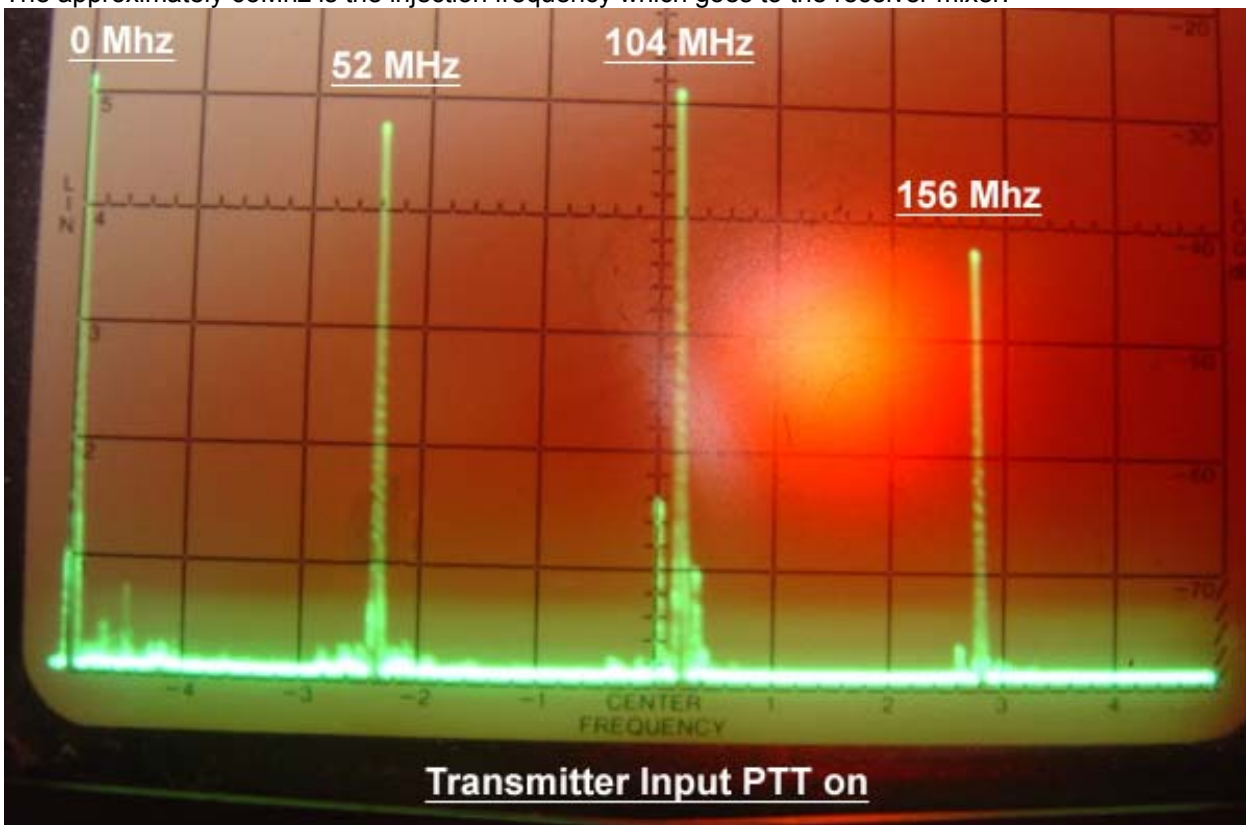




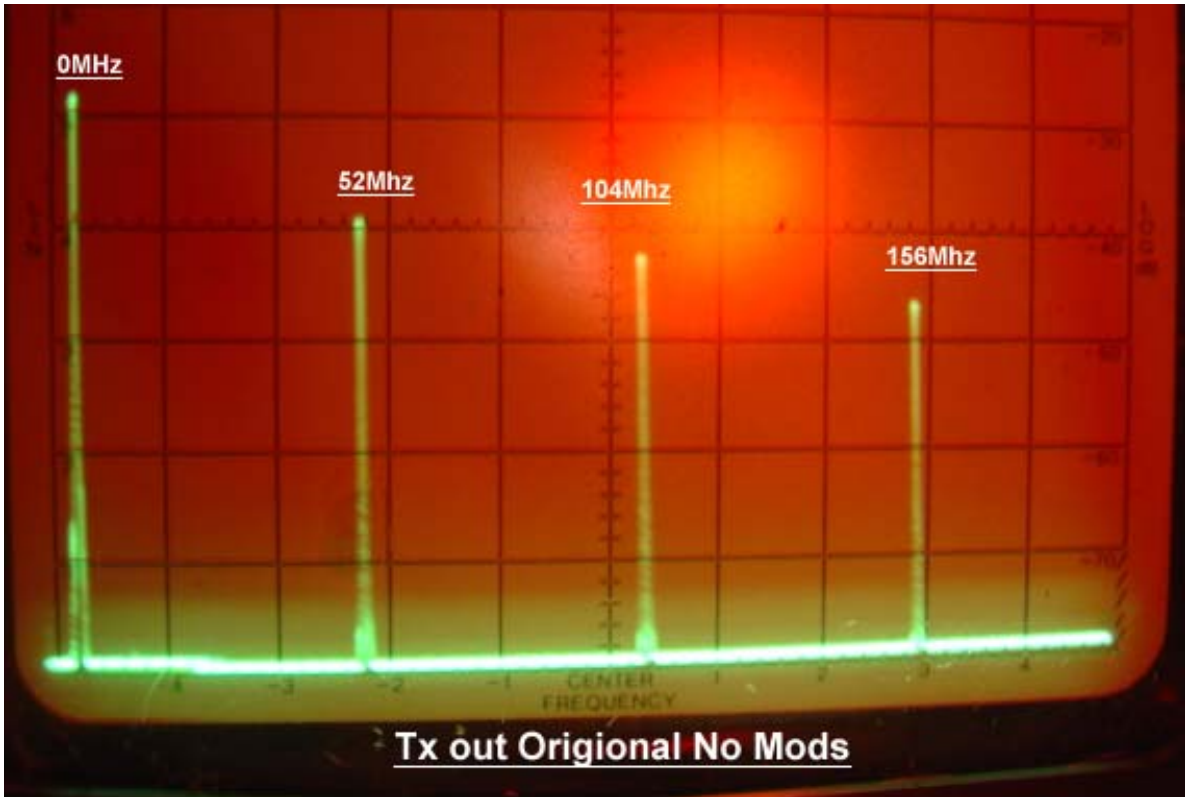
FM91E PA Board above view.

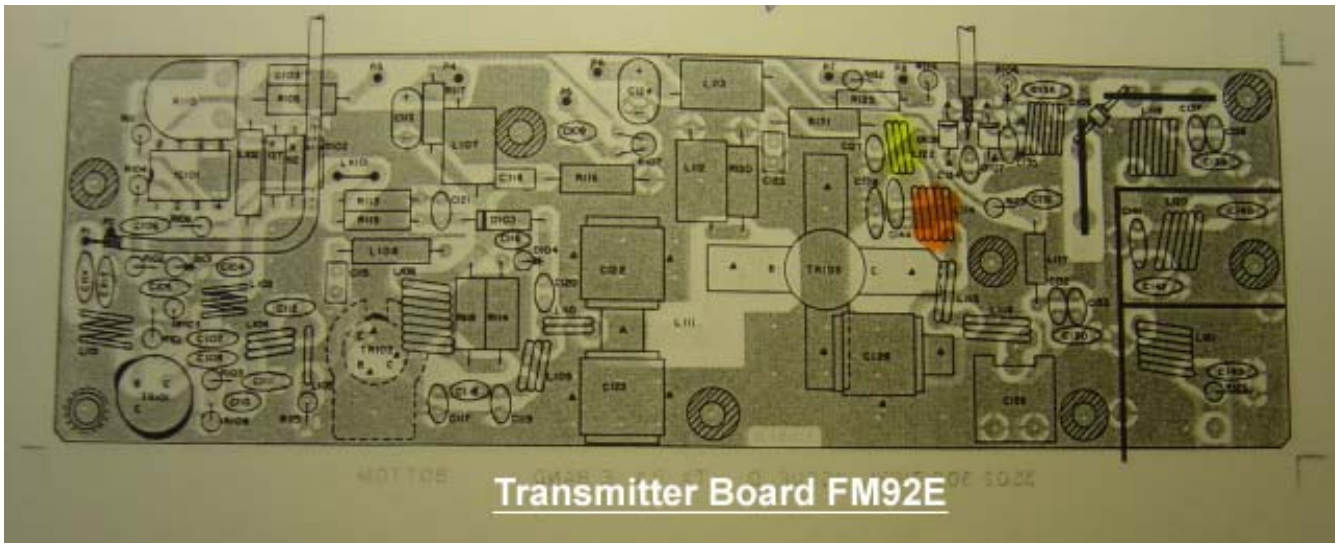


Above is a spectrum analysis of the signal from the FM92E Synthesiser VCO Buffer Output [pin 18] to the transmitter board input via coax with the Transmitter OFF and the Receiver on. The IF in this case is 10.5 MHz. The approximately 65Mhz is the injection frequency which goes to the receiver mixer.



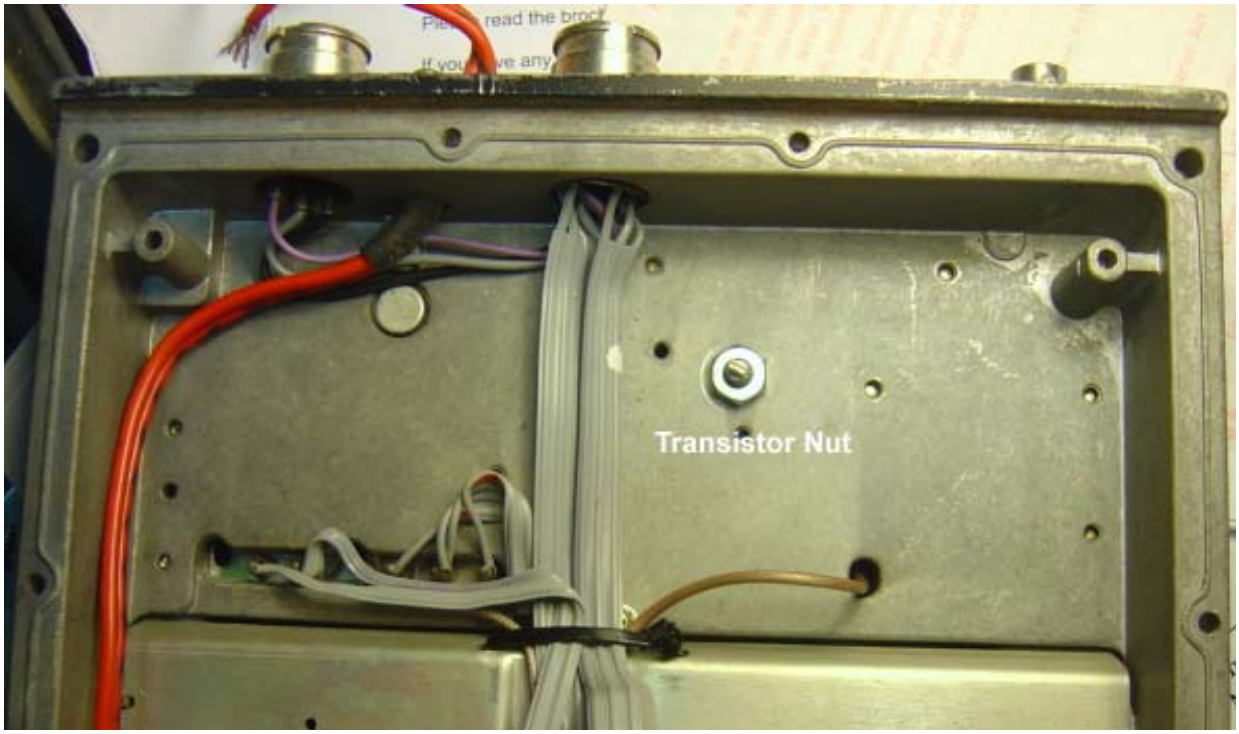
Above is a spectrum analysis of the signal from the FM92E Synthesiser VCO Buffer Output [pin 18] to the transmitter board input via coax, with the Press To Talk key ON, ie the transmitter was keyed. Note the harmonics of the originating approximately 52Mhz frequency.


0MHz
52Mhz
104Mhz
156Mhz
CENTER
FREQUENCY
Tx out Origional No Mods


Transmitter Board FM92E

Release PA Board from Chassis by:

- Remove PA transistor nut from rear of PA Board.

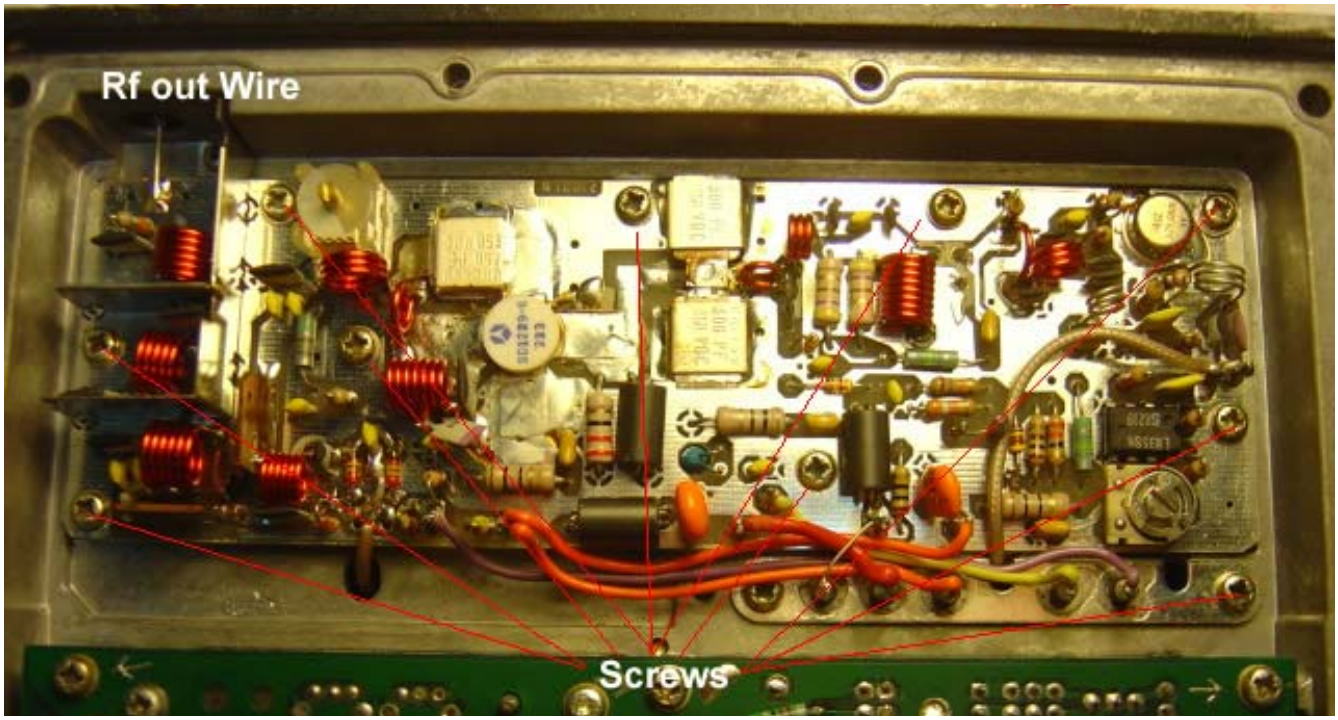


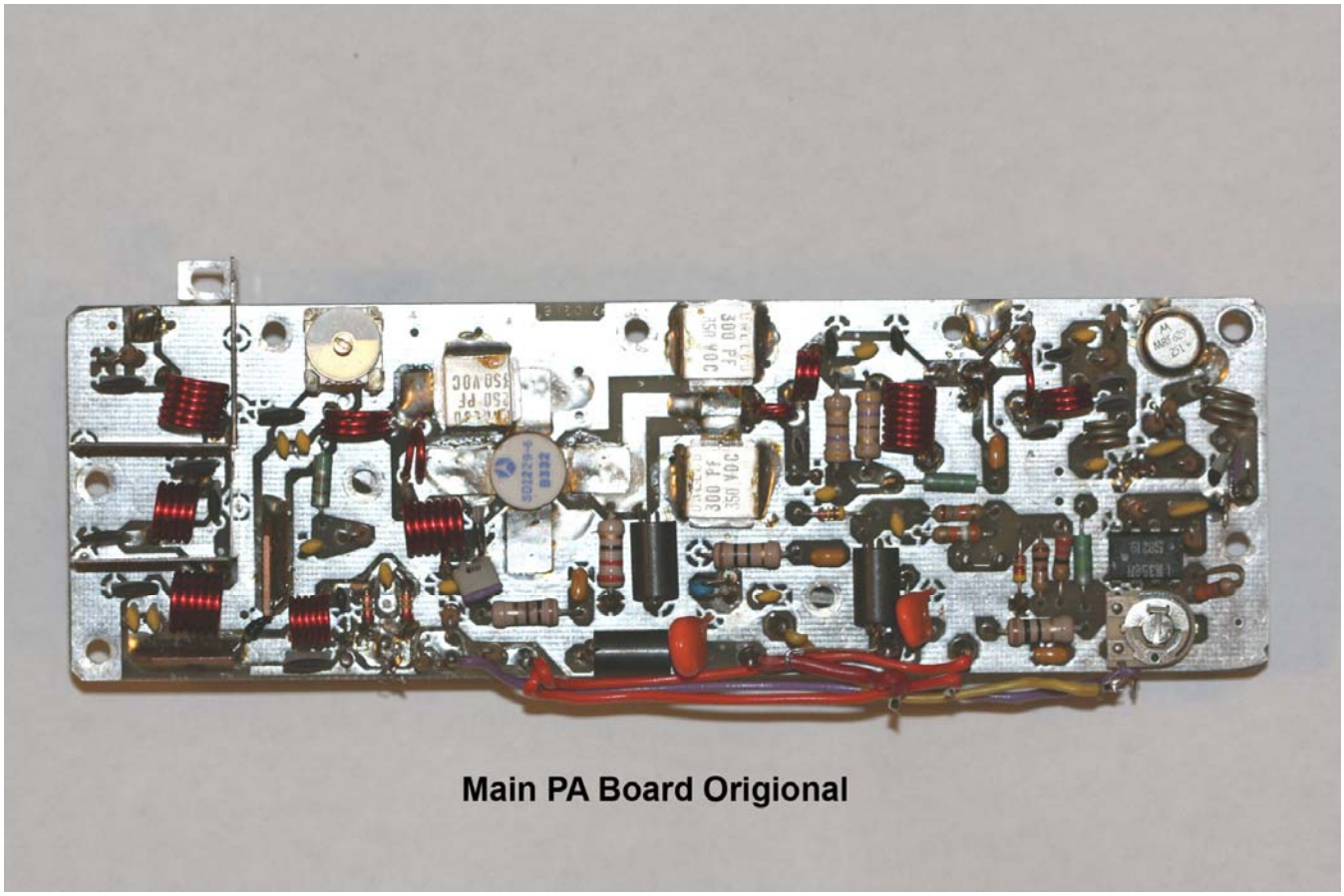
Unsolder RF output wire from PA Board.

Unsolder five of wires from chassis feed through pins.

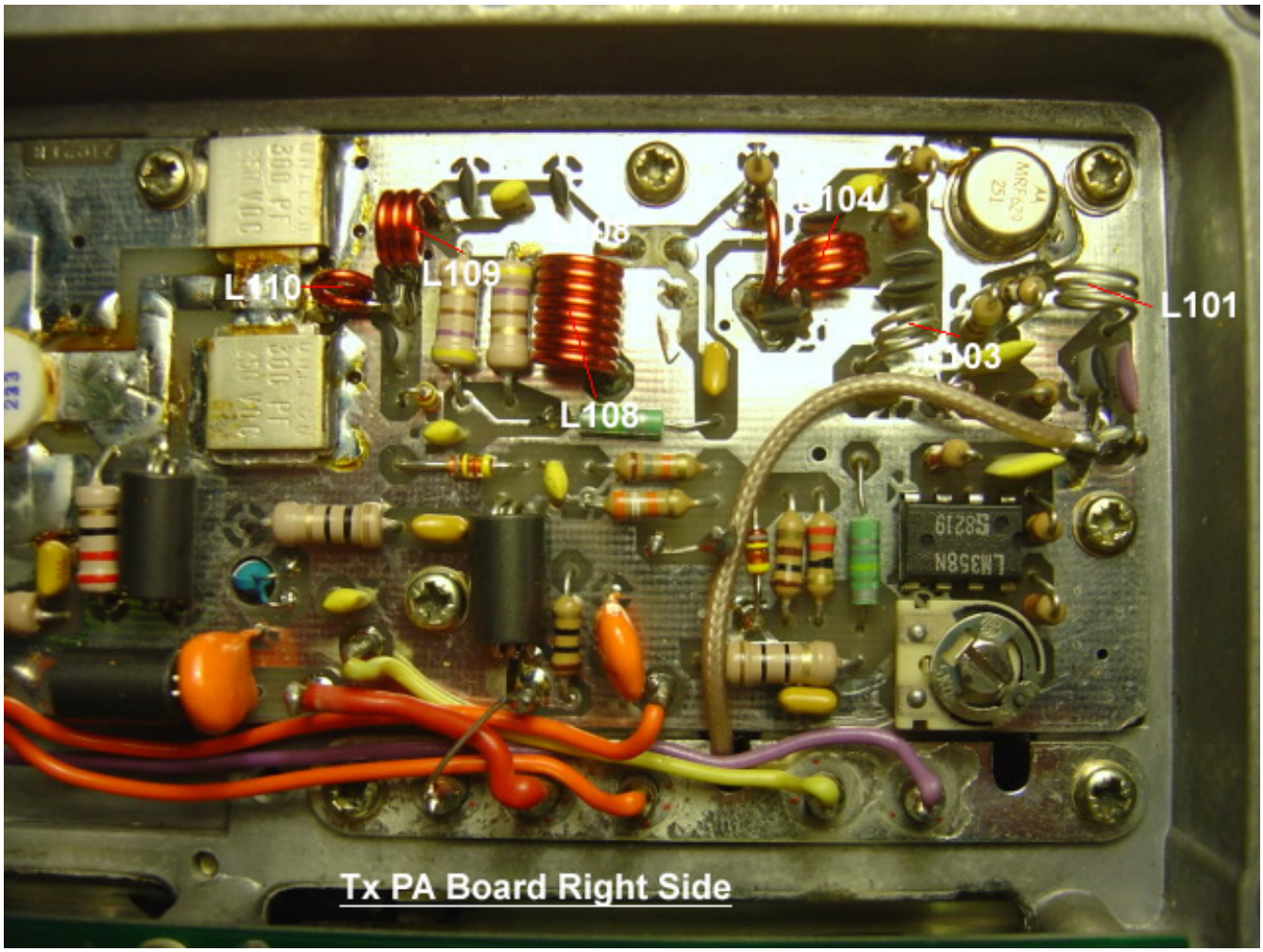
Unsolder two of coax lines going to PA board from PA board.

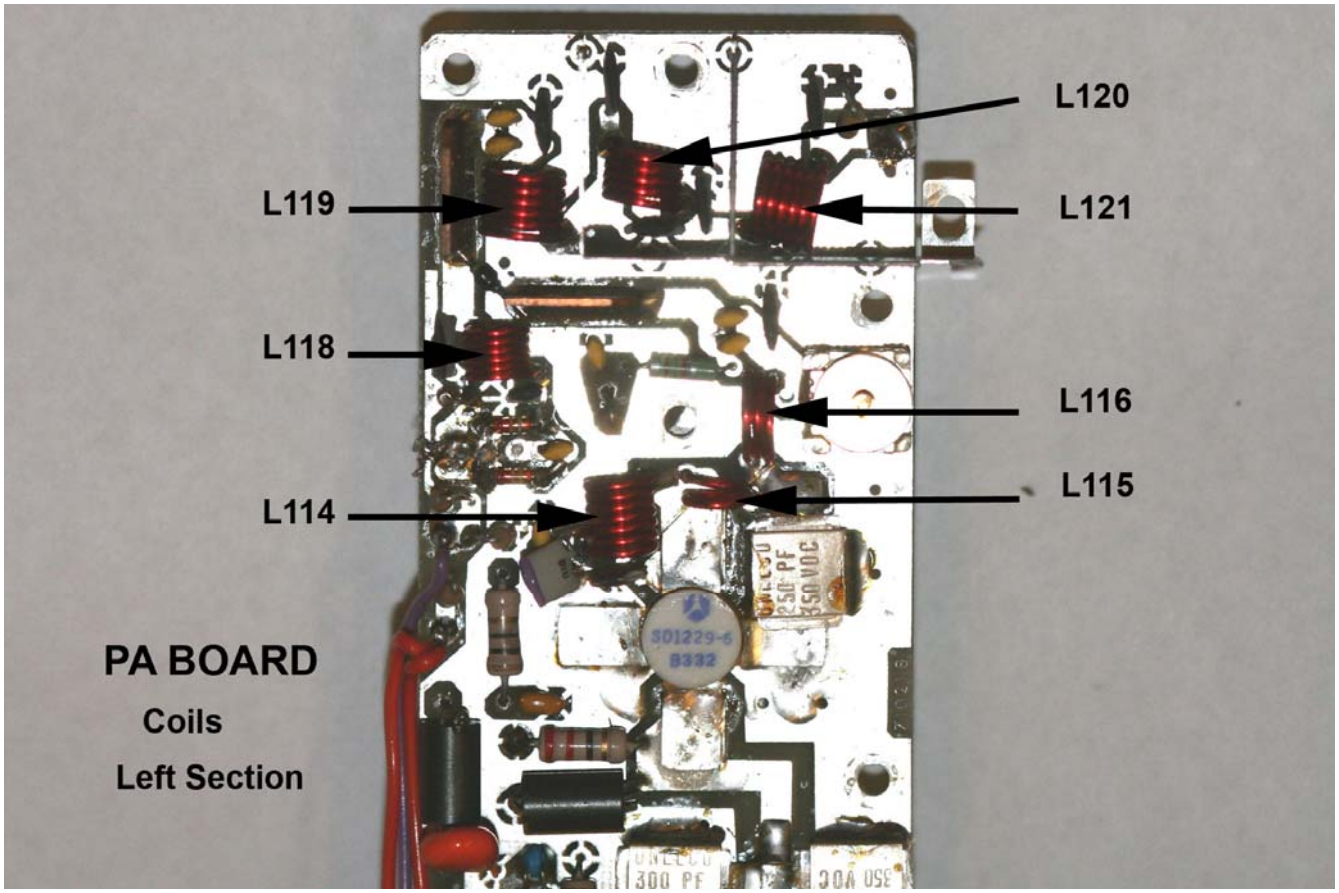
Unscrew screws holding PA board to chassis.

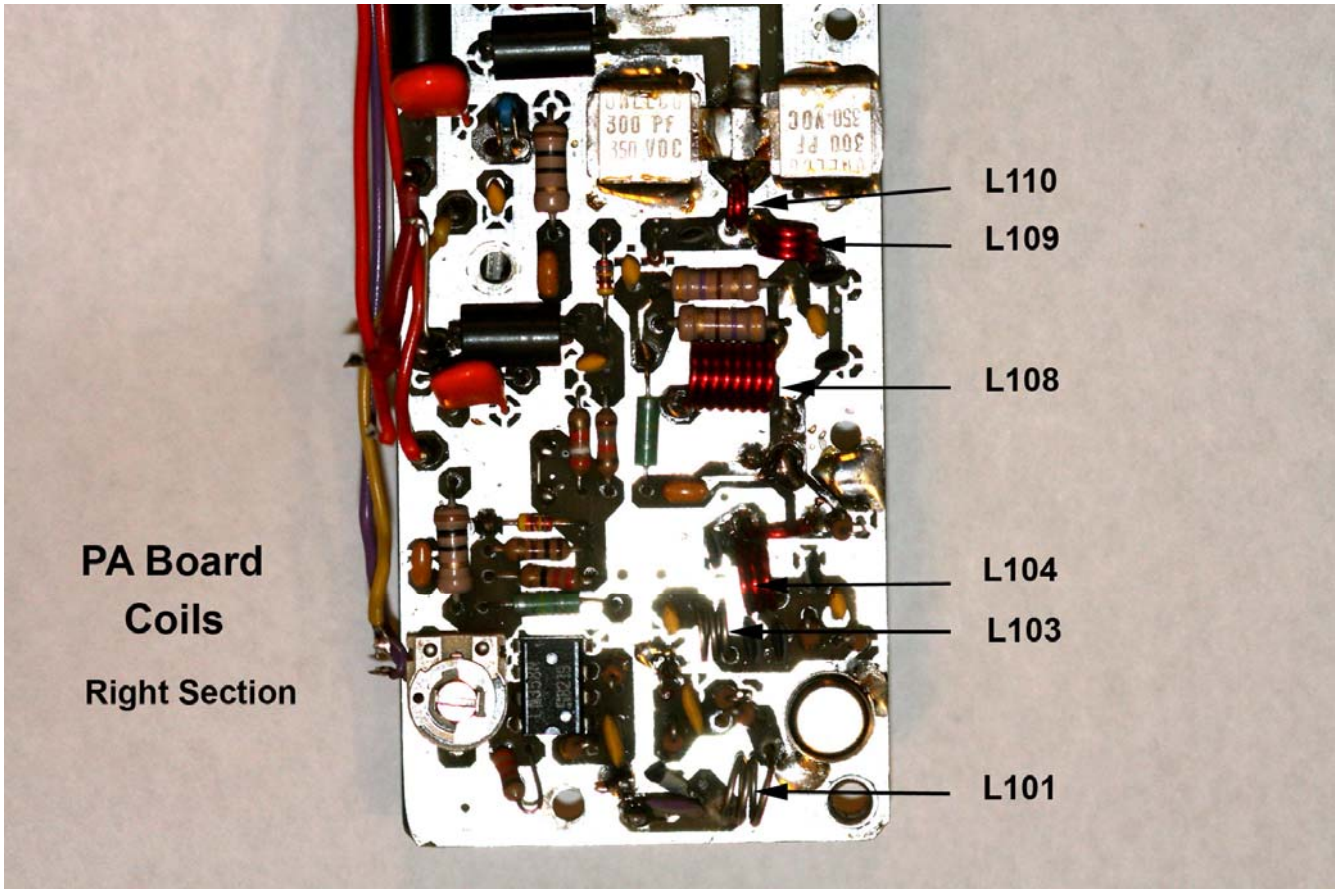


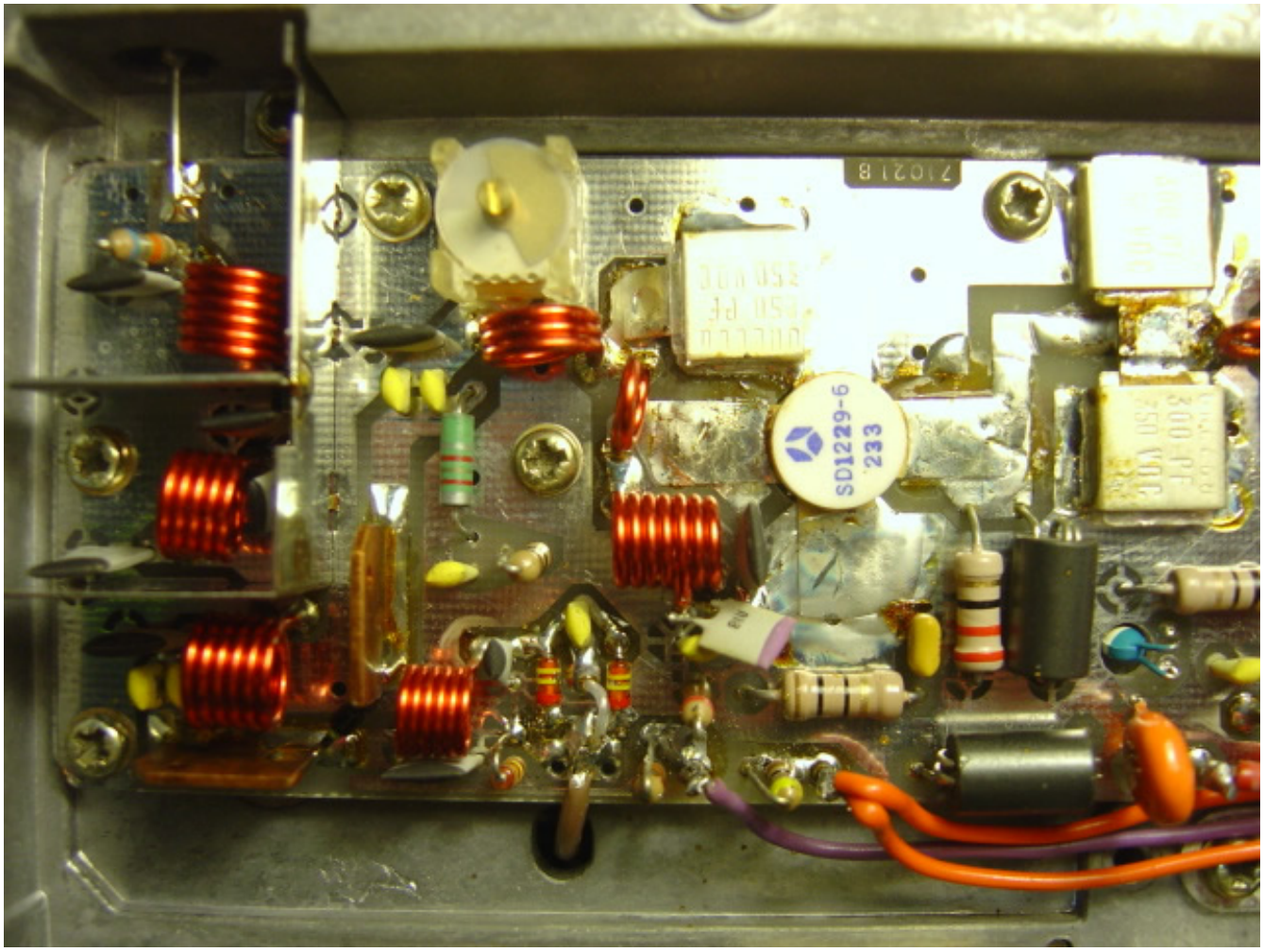


Main PA Board Origional




L120
L119
L121
L118
L116
L115
L114
PA BOARD
Coils
Left Section


L110
L109
L108
L104
L103
L101
PA Board
Coils
Right Section



Coil Upgrade:

Remove by de-soldering and upgrade the following coils to the new turns, with the same diameter on the PA board.

Coil	Original Turns	New Turns	Wire Size	Drill Size for winding coil
L101	2.5	3.5	0.63mm	
L103	2.5	3.5	0.63mm	
L104	2.5	3.5	0.63mm	
L105	0.75	1.0	0.63mm	
L106	No change			
L107	No change			
L108	8.5	10.5	0.80mm	
L109	2.5	3.5	0.80mm	
L110	1.0	1.5	0.80mm	
L111	No change			
L112	No change			
L113	No change			
L114		2.5	0.80mm	
L115	1.0	1.5	0.80mm	
L116	2.5	3.5	0.80mm	
L117	No change			
L118		6.5	0.80mm	
L119	5.5	6.5	0.80mm	
L120	4.5	5.5	0.80mm	
L121	5.5	5.5	0.80mm	
L122		5.0	0.63mm	

Capacitor Upgrade:

The following capacitors have to be purchased. *If original capacitors reused, then the following are necessary.*

		Jacar Part No	Dick Smith Part No
1 of 18 pF	*1 of 18 pF*	RC5315	R2241
3 of 33 pF	*3 of 33 pF*	RC5318	R2247
2 of 47 pF		RC5320	R2251
4 of 56 pF	*4 of 56 pF*	RC5321	R2253
2 of 68 pF		RC5322	R2257
5 of 82 pF	*3 of 82 pF*	RC5323	R2259
1 of 100 pF		RC5324	R2285
3 of 120 pF	*3 of 120 pF*	RC5325	R2287
1 of 220 pF		RC5328	R2293
1 of 270 pF	*1 of 270 pF*	RC5329	R2295

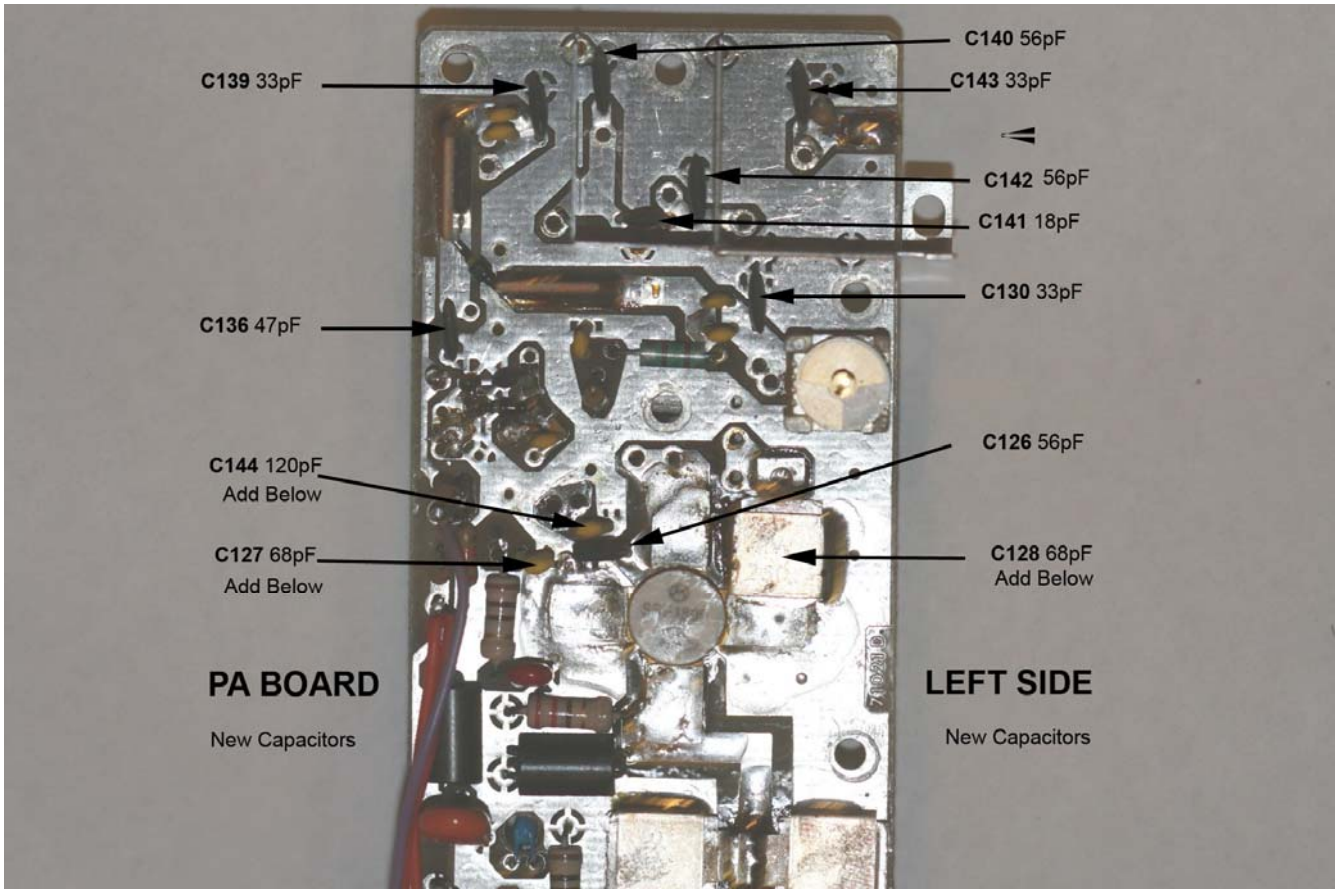
Remove the following capacitors from the PA board, then, replace with capacitors.

Original Capacitor		Replacement
C101	220 pF	270 pF
C102	68 pF	82 pF
C107	100 pF	120 pF
C108	82 pF	100 pF
C111	68 pF	82 pF
C112	82 pF	120 pF
C117	47 pF	56 pF
C118	180 pF	220 pF
C119	68 pF	82 pF
C126	47 pF	56 pF
C130	27 pF	33 pF
C135	39 pF	47 pF
C136	39 pF	47 pF
C139	27 pF	33 pF
C140	47 pF	56 pF
C141	15 pF	18 pF
C142	47 pF	56 pF
C143	27 pF	33 pF

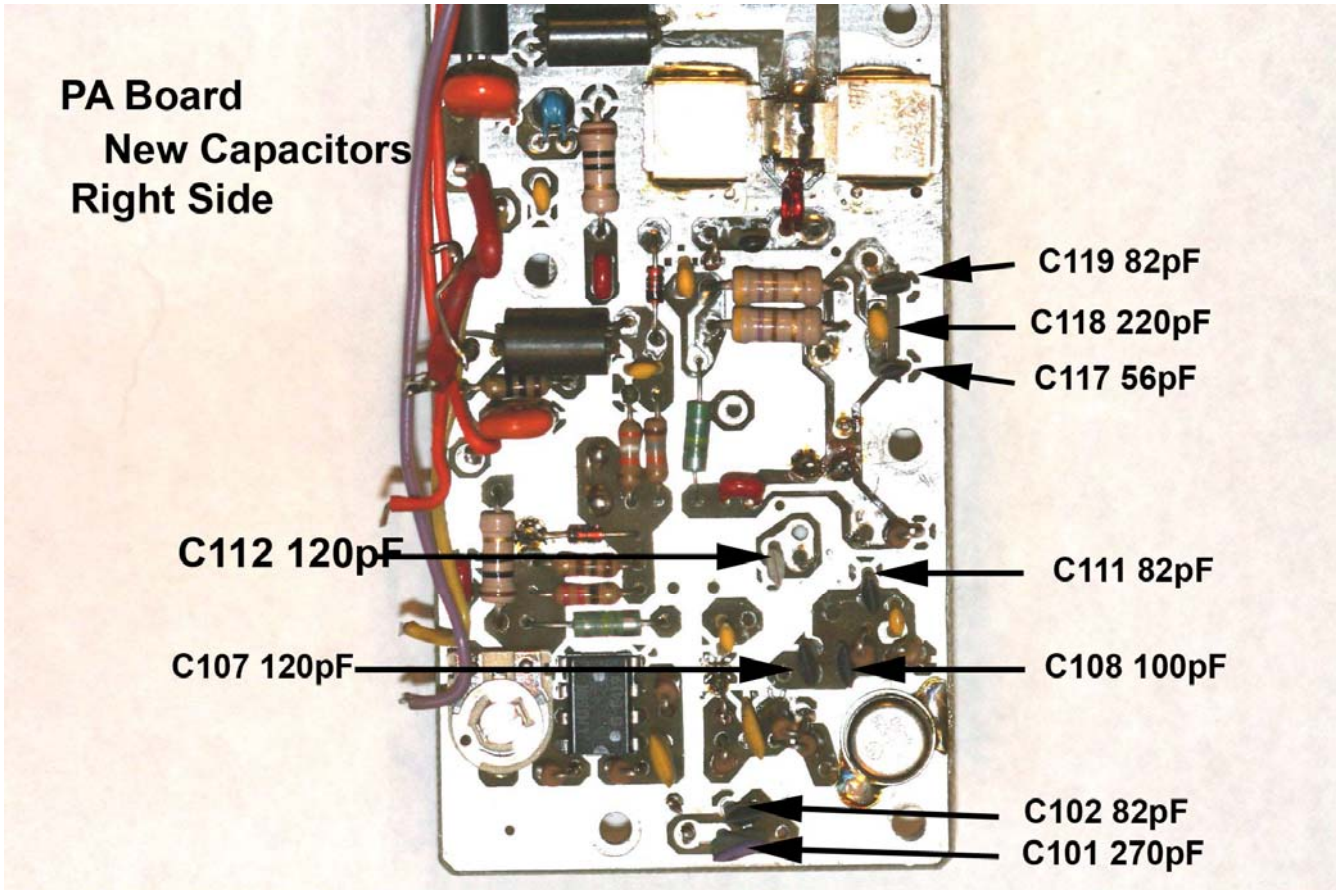
Add the following capacitors across the original capacitors. [ie 2 capacitors in parallel]

Original Capacitor		Add Capacitor	
C122	330 pF	82 pF	
C123	330 pF	82 pF	
C127	270 pF	68 pF	solder beneath board; lay flat.
C128	250 pF	68 pF	
C144	470 pF	120 pF	solder beneath board; lay flat.

REMEMBER: Do not remove capacitors **C128**, **C144** and **C127**; you place the extra capacitors underneath the PA board.



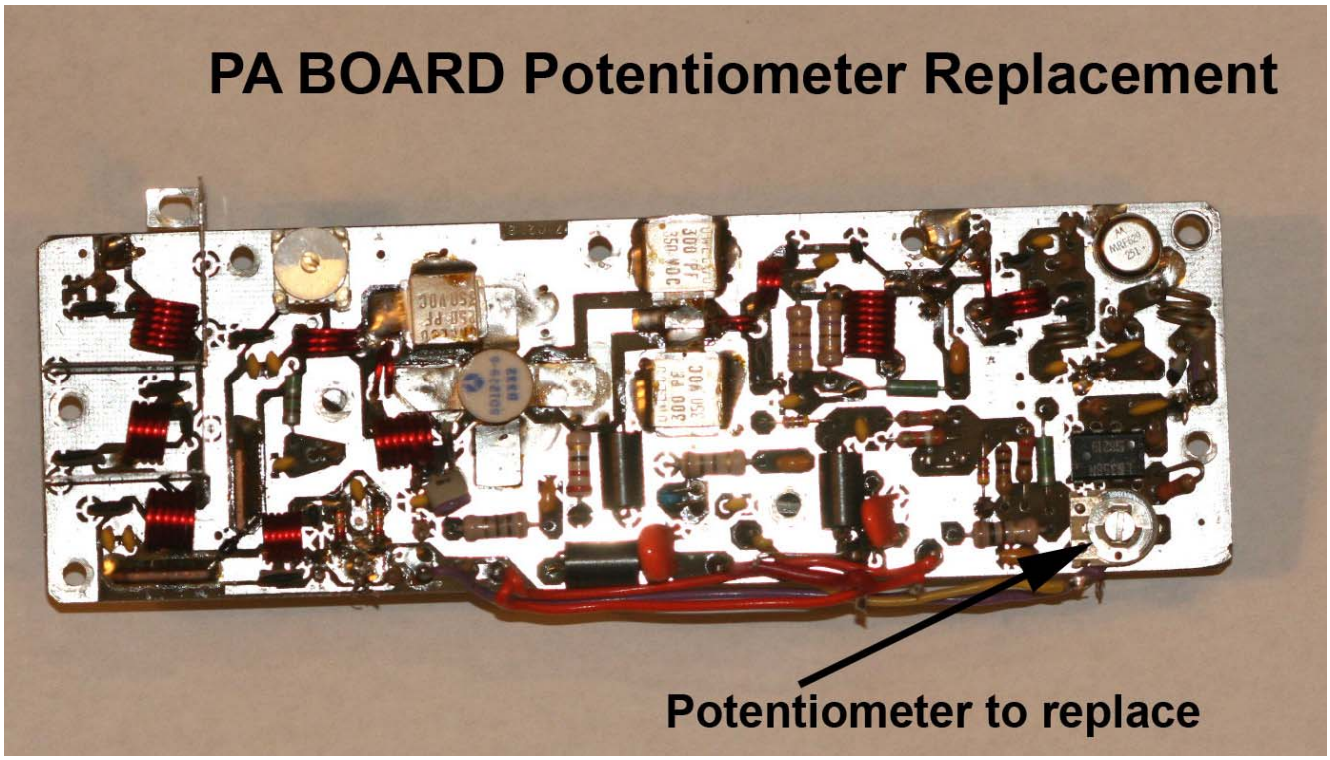
Above is PA Board - left section, at bottom of Screen: **New Capacitors:**



Above is PA Board - right section, at bottom of Screen: **New Capacitors:**

PA Deviation Potentiometer Replacement:

Needed – 1 of 100K ohm potentiometer. Jaycar # - **RT-4366**



Remove the 50K ohm potentiometer and replace with a **100K ohm** potentiometer.

The original FM92E transceiver converted to 6 meters, lacked deviation adjustment and for those who know the circuit, this increases the loop gain of the exciter.

PA adjustment:

- Open up the lid on the PA stage (other side of the radio) and adjust R102 (the only pot in the PA) for maximum power.
- Key the radio and adjust C111 and C124 for maximum power. Check the supply current is less than 6.5Amps. What you want is maximum power for least current.
- If you have a deviation meter, connect it and speaking very loudly into the microphone (counting to 10) adjust R367 for a maximum of 5khz.
- If you have a channel programmed as low power, the adjustment is R403

24: Removal of the annoying BEEP from the FM92E:

One of the first things that will strike you when you operate a Philips FM92E is the annoying beep when you change channels.

Remove the covers so the bottom (solder) side of the Synthesizer board can be seen. This is the board under the plastic sheet.

Position the unit with one side towards you such that the front panel is on the left and the rear panel on the right.

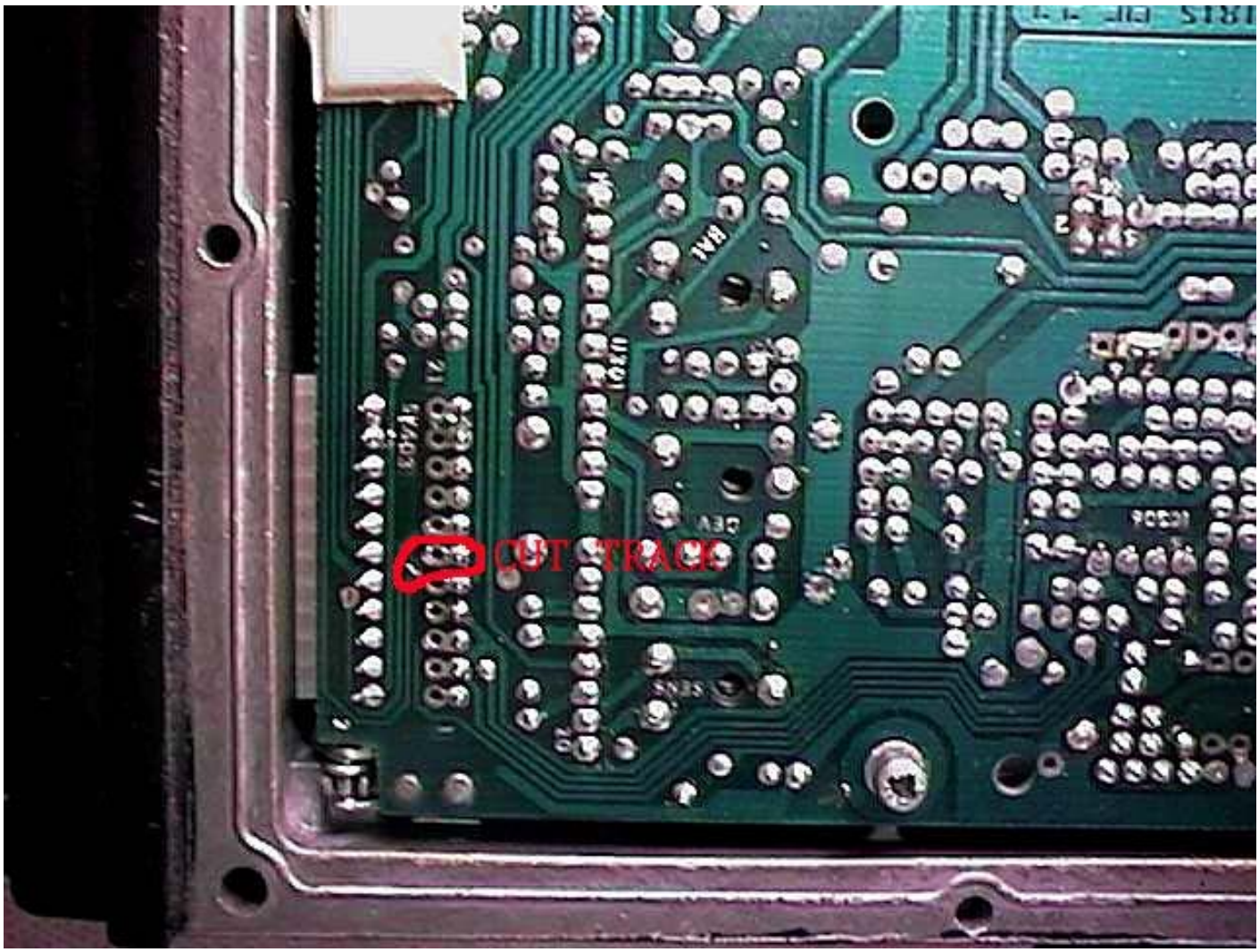
Locate the dual 11 pin solder rows on the left corner nearest yourself, this is labelled SK403.

Looking at the row which is further from the front panel, locate the solder point which is in the middle (6th pin from either end). ie resistor R254.It also can be found on the receiver board, the smaller of the two hinged boards. With the board in a vertical position (the only way to see the components) the resistor is located in the bottom right corner. It is the closest resistor to SK203, a ribbon cable termination device. The resistor is mounted vertically and has a value of 1.5 M ohms. It is also surrounded by capacitors. Second tip about locating this resistor is that it is dead in line with leg number 8 of IC202.

The cut, will not kill the beep completely, but it will certainly make it quieter than it was.

Cut the track joining to this pin with a sharp blade.

No more BEEP!



Synthesiser Board [Bottom View]

25: FM92R Source Listing:

```
1BE8 - Ch 1 frequencies (1st 3 bytes RX - 2nd 3 bytes TX)
1BF0 - Ch 2 frequencies, through to 1EF9 - Ch 99 frequencies.

1F31 - Default Power-up Channel - defaults to Lowest Channel programmed.
1F3E - Ch No (in Hex) of Highest Channel programmed
1F3F - Ch No (in Hex) of Lowest Channel programmed

1F40 - Lists Scan Group 1 Priority Channel.
1F4C - Lists Scan Group 2 Priority Channel.

1F41-1F4A - Scan Group 1 Channels
1F4B - No of Channels in Scan Group 1

1F4D-1F56 - Scan Group 2 Channels
1F57 - No of Channels in Scan Group 2

1F58 (01) and 1F59 (02) are the UP and DOWN buttons on the front panel.

1F64 - Eprom Checksum

1FE2-1FE4 - Date binary file created - stored in DDMMYY format (in decimal).

1FEB-1FEF - Indicates which Master Binary File was used.

eg.      30 30 32 30 3A - (0020:)  FM91   (Hardware Code:9502 97340000)
         30 30 32 38 3A - (0028:)  FM92L  (Hardware Code:9502 92330000)
         30 30 32 39 3A - (0029:)  FM92R  (Hardware Code:9502 92340000)
```

26: Disable "scan inhibit" with microphone off-hook:

Precautionary note: We are informed that this mod will disable some functions such as CTCSS, etc.

Normally, the FM92E will not go into "scan" mode unless the microphone is in its cradle. A magnet in the cradle latches a reed switch in the microphone housing, to enable the scan feature.

To enable scanning without hanging up the microphone, locate SK405 at the hinge end of the Synthesiser board, and short pins 1 & 3.

27: Emergency "umbilical cable" replacement:

The multi-pin cable which connects the control unit to the transceiver unit can be replaced by a standard IBM-PC "joystick" extension cable, costing between $10 and $25. Be aware, though, that some cheap computer cables are poorly shielded, and replacement with one of these is likely to result in an audible "whine" resulting from the display switching circuitry pulses getting into the audio. You get what you pay for. If your replacement cable doesn't have metal around the end connectors, it is almost certain to be noisy. I have overheard that cables from DSE are okay.

28: Fixing "tinny" transmit modulation by adjustment:

Some FM-900's sound fine on air, some have a quite "tinny" sound. The simplest way to cure this is by slight adjustment of the "modulation Balance" potentiometer (R371) on the Synthesiser Board. It is the innermost of three potentiometers, and looking from the "track" side of the board, adjust slightly anticlockwise (only a few degrees) to increase both bass response and overall deviation.

Warning!. Do not make large adjustments here, as you could cause over-deviation!

29: Fixing "tinny" transmit modulation by microphone modification:

Some FM-900's sound fine on air, some have a quite "tinny" sound. Another way to fix this is to replace the standard microphone insert with an electret insert, available from various electronics suppliers.

30: Fixing "tinny" transmit modulation by microphone modification:

Simply "flip over" the microphone insert. Truly! This places a "baffle" (the back of the insert) between you and the microphone element, which helps to reduce response peaks.

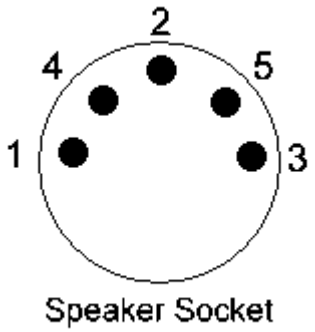
31: FM900 12 Digit Hardware codes:

(A 12-Digit Product Code Is Derived From The Chart Below The First 4 Digits (9502) Denote Australia And Are Fixed. The Remaining 8 Digits of the code detail the type of Equipment And Frequency Band (center 3 Digits- Columns 5,6,7), And Optional HardWare Facilities (last 5 Digits - Columns 8 Thru 12)

1 2 3 4
9 5 0 2

	5 Series	**6** Model	**7** Freq/Band	**8** Local/Remote Spacing	**9** Temp/Range Stability	**10** Signaling Hardware	**11** Selcall System	**12** Option
0					10 ppm Standard	STD (none)	STD No S/Call	STD
1					5ppm Xtal/Heater	CTSS Enc Only	CCIR	Rev/Tone Burst
2		SimPlex	E			CTSS ENC/DEC	ZVIE	
3			B	Local 20/25/30		SelCall Type 1 No CTSS	EEA	
4			A	Remote 20/25/30		CTSS ENC+ SelCall T1	EIA	Ignition Blanker
5			T		2.5ppm TXCO Kit	CTSS ENC/DEC SelCall T1	I/O Exp No S/Call	
6			U			SelCall Type 2 No CTSS	CCIR T2 ENC Only	
7			W1	12.5 Local		CTSS ENC SelCall T2	ZVEI T2 ENC Only	
8			W2	12.5 Remote		CTSS ENC/DEC SelCall T2	EEA T2 Enc Only	
9	Mobile						EIA T2 ENC Only	

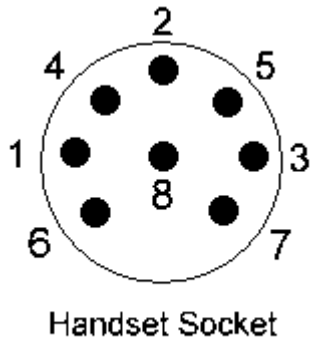
32: FM92E Rear Connectors:



Speaker Socket

Speaker Socket
1 Cradle
2 Ground
3
4 Speaker
5

Pinout are as viewed from rear of plug.

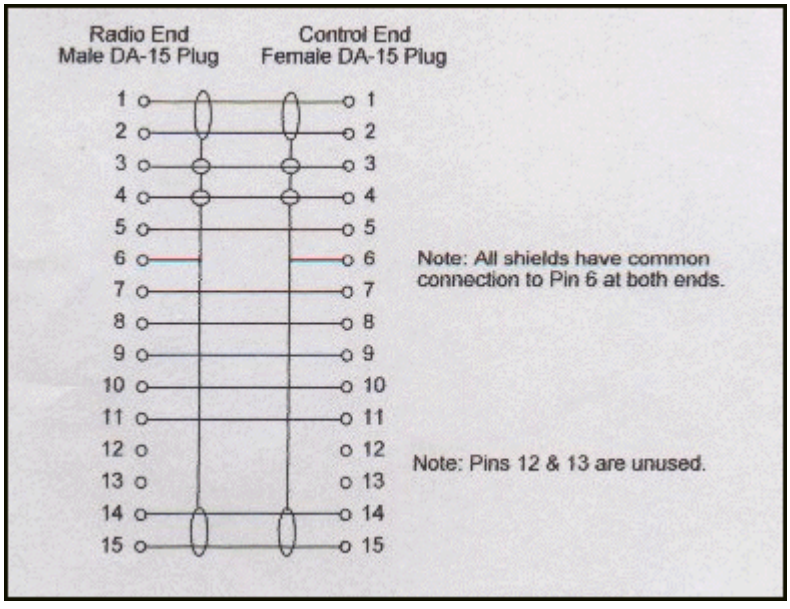


Handset Socket

Handset (Microphone) Socket
1 Cradle
2 Power Ground
3 Mike Ground
4 Speaker
5 Mike Active
6 RX Audio out
7 PTT
8 Not used

Pinout are as viewed from rear of plug.

33: 15 Pin Cable:



34: Philips FM 900 Series Service notes: [compiled by Andy Beales VK4KCS]

Symptom:

Radio keys up when P.T.T. is operated but although the "Tx" light stays on, only a brief moment of power out is observed.

The loudspeaker makes a faint "squeak" whilst this output power is present.

Check:

Using an oscilloscope check to see if Pin 6 of I.C. 201 on the R.F. board shows signs of oscillations while the P.T.T. is operated.

This I.C. takes care of switching the radio from Rx to Tx and some other housekeeping chores.

If oscillations are observed then this is what is causing the problem.

Fix:

1/. Fit a 0.001uF ceramic or polyester capacitor from Transistor Q 203 B to E. This is successful about 80% of the time.

2/. Fit a 0.1uf polyester capacitor from I.C. 201 Pin 6 to Pin 16 (ground). This is almost always successful.

Only one radio I ever encountered needed both these fixes added.

Symptom:

No Transmitter.

Check:

After eliminating all the usual causes for no power out, check that there is switched supply voltage at the Emitter of Q 209 on the R.F. board.

Voltage should be present while P.T.T. is operated.

Fix:

If this transistor is faulty then either leave it in place and short-circuit Collector to Emitter or if a replacement is available then change the device.

General Notes.

Anytime that the lids are off an FM 900 then check the condition of C 432 and C439, 10uF 50v. electrolytics in the Voltage doubler circuit on the Logic board.

This part of the circuit supplies about 17 volts to the P.L.L. circuit, this voltage eventually being used as the Steering voltage to control the frequency of the V.C.O.

All manner of strange symptoms can be observed if this voltage is not stable and fed from a low impedance source.

These capacitors operate at an elevated temperature in the radio and are prone to dry out long before any other components show signs of distress.

I have measured caps removed from a faulty radio as having a capacity as low as 0.5uF.

One of these capacitors is inside the metal shield and one just outside.

Quite often the P.C.B. is is coloured in the vicinity of these capacitors.

V.C.O. and Rx. Front end Modules.

These two modules both suffer a common manufacturing problem in that when the module housing was being plated, they were not cleaned properly of the acid used to initially clean the base metal.

This acid slowly reacts with the plating and over a period of time, Dendrites or very fine metallic whiskers can grow inside the module and eventually short out to the pins on the edge of the ceramic substrate that supports the components or to the components themselves.

These whiskers can be seen with the naked eye if observed under a strong light source.

The usual symptoms of the presence of these whiskers is the Receiver goes deaf, or the V.C.O. becomes intermittent or stops altogether.

A quick fix is described as "Percussive Maintenance", which consists of elevating the radio to a height of 18 inches above a hard surface covered in newspaper or other covering (we don't want to scratch the radio do we ?) and then letting Sir Isaac Newtons' invention do its thing.(drop it !).

This brutal but effective method should only be done to somebody else's radio preferably when the owner is out of sight and hearing range.

The second method is to laboriously, and with a very big soldering iron, remove the offending module from the circuit board, prise the side cover off with a fine bladed screwdriver and then use a small paint brush to thoroughly clean out the inside of the module.

That’s the easy part.

Position the side cover back on the module, tack solder it at a number of points around the edge, using a screwdriver to press the cover into place, then flow solder as neatly as possible to reinstate the shielding effect of the cover.

Its impossible to not dramatically heat the module during this process but it doesn't seem to worry it, because the "Circuit Board"' is ceramic.

If you are working on the V.C.O. module check while the cover is off, that the ceramic substrate has not broken free from one or more of the mounting posts inside the module.

This fault is typified by the radio being extremely microphonic on both Rx and Tx.

If this is the case don't try to force the substrate back down, just apply lots of heat to the outside of the module adjacent to the post in question and allow solder to flow into and fill the gap.

Allow the module to cool and then refit to the radio, not forgetting to solder both ends of the V.C.O. module to the P.C.B. ground plane.

Although this procedure seems fairly drastic, the module was faulty anyway and attempting this repair is definitely worthwhile, compared to parting with the hard-earned for a new module, if you can even source one now.

Other problems that can occur include the flexible printed circuits that join the circuit boards at the front of the radio becoming intermittent, especially in early model radios that used paper insulated flexible P.C's.

If they are in a bad way they can be replaced with ordinary ribbon cable, just keep the lengths about the same and maybe apply hot-melt glue to the solder joints after it has been fitted in place, to provide some mechanical support.

The volume and squelch controls have a printed carbon track and due to the cunning design of the adjustable part of the control, the cup that keeps the dust out of the control will wear away the carbon and cause it to go open circuit. GOOD ONE, Mr. P !!

I have had no luck finding an "off the shelf"' replacement that doesn't involve a bit of hacking and gouging to make it fit, but a reasonable job can be done with a "gas axe" and only 2 Kilo's of body filler !!

If water has made its unwanted way into the control head on either the local or remote control versions, the P.C.B. can be removed from the housing, successfully cleaned and repaired.

I use a cut down artists paint brush about 6mm wide to clean around the components with either metholated spirits or preferably one of the low evaporation circuit board cleaners available from Woolietronics or other suppliers.

The plastic housing is best just washed in soapy water and thoroughly dried.

A quick squirt of Mr Sheen restores that "just like a bought one" finish to your pride and joy. Be careful though, not to scratch the red plastic lens that covers the 7 segment displays.

The push-button switches can be carefully prised apart and the contacts cleaned.

It helps to have obtained a rubber pencil" (made by Stadtler for correcting typing errors) from the stationers to achieve this.

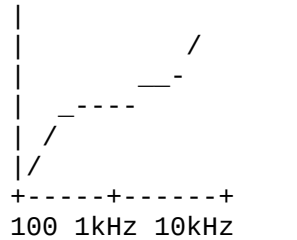
Most solvents/cleaners available will take the carbon film from the moving contact if applied, so use the pencil and then brush the residue out with a small artists brush (the adjective refers to the brush not the artist.)

35: Making my FM92 sound better: (transmit audio) [by Mal VK2TMF]

OK, after some fiddling about I think I now have a nice (well OK at least) sounding FM92.

I modelled the preamp in Electronics Workbench and ran some bode plots with and without the 4.7uF electrolytic capacitor (c3) in the microphone preamp.

With the electrolytic, the preamp boosts treble and cuts base. ASCII art time...

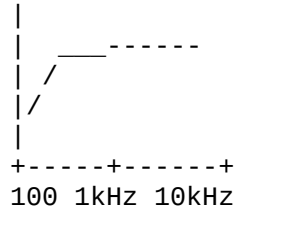



```
at 100Hz, the gain is about 8db
at 1kHz, the gain is about 12db
at 10kHz, the gain is about 21db

the -3db point (9db gain) is at about 110Hz
the +3db point (15db gain) is at about 3.6kHz
```

This explains the 'tinny' sound

Without the electrolytic, you get a reasonably flat response with 12db gain.




```
at 100Hz, the gain is 8.7db
at 1kHz, the gain is 12db
at 10kHz, the gain is 12db
```

the -3db point (9db) is at about 110Hz

So to make an FM92 sound better...

1. Remove the 4.7uF electrolytic capacitor from the microphone preamp.
2. Adjust R365 (microphone sensitivity) on the synthesiser board about 90 degrees, or until it sounds about right anticlockwise with the PCB in place, or clockwise from the component side.

36: Tuning up an FM900:

To tune an FM 900 radio you will need:

- A power meter of some kind (a VSWR meter into a Dummy load will do)
- Insulated tuning tools to suit VCO slugs. Ceramic works best. Available from Dick Smith Electronics at around A$8
- Voltmeter 0-20v Analogue is better.
- Signal Generator.

That's the bare minimum required. If you have access to a FM Deviation meter that will be handy.

Undo the screws holding both lids on noting which side they came from.

Identify the Synth/ Controller PCB . It's the larger of the two PCB's and on the side of the radio opposite the antenna connector. Undo the screws marked with an arrow and hinge the PCB out. Undo the 4 screws holding the shield over the EPROM (its the larger one).

Earth yourself to the radio with one hand or use a anti-static wrist strap. Carefully remove the EPROM. Its the chip with a paper label on it, noting its position. If you don't have a IC tool, carefully lever it up with a screw-diver at each end.

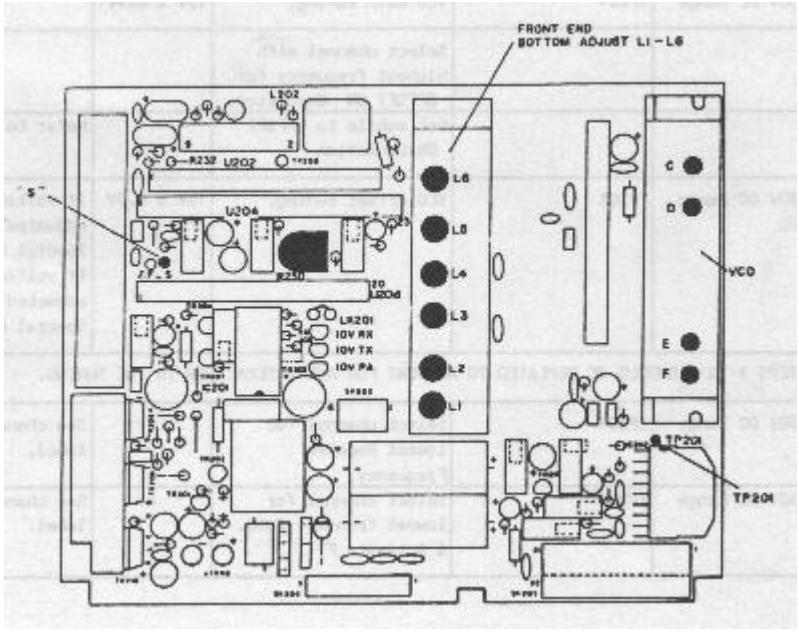
Re-insert the EPROM with your new frequencies

Carefully power up the radio and see if the channel numbers come up on the display. If so good, if not check the EPROM is in correctly and that it is actually a good EPROM. Sometimes they 'give up' while being reprogrammed.

Wind the mute to minimum. If the set is 'ticking' the VCO is out of lock.

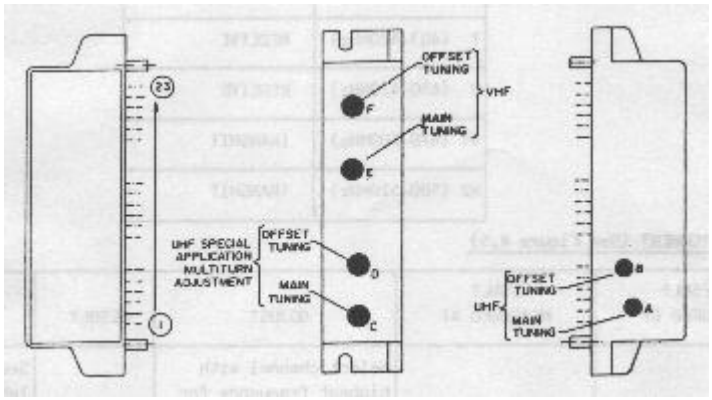
Undo the screws holding the radio board in and hinge it out.

Find test point 201



Connect your volt meter between TP201 and ground.

Identify the VCO slugs.



Carefully- if you force a slug and break it you may as well throw the radio in the bin as changing a VCO is a painful exercise. They do break easily as they are only ferrite cores. Although the UHF ones seem to be more robust than the VHF ones - adjust the VCO for 12v in receive mode and then for 12v in transmit mode. See the following tables for more details. When adjusted the dot in the display should not stay on after the TX is released. If you're lucky it won't come on at all. You may have to turn the set on and off to reset the dot.

FREQUENCY BAND	Mobile mode for **OFFSET OFF** Conditions
E (68- 88MHz)	RECEIVE
A (148-175MHz)	TRANSMIT
T (403-420MHz)	RECEIVE
U (450-470MHz)	RECEIVE
W1 (470-500MHz)	TRANSMIT
W2 (500-520MHz)	TRANSMIT

Note: Switching from OFFSET OFF to OFFSET ON, lowers VCO frequency.

4.6 **VCO ALIGNMENT (See Figure 4.5)**

STEP	RESULT MEASURED BY	RESULT MEASURED AT	ADJUST	RESULT	NOTES
1			Select channel with highest frequency for **OFFSET OFF** Condition		See channel frequency label and above table.
2			Set mobile to **OFFSET OFF** Condition		See offset table.
3	AVO 30V DC Range	TP201	VCO main tuning.	12V ± 0.5V	
4			Select channel with highest frequency for **OFFSET ON** Condition.		
5			Set mobile to **OFFSET ON** Condition		Refer to above table.
6	AVO 30V DC Range	TP201	VCO offset tuning.	12V ± 0.5V	If voltage cannot be adjusted below 13V go to Special Procedure No. 1. If voltage cannot be adjusted above 11V go to Special Procedure No. 2.

To adjust the receiver front end inject a 1 khz tone of 5khz deviation (2.5khz deviation for narrow band sets) at your centre frequency and adjust the tuning slugs starting at L1 (the slug in the centre of the board) through to L6 until the receiver is sensitive enough. *See previous picture for location* .You can use a SINADDER or by ear.

I will assume if you have a SINADDER that you can use it else use the ear method.

EAR METHOD OF TUNING A RADIO

1. Apply enough signal to open the set up (this may be a few volts if the set is really 'deaf')

2. Wind back the level until you can just hear the tone above the hash.

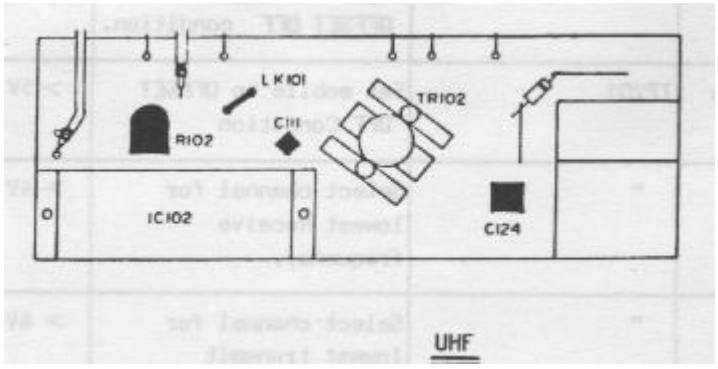
3. Adjust the tuning slugs until the tone is loud and clear.

4. Repeat steps 2 and 3 until the level is less than 1 microvolt.

5. Turn the signal right off, make sure the mute is at minimum (anticlockwise) then slowly wind up the signal until the set opens. If the level to make it open is anything less than .5uV well done. If its less than .3uV go and buy a lottery ticket. If not repeat steps 2 to 5 until it does.

PA adjustment

Open up the lid on the PA stage (other side of the radio) and adjust R102 (the only pot in the PA) for maximum power.



Key the radio and adjust C111 and C124 for maximum power. Check the supply current is less than 6.5Amps. What you want is maximum power for least current.

If you have a deviation meter, connect it and speaking very loudly into the microphone (counting to 10) adjust R367 for a maximum of 5khz.

If you have a channel programmed as low power, the adjustment is R403

37: FM900 Field Personality Programmer (FPP) Guide:

This section has been added for remote users who have an EPROM burner and need to setup their own EPROM for 6 Meters, 2 Meters or 70 CMeters. The information below is applicable to FPP Version 3.0 – the altered version.

Hardware Code made up of eight digits: (ie: ABCDEFGH)

The hardware code for a FM92E with remote head for 6 Meters = **92240000**

A - always **9** for FM900 mobiles

B – **1** = High Simplex (FM91)
2 = Medium Simplex (FM92)
3 = Low Simplex (FM93)
7 = FM92 with FM91 PCBs (FM97)

C - **Band** **2** = E band (66- 88 MHz)
3 = B band (132-153 MHz)
4 = A band (148-174 MHz)
5 = T band (403-440 MHz)
6 = U band (440-470 MHz)
7 = W1 band (470-500 MHz)
8 = W2 band (500-520 MHz)

NOTE: These are the suggested Transmitter limits, however Receiver limits can be programmed up to 30 MHz either side of the specified band edge. No guarantee of PLL lock, however.

D – Radio Type
1 = local control 25.0 KHz Channel Spacing 10.7MHz IF
2 = remote control 25.0 KHz Channel Spacing 10.7MHz IF
3 = local control 25.0 KHz Channel Spacing 21.4MHz IF
4 = remote control 25.0 KHz channel spacing 21.4MHz IF
5 = local control 12.5 KHz Channel Spacing 10.7MHz IF
6 = remote control 12.5 KHz Channel Spacing 10.7MHz IF
7 = local control 12.5 KHz channel spacing 21.4MHz IF
8 = remote control 12.5 KHz channel spacing 21.4MHz IF

E – Mechanical Construction
0 = standard
1 = extended temperature, 5ppm (crystal heater)
2 = waterfroof
3 = extended temperature & waterproof
5 = extended temperature TXCO
7 = extended temperature range, waterproof TXCO

(when using a waterproof radio, a waterproof option in the hardware)

F – Signalling Hardware
0 = none
1 = CTCSS encode only
2 = CTCSS encode/decode
3 = No CTCSS, Selcall type 1(G=>1)
(Selcall type 1 = encode only)
4 = CTCSS encode, Selcode type 1
5 = CTCSS encode/decode, Selcode type 1
6 = No CTCSS, Selcode type 2
(type 2 = encode/decode selcall)
7 = CTCSS encode, Selcode type 2
8 = CTCSS encode/decode, Selcode type 2

Note on Selcall encode & decode: Selcall type 1 can both send selcalls and respond to only one received selcall code and group call if applicable. Selcall type 2 can both send selcalls and respond to and

receive multiple and variable selcalls (used for sending status, etc)

G – Selcall Options

0 = none
1 = CCIR selcall
2 = ZVE 1 selcall
3 = EEA selcall
4 = EIA selcall
5 = No selcall with I/O expansion kit
6 = CCIR selcall, encode type 2
7 = ZVE selcall, encode type 2
8 = EEA selcall, encode type 2
9 = EIA selcall, encode type 2

Note that most selcall hybrids can only respond to CCIR selcall tones, the other selcall modes were not well supported.

H – Other Options

0 = standard version
1 = reverse tone burst
2 = dual front end
3 = ignition noise blanker

Software Code made up of 15 Digits:

The software code for a FM92E with remote head for 6 Meters = **110404200000000**

A – Channel Display

0 = intermittent display
1 = continuous display
2 = intermittent + 8 hour timer (auto power off)
3 = continuous display + 8 hour timer

B – Mute

0 = fixed, with defeat
1 = variable with defeat
2 = fixed, no defeat
3 = variable, no defeat

C – Economiser Options

D – Tx Timer

0 = Off
1 = 1 minute
2 = 3 minutes
3 = 5 minutes
4 = 10 minutes

E – Tx Inhibit

0 = Off
1 = Tx inhibit all channels

F – Tx power

0 = 25 watts
1 = 5 watts
2 = 1 watt
3 = Operator selectable
4 = per channel option
6 = 5-25 watts
7 = 1-25 watts adjustable

G – Scanning

0 = none
1 = scan vacant per programmed blocks
2 = scan occupied per programmed blocks
3 = scan occupied, operator selectable, per programmed blocks

H – Voting

0 = No voting

1 = Enabled

I – CTCSS Options
0 = no CTSS
1 = CTSS variable per channel
2 = variable via keyboard
3 = variable per channel, RTB (Reverse Tone Burst)
4 = variable via keyboard, RTB
5 = Community Repeater (CR) per channel
6 = via keyboard and CR
7 = CR per channel and RTB

J – Tone Period
0 = no tone period
1 = 100 ms
1 = 40 ms
1 = 70 ms
1 = 20 ms
1 = 33 ms

K – Simplex Mode **0** = simplex mode (no other option selectable)

LL - Selcall encode (only selectable if selcall is fitted, otherwise program 00)
Selections available: 00-05, 07-09, 11, 12, 15, 16 ,27 ,28, 35 ,36 ,43 ,44 ,51-53

MM - Selcall decode (only selectable if selcall is fitted, otherwise program 00)
Selections available: 00-18 for selcall type 2
00, 31-46, 50, 51 for selcall type 1

Note on Selcall software selection codes: selection of all codes is not available to all radios. See selcall selection chart for the type of radio you have. (not included here due to large size & complexity)

Common error messages:

* Selecting CTCSS tones without the correct hardware being installed at the hardware code and the like will cause error messages. This applies to things like selcall setups etc.
* TX inhibit cannot be selected with Community repeater
* Tone periods 33 & 20 ms are incompatible with type 2 selcall.
* Voting cannot be selected with mute defeat.
* Selcall decode (in software code) must =00 if encode =03 or =07
* Selcall encode 04 & 08 with decode 00 is an invalid combination if type 2 selcall is fitted
* Selcall software code 0400 & 0800 can still be used for special applications (no decode, ident with PTT, and base selcall encode when operating ‘send’ button)
* Community repeater must have CTCSS encode/decode hardware available, and requires CTCSS tone to be entered.
* Voting isn't selectable without NSC810 I/O expander hardware
* Can only have duplicate channel numbers with voting selected on those duplicate channel numbers
* Cannot select economiser and voting on same channel
* Cannot select economiser and selcall on same channel
* Cannot select selcall and scanning on same channel
* Must have more than one channel in a vote group

Programming notes:

NEVER program in XXXXXXX for TX or RX frequency (VCO goes unlocked no matter what)

NEVER program in 000.000 for TX or RX frequency (TX will be *very spurious* no matter what)

IF you want to create a channel as RX only, program in a TX frequency that you are licensed for or a UHF CB channel if in U, W1 or W2 band.

BE VERY CAREFUL when programming in frequencies - there should only be 4 digits after the decimal point and no spaces to pad out the last position, and no leading zeros in the digits before the decimal point in the case of E band VHF.
Incorrect programming will have one of three effects:
radio will not power up (most common)
radio will have unlocked VCO on TX or RX or both (radio inoperable)
radio will cause severe interference due to the transmission of spurii.

3 scan groups available for FM91, FM97
2 scan groups available for FM92, FM93 and all waterproof radios
10 channels per scan group, with one of these only available as a priority channel
Channel number range for FM91, FM97 is 1-120; FM92 is 1-99; FM93 is 1-10
Power selection entry is by the letters H, M, L & S (for high, medium, low and selectable) and must be consistent with the type of radio.

CTCSS Tone selection table

67.0	Q	88.5	A	114.8	D	146.2	P	186.2	3	241.8	*
71.9	R	91.5	W	118.8	M	151.4	H	192.8	4	250.3	#
74.4	S	94.8	J	123.0	E	156.7	X	203.5	5	97.4	=
77.0	T	100.0	B	127.3	N	162.2	Y	210.7	6		
79.7	U	103.5	K	131.8	F	167.9	0	218.1	7		
82.5	I	107.2	C	136.5	0	173.1	1	225.7	8		
85.4	V	110.9	L	141.3	G	179.9	2	233.6	9		

Community repeater: select tone Z

Notes on FPP Version 3.0 (altered)

Normally, FPP will restrict programming of the radio in two ways: first, transmission and reception was fixed within the one band ie. If you had a U band radio 450 to 470 MHz, you could not program in any frequency for outside this band. In this altered version of FPP however, this restriction is lifted. Bear in mind that the VCO has its limits, however. Depending on your setup and tuning, up to 20 MHz reception outside of the band your radio was designed for may be possible. The second restriction was to only allow the selection of low or medium transmit power (1 or 5 Watts) on UHF CB. User selectable power levels was not an option. In this altered version of FPP, this restriction is also lifted, and as a by-product will allow high power selection for UHF CB as well. This altered version of FPP does have its bugs, too. If you try to program in a RX only channel by programming in xxxxxx or 000.0000 as the TX frequency, the radio will either not operate at all, go VCO unlocked on RX or transmit large amounts of spurious signals. BE WARNED!

Help! I get an error message saying the EPROM programmer is not initialised! That is OK. FPP was designed to work with an internal EPROM programming card specific to FPP. What you will need to do is to take the file that FPP saves to disk, and write that file as it is to an EPROM. If you create a personality called, say, “myfreq”, you will find a file called myfreq.bin in the directory x:\fpp\fm900\jobs\myfreq\ Take this file, and copy it to your root directory and then using whatever EPROM burning equipment/software you have, write this file to the EPROM.

Notes for Programming EPROMS using Electronics Australia kit (Dick Smith kit) with Eprom.exe software:

Eprom types suitable for use: 2764 (21-25 V pgm), 2764A (12.5 V pgm) or other 27x64-x EPROMs.

Set device switch on programmer
Set Vcc = 5V
Connect programmer, insert EPROM and turn on.

Select:

Vpp = 12.5V, write pulse 1ms, 1 pulse per byte, final pulse = 0, EPROM multi pulse. If this does not work, then try:

Vpp = 12.5V, write pulse 10ms, 1 pulse per byte, final pulse = 0, standard EPROM single pulse. If this does not work, then try:

Vpp = 12.5V, write pulse 50ms, 1 pulse per byte, final pulse = 0, standard EPROM single pulse. If this does not work, then try:

Vpp = 21V, write pulse 1ms, 20 pulses per byte, final pulse = 0, EPROM multi pulse. If this does not work, then try:

Vpp = 21V, write pulse 50ms, 1 pulse per byte, final pulse = 0, standard EPROM single pulse. If this does not work, then try:

Vpp = 25V, write pulse 50ms, 1 pulse per byte, final pulse = 0, standard EPROM single pulse.

When comparing binary files using DOS command ‘FC’, don’t forget the binary code output from FPP contains the date; two identical personalities may not match using FC due to the different date encoded into the binary file, at locations 1FE2, 1FE3 & 1FE4 from the start of the file, and the checksum, see note below.

Location 1F64 appears to be a data checksum for the whole data stored on the EPROM.

38: FM92E Test EPROM 68-88 Mhz Channel verses Frequency List:

6Mtest1L.bin	= Local Head	Test EPROM	10.7 MHz IF.
6Mtest2R.bin	= Remote Head	Test EPROM	10.7 MHz IF.
6Mtest3L.bin	= Local Head	Test EPROM	21.4 MHz IF.
6Mtest4R.bin	= Remote Head	Test EPROM	10.4 MHz IF.

Channel	Frequency
1	68 Mhz
2	69 Mhz
3	70 Mhz
4	71 Mhz
5	72 Mhz
6	73 Mhz
7	74 Mhz
8	75 Mhz
9	76 Mhz
10	77 Mhz
11	78 Mhz
12	79 Mhz
13	80 Mhz
14	81 Mhz
15	82 Mhz
16	83 Mhz
17	84 Mhz
18	85 Mhz
19	86 Mhz
20	87 Mhz
21	88 Mhz

39: Repairing FM92 VCOs by VK7ZRO

Over the last year, I have been involved in servicing about 70 FM92 E band radios. It soon became clear that these radios have a major problem with their VCO's. From my observations, the majority of radios have a VCO problem to some degree, particularly the older radios. Symptoms are varied but generally involve noise, cracking, microphony and in bad cases, the VCO dropping out of lock.

I test the VCO in the following way. Open the case and hinge open the receiver board. Power up the radio and connect a power meter. Monitor the radio on another receiver. Connect a voltmeter to the VCO test point. Press the PTT and note the power reading and the reading on the VCO test point. Gently tap the VCO with the back of your knuckle. With a good VCO there should some microphony but there should not be any cracking or frying noises. Bad VCO's are usually highly microphonic. Watch the meter on the VCO test point. It should remain quite stable, if not you have a faulty VCO. Watch the power meter. If the output drops to zero, the VCO has temporally gone out of lock and is definitely faulty.

After hearing various stories about the repair of the VCO's, I decided to develop my own method. I was in the fortunate position of having a number of radios that I could cannibalise for VCO's should my efforts fail. So far I have repaired about 15 VCO's with a 100% success rate. This includes both 10.7 and 21.4 IF VCO's. I still have a heap more to repair. The repair takes about half an hour including taking the VCO of the board, which is the hardest part. The repair method is quite drastic and is probably not for the faint hearted although it is quite easy to do.

Slide a sharp trimming knife under the label and remove it. Place the label to one side. The cover plate under the label has a number of notches in it. Place the point of the knife in one of the notches and slide the knife along under the cover, breaking the solder joint as you go. The cover plate should come off easily. Make sure the cover is flat and straight and place it to one side. Examine the VCO. Using a small screwdriver VERY gently lever the circuit board upward in the area of the coil on the end of the board. Look very carefully at the soldered joint under this coil. The odds are you will see movement indicating that the soldered joint has broken. Do this very carefully. On all boards that I have diagnosed as needing repair I have found that this joint has broken. I have also found other joints broken but not as often. I have had two cases where the board has been completely loose in the case when I have removed the VCO from the receiver board. I'll leave it to the experts to determine why this joint breaks.

Now for the interesting part. Clamp the VCO in a vice. Clamp it between the ends with the exposed board upwards. Only clamp the VCO lightly as it possible to distort the case. Now heat the VCO using a hot air gun on high. Use an ordinary heat gun, the type used for paint stripping available from most hardware stores. You'll have one for shrinking heat shrink tube anyway. Aim the gun at the VCO at an angle from underneath and at the side away from the pins. This is done to minimise heating of the board. Gently touch the pins with a small screwdriver. When the case is hot enough, you will see the board move as you touch the pins. This takes about 30 seconds to 1 minute. Remove the heat gun. Lift the board out of the case with a pair of long nosed pliers. This is a bit tricky. Be very careful, as it is quite likely that you will have melted the solder on the board and the components will be loose. Don't worry, they seem to survive. Put the board aside to cool. Immediately re-tin the solder pads in the case while the case is still hot. Leave plenty of solder on the pads. When the board has cooled, turn it over and re-tin the board where it solders to the pads in the case. Leave plenty of solder on the board. You could find that some of the silver plating has come off the board. This is why I recommend using plenty of solder. The aim is to get the board soldered to the case in as many places as possible, particularly the pad under the end coil. Put the board back in the case. Position the top edge of the board against the top of the case. Position the board laterally so that the gap between it and the case is the same both ends. This is important, as you won't be able to mount the VCO onto the receiver board unless this alignment is correct. It is critical so make sure its right. Re-heat the case. After about 30 seconds gently push down on the centre of the board clear of any components. When the board is hot enough, the board will sink down onto the pads as you push gently on it. Remove the heat gun. It is important that the board is pushed down otherwise the alignment won't be correct. Be careful not to touch any components as you may have melted the solder on the board. Now place the cover onto the case while the solder is still melted. Hold the cover down onto the case as the solder sets. Wait a couple of minutes and while the case is still warm (not hot) put the label back on. Let it cool. Done properly, the VCO should not look like anything has been done to it.

Put the VCO back onto the receiver board, align the VCO coils and retest as above. I think you will be amazed at the difference.

Have fun. Ron VK7ZRO.

P.S. Since writing the above, I have had one repaired VCO become very microphonic after about 6 months. Running the tests that I outlined above showed that the VCO was OK apart from being very microphonic. It was so microphonic that mechanical feedback from the speaker produced an audio howl. The most sensitive area was around the Rx coil. When I removed the slug an extremely small piece of broken slug fell out. This fixed the problem. There must be a moral here somewhere.

40: 6 Meter Antenna:

41: Weekly Procedures:

6th March 2007 - Lecture covers this document; what is needed, what we shall do, brief description of what will be done, etc.

13th March 2007 – Upgrade your soldering techniques by Bruce VK2ZAD. This lecture will cover the file **solderin.ppt**.

20th March 2007 – Tonight will cover the following procedures:->

- Members to bring along:-
    1. Small & A4 GLAD bags.
    2. No 1 Philips Screwdriver to remove FM92E covers.
    3. In-Line Fuse holder – VK2ZAD will be supplying.
    4. Fuse 8 or 10 Amp. For fuse holder.
    5. 12 Volt 10 Amp power cable.
    6. Old tooth brush for cleaning.
- Transceivers will be given out.
- Remove cover on Synthesiser board side from the FM92E Transceiver, this is the side closest to the two sockets at the rear of the FM92E transceiver. Then mark the cover with your call-sign, then place it & the cover's screws into their GLAD bags & mark your Glad bags with your call sign. Should be a minimum of 4 screws.
- Note the center frequency or single frequency your transceiver was working on.
- Undo screws, holding the Synthesiser board down. Swing Synthesiser board out.
- Remove can covering the EPROM socket. [4 screws] Place Can & Screws into Glad bag.
- Solder single fuse holder onto positive power lead [red cable from transceiver; **NOT** red lead with black dashes]. Or; remove fuse section on earth lead from transceiver, then connect this section via fuse to the positive section of the red wire from the transceiver. * This can be done before EPROM can removal.
- Solder 12 Volt black cable power cable onto the earth lead from the transceiver [red with black dashes] and red onto end of fuse. * This can be done before EPROM can removal.
- Plug in Test EPROM **6mtest1l.bin, 6mtest2r, 6mtest3l or 6mtest4r** see page 3 on "Files associated with this project". For testing the FM92E Transceiver from 68-88 MHz. EPROM notch points away from hinges of Synthesiser board.
- Apply 12 Volts to the FM92E Transceiver. Check that audio emanates from the speaker
- With Marconi VHF Signal Generator check that FM92E receiver mute works under 4 micro Volt on test frequency and note this reading.
- Connect VHF Power Meter and press PTT button so a power out is registered on the Power Meter. Note the power reading.
- Disconnect Power.
- Remove EPROM.
- Plug in EPROM **6m1loc.bin**, **6m2rem.bin**, **6m3loc.bin, 6m4rem.bin, 6m5.bin, 6m6.bin, 6m7loc.bin or 6m8rem.bin** See page 3 on "Files associated with this project". to EPROM socket. EPROM notch points away from hinges of Synthesiser board.
- Replace EPROM tin box over EPROM, 4 screws.
- Screw down the Synthesiser board.
- Replace cover for Synthesiser board side.
- Clean up FM92E transceiver. * This can be done at any time.

27th March 2007 – Break, but group can continue on with last week's procedure, if not finished, eg fuse holder.

03rd April 2007 –

- Recheck that your EPROM receiver frequency is correct, NOT off frequency as some people had last Tuesday nite. If they were off-frequency [Ken (VK2KJ) will recheck your FM92E transceiver again], we will change your EPROM. Those that were given the short FM92E, we will change them back to the long FM92E that was converted from the batch of FM91E's.

  Ken, VK2KJ will recheck that your FM92E is on the frequency, again; then

10th April 2007 – Easter Break. The Rev Bruce, VK2ZAD has to recover from chocolate over eating.

17th April 2007 –

- Members to bring along:-
  1. Soldering Iron. [only if you have one]
  2. De-Soldering Iron. [only if you have one]
  3. 0.63mm [B&S 22] Enamelled Copper Winding Wire. [Leon has some spare wire!!!!]
  4. Paper if they want a printout of tonight's operation. [approximately 10 pages]
- Remove cover on Receiver board side from the Philips FM92E Transceiver, this is the side opposite from the two sockets at the rear of the FM92E transceiver. Then mark the cover with your call-sign, then place it & the cover's screws into their GLAD bags & mark your Glad bags with your call sign. Should be a minimum of 4 screws.
- From **http://vk2afy.dyndns.org/6mconv/6mfiles.htm** site, download the document pages from sections 19 & 20 of file **6m_con.pdf**, or from the directory "**D:FM92E Conversion to 6 meters**" of the main computer in the operations room.
- Do **VCO Change** – See section 19 & 20, starting on page 20 of this document.

24th April 2007 – Continue with week 17th April 2007.

01th May 2007 – Video Movie on climate change. To give Leigh a rest for his eyes.

08th May 2007 – Start Receiver/Mixer Module change.

- Members to bring along:-
  1. 4 of capacitors.
  2. 0.80mm [B&S 20] Enamelled Copper Winding Wire. [Leon has some wire!!!!]
- From **http://vk2afy.dyndns.org/6mconv/6mfiles.htm** site, download the document pages from sections 21 & 22 of file **6m_con.pdf**, or from the directory "**D:FM92E Conversion to 6 meters**" of the main computer in the operations room.
- Do **Rx RF Amp/Mixer Change** – See section 21 & 22, starting on page 29 of document **6m_con.pdf**.

15th May 2007 – Continue with week 08th May 2007.

22th May 2007 – Continue with week 08th May 2007.

29th May 2007 – Continue with week 08th May 2007.

05th June 2007 - Video Movie.

12th June 2007 - Start Tx PA Change.

19th June 2007 - Continue with week 12th June 2007.

26th June 2007 - Continue with week 12th June 2007.

03rd July 2007 - Continue with week 12th June 2007.

10th July 2007 - Continue with week 12th June 2007.

17th July 2007 - Continue with week 12th June 2007.

24th July 2007 - Continue with week 12th June 2007.

31st July 2007 - Continue with week 12th June 2007.

42: Philips FM91 Schematic Diagrams:

43: 6 Meter Repeaters:

Output	Input	Call	Service Area	Status	Grid Sq	Date last confirmed	
53.550	52.550	VK2RAY	Albury	unknown	QF33	Jan99	
53.550	52.550	VK2RIC	Lismore-Casino	Op	QG61	Jan 2007	
53.550	52.550	VK2RSJ	Sydney West	scrapped, defunct			
53.550	52.550	VK3RMH	Melbourne N.E.	Operational	QF22	Dec 2006	Wattle Glen (30km NE of Melb)
53.575	52.575	VK2RPW	Grundy Mtn,Walcha	UC	QF59	Feb98	
53.575	52.575	VK2RSM	Tumut, Snowy Mtns	Operational	QF43	Jan 2006	
53.575	52.575	VK3RDD	Melbourne, S.E.	Operational	QF21	Feb 2007	Cockatoo (40km SE Melb)
53.600	52.600	VK2RNW	MtKaputar,Narrabri	Operational	QF40	Jun 2003	
53.600	52.600	VK3RMR	Gippsland	Operational	QF32	Dec 2006	Aberfeldy, nth of Morwell
53.625	52.625	VK2RSN	Newcastle	Operational	QF57	Jan 2006	
53.625	52.625	VK3RHF	Melbourne East	Operational	QF22	Dec 2006	Mt Dandenong/Olinda . 53.625/ 52.625 & 29.640/29.540 & 1273.400/1293.400 linked together, access via 438.750/ 433.750 using 88.5 Hz CTCSS
53.625	52.625	VK4RXD	Sunshine Coast	Operational	QG62	Dec 2006	
53.625	*52.625*	--	*Otago, N.Z.*	*Operational*		*Jan 2007*	*central south island. ZL rptrs are licensed, but dont have callsigns*
53.650	52.650	VK2RMP	Wollongong	Operational	QF55	Feb 2007	50w, split 1/2 wave verticals
53.675	52.675	VK2RMB	Sydney, Nth East	being rebuilt	QF56	Dec 2006	Terrey Hills
53.675	52.675	VK3RAD	Melbourne East	Planned			Mitcham
53.675	52.675	VK3RTN	NE Vic.	Operational	QF22	Oct99	Mt.Gordon, Marysville, 80 km NE Melb. linked to VK3RAG 146.775 and VK3RAD 438.525 VK3RAD/RAG has 123 Hz CTCSS access.
53.700	52.700	VK2RGN	Goulburn	Operational	QF45	Feb 2007	idents every 10 mins, split site.
53.700	52.700	VK4RGY	Sunshine Coast	Operational	QG63	Feb 2007	Maroochydore, 50w
53.725	52.725	VK4RGA	Gladstone	Operational	QG56	Dec 2006	
53.725	52.725	VK4RIK	Cairns	scrapped, defunct			
53.725	52.725	VK4RLB	Brisbane Sth	Operational	QG62	Jan98	Logan. linked to 70cm rptr
53.725	*52.725*	--	*Auckland N.Z.*	*Operational*	*RF73*	*Jan 2007*	*ZL rptrs are licensed, but dont have callsigns*
53.750	52.750	VK5RDX	Adelaide Sth	now moved to VK5RSB			VK5RDX 53.750 and 438.075 are to be linked.
53.750	52.750	VK5RSB	Adelaide Sth	Operational		Dec 2006	Summertown.
53.750	*52.750*	--	*Wellington, NZ*	*off air*	*RE78*	*Jan 2007*	*ZL rptrs are licensed, but dont have callsigns*
53.775	52.775	VK4RAF	Mackay	scrapped, defunct			
53.775	52.775	VK4RBP	Atherton Tablelands	Op	QH22	Dec 2006	Near Butchers Creek, 3,500ft ASL, 60w 1/2wav vertical
53.775	52.775	VK4RRC	Redcliffe, N.Bris.	scrapped, defunct			
53.775	52.775	VK5RAD	Adelaide North	Op 30w	PF95	Jan 2007	
53.800	52.800	VK6RAP	Perth	Op		Nov98	
53.800	*52.800*	--	*Mt.Climie, NZ*	*OFF AIR*		*Jan 2007*	*n.e. of Wellington. ZL rptrs are licensed, but dont have callsigns*
53.825	52.825	VK2RMS	Cooma, SE NSW	cancelled		Jan 2003	
53.825	52.825	VK7RAD	Hobart	Op	QE37	Jan 2007	
53.825	52.825	VK7RNW	Lonah, NW Tas	moved, now VK7RMD			
53.825	52.825	VK7RMD	Mt.Duncan,NW Tas	Op		Dec 2006	

Output	Input	Call	Service Area	Status	Grid Sq	Date last confirmed	
53.850	52.850	VK2RWI	Dural, NW Sydney	Operational	QF56	Feb 2007	
53.850	*52.850*	*--*	*Christchurch, NZ*	*Operational*	*RE66*	*Jan 2007*	*ZL rptrs are licensed, but dont have callsigns*
53.875	52.875	VK2RBM	Lawson, Blue Mtns	Op	QF56	Feb 2007	123Hz access
53.875	52.875	VK7RAA	Mt.Barrow, NE Tas	Op	QE38	Jan 2007	
53.875	52.875	VK7RNT	Tamar Valley	defunct			
53.900	52.900	VK3RMS	Melbourne East	Operational	QF22	Nov99	Mt Dandenong/Olinda (30km East) Receiver site Ferny Creek.
53.925	52.925	VK1RGI	Mt.Ginini	planned	QF44		
53.925	52.925	VK4RBX	The Knobby	unknown			
53.925	52.925	VK8RDX	Darwin	defunct, scappped		Jan 2007	CTCSS 123 Hz
53.950	52.950	VK4RBL	Brisbane sth	?	QG62		Bardon
53.975	52.975	VK3RGM	Mt.Buller NE Vic.	?			VK3RGM 53.975 and VK3RUG 438.175 are linked.
53.975	52.975	VK4RBR	Brisbane	?	QG62		

FM Voice simplex:

52.525 International FM voice Call Freq. (everybody hangs out here)
53.500 VK National FM voice Call Freq. (nobody uses this)

Packet simplex:

53.000 VK2RAG	BBS	Gosford	1200bps
53.000 VK7RPU			1200bps
53.025 VK3RPK			1200bps
53.050 VK2IO	BBS	Sydney	1200bps
53.100 VK2RPH		Sydney Nth	1200bps
53.100 VK3RUG			1200bps
53.100 VK3RMG		Seymour/Yea	1200bps
53.100 VK4BOO-1	BBS	Sunshine Coast	1200 & 2400bps
53.100 VK4RBP		Atherton Tablelands	1200/2400

44: Version Amendments:

3.5 - Circuit heading added.
3.6 - Weekly procedure, front page added for printing, VCO update, PA capacitors & coils added.
3.7 - Version amendment added, PA Deviation Potentiometer Replacement now included.
3.8 - PA Board - capacitors under board, capacitors – Dick Smith part numbers; amended number of screws holding down the receiver board, changed VCO de-soldering section, added tooth picks to de-soldering section.
3.9 - VCO change – coil placement, page 31. VCO change – unsoldering coils, page 30. VCO change – Removing glue from coil formers, page 30. VCO change - Scrape off old glue from former with scalpel and fine file, page 30. VCO change - [or 146mm of Leon's enamel coated wire, 0.??mm] , page 30. VCO change - Don't forget to use a hot glue gun or, page 30. VCO test – check with signal generator, page 36. VCO Tune - VCO Tune Check References: added & pin 19 check of VCO, page 37. Renumber of all pages after page 36.

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